Directorate General of Renewable Energy and Energy Conservation, Ministry of Energy and Mineral Resources

United Nations Development Programme
Country: INDONESIA
PROJECT DOCUMENT

Project Title: Market Transformation through Design and Implementation of Appropriate Mitigation Actions in the Energy Sector (MTRE3)

UNDAF 2016-2020 Outcome(s):	Outcome 3: By 2020, Indonesia is sustainably managing its natural resources, on land and at sea, with an increased resilience to the effects of climate change, disaster and other shocks.
UNDP Strategic Plan 2014-2017 Primary Outcome:	Outcome 1: Growth and development are inclusive and sustainable, incorporating productive capacities that create employment and livelihoods for the poor and excluded
UNDP Integrated Results and Resources Framework 2014-2017 Outputs:	Output 1.5: Inclusive and sustainable solutions adopted to achieve increased energy efficiency and universal modern energy access (especially off-grid sources of renewable energy).
Expected CPD 2016-2020 Outcome(s):	3. Sustainable natural resources management and increased resilience.
Expected CPAP 2011-2015 Output(s):	2.2.1. National energy policies and guideline developed and integrated into sub-national development plan. 2.2.2. Sub-national authorities and key partners are able to implement programmes, mobilize resources and develop public-private partnership for RE/EE, which will contribute to the reduction of national greenhouse gases emission.

Implementing Entity/Responsible Partners: Directorate General of Renewable Energy and Energy Conservation, Ministry of Energy and Mineral Resources (MEMR)
Executing Entity/Implementing Partner: Ministry of Energy and Mineral Resources (MEMR)

Brief Description

The objective of *"Market Transformation through Design and Implementation of Appropriate Mitigation Actions in the Energy Sector (MTRE3)"* project is to support the design and implementation of appropriate climate change mitigation actions in the energy generation and energy end use sectors. It is intended to incrementally support Government of Indonesia to achieve the voluntary GHGs emission target by supporting effective implementation of RAN-GRK and RAD-GRK in Energy Sector. During 5 years of its implementation, the project will put in place enabling environment and removing barriers to sustainable market of renewable energy and energy efficiency. To achieve this purposes, the project will focus its interventions in renewable-based electricity generation and energy efficiency in commercial building interventions. The project will consist of 3 main components: (1) Climate change mitigation options for the RE-based energy generation and energy efficiency; (2) Market transformation through implementation of appropriate mitigation actions; (3) Measurement, Reporting and Verification (MRV) system and national registry for mitigation actions. The implementing partner of the project is Ministry of Energy and Mineral Resources.

Programme Period:	2016-2020	Total Allocated Resources:	USD	68,125,000
		• GEF:	USD	8,025,000
ATLAS Award ID:	00086173	• UNDP	USD	100,000
Project ID:	00093506	• MEMR	USD	8,000,000
PIMS #:	4673	• **Private Companies**	**USD**	**52,000,000**
Start Date:	June 2016	○ PT. Multi Fabrindo Gemilang	USD	2,000,000
End Date:	March 2021	○ PT. Pasadena Engineering Indonesia	USD	10,000,000
Management Arrangement:	NIM	○ PT. Daun Biru	USD	40,000,000
PAC Meeting Date:	April 2016			

Agreed by Director General of Debt Management, Ministry of Finance — Date/Month/Year — Signature

Agreed by Director General of Renewable Energy and Energy Conservation, Ministry of Energy and Mineral Resources — Date/Month/Year — Signature

Agreed by Country Director, UNDP Indonesia — Date/Month/Year — Signature

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List of Acronyms

APR/PIR	Annual Project Reports and Project Implementation Review reports
AWP	Annual Work Plans
BAU	Business-as-Usual
BKF/FPO	*Badan Kebijakan Fiskal/* Finance Policy Office
BRH	Bangkok Regional Hub
CO_2	Carbon dioxide
CTA	Chief Technical Advisor
DAK	Special Allocation Fund
DEN	National Energy Council
DGNREEC/EBTKE	Directorate General of New-Renewable Energy and Energy Conservation
DME	*Desa Mandiri Energi*/ Self-sustained Energy Village
DNPI/NCCC	Indonesia National Council of Climate Change
DPR	*Dewan Perwakilan Rakyat* / Parliament
DPRD	Dewan Perwakilan Rakyat Daerah / Local Parliament
DPR (bank)	Dividend Payout Ratio
DJK	Directorate General of Electricity (MEMR)
EC&EE	Energy conservation and energy efficiency
EE	Energy efficiency/energy efficient
ES&L	Energy Efficiency & Labeling
ESCO	Energy Service Company/Energy Saving Company
EECCHI	Energy Efficiency and Conservation Clearing House Indonesia
FiT	Feed in Tariff
GBCI	Green Building Council Indonesia
GEF	Global Environmental Facility
GHG	Greenhouse gas
GoI	Government of Indonesia
GBCI	Green Building Council Indonesia
HPMP	HCFC Phase-out Management Plan (HPMP)
IGA	Investment Grade Audit
IKE	Energy Consumption Intensity
IOCs	Incremental operating costs
IPMVP	International Monitoring and Verification Protocol
KEN	National Energy Policy
kt	kilo tons
LEAP	Long-range Energy Alternative Planning
LFA	Logical Framework Analysis
M&E	Monitoring and evaluation
MACC	Marginal GHG Abatement Cost Curve
MEMR/ESDM	Ministry of Energy and Mineral Resources
MEPS	Minimum energy performance standards
MLF	Multilateral Fund
MMT	Million metric tons
MoF	Ministry of Finance
MoEF	Ministry of Environment and Forestry
MoPW	Ministry of Public Works
MRV	Measurement, Reporting, Verification
MTRE3	Market Transformation for Renewable Energy and Energy Efficiency
NAMA	Nationally Appropriate Mitigation Action
NEX	Nationally-executed
NPD	National Project Director
NPM	National Project Manager
ODS	Ozone depleting substances
OJK	*Otoritas Jasa Keuangan* (Financial Services Authority)
PIF	Project Identification Form
PLN	National Electricity Company

PMU	Project Management Unit
PPG	Project Preparation Grant
PPM	Project Planning Matrix
PSC	Project Steering Committee
QPR	Quarterly Progress Reports
R&D	Research and development
RAN-GRK	National Action Plan to Reduce Green House Gases
RAD-GRK	Regional Action Plan to Reduce Green House Gases
RCU	Regional Coordination Unit
RE	Renewable Energy
RIKEN	National Master Plan of Energy Conservation
RUED	Local Energy Plan
RUEN	National Energy Plan
SNI	Indonesia National Standard
TWG	Technical Working Group
UNDP	United Nations Development Programme
UNFCCC	United Nations Framework Convention for Climate Change

I. SITUATION ANALYSIS

1. National and Global Context

1. Indonesia's primary energy supply mix is dominated by crude oil (47%), followed by coal (27), natural gas (21%), hydropower (2.5%) and geothermal (1%) and biofuel (0.19%) excluding biomass for firewood and charcoal[1]. The country has abundant renewable energy resources potential i.e. 29 GW geothermal, 75 GW hydropower, 50 GW bioenergy, 49 GW ocean energy, solar energy (solar insolation of 4.8 kWh/m^2/day), and wind energy (average wind speed 3-6 m/s)[2]. Indonesia is facing long-term challenges to its energy security. The expected growth in electricity demand in the coming years is 8% annually, the current energy mix leaves Indonesia vulnerable to the price of imported oil due to subsidizes. Diversification of primary energy sources is thus important.

2. The country's Law No. 30/2007 on Energy states that new energy and RE resources shall be managed by the state and utilized in a just, sustainable, rational, optimum and integrated order for the greatest welfare and prosperity of the people[3]. According to this law, the control and management of energy resources is by both the central and local governments, depending on the jurisdiction. Moreover, the state owned electricity enterprise (PLN) is the sole purchaser and provider of electricity in the country. This ensures that all electricity is controlled by the state.

3. The Government of Indonesia (GoI) has enacted number of sustainable energy policies and regulations in response to the Energy Law. These include, Government Regulation No.79/2014 on National Energy Policy, which sets a target by 2025 of a 23% contribution from renewable energy (RE) in the national primary energy mix and average of 1% annual reduction in final energy intensity through various energy efficiency and energy conservation measures.

4. The recently issued Presidential Regulation No. 02/2015 on the Medium-term National Development Plan (RPJMN) 2015-2019 also emphasizes the importance of the contributions of renewable energy, energy efficiency and access to energy to support the national energy sovereignty agenda. The baseline situation showed 4% contribution of RE in the country's primary energy mix; 500 BOE/IDR billion primary energy intensity; and, 81% electrification ratio. Based on the RPJMN, by 2019, RE will account for 10% to 16% of the country's primary energy mix, and the installed RE-based power generation capacity would be about 7.5 GW. In terms of energy efficiency, the RPJMN targets a primary energy intensity of 472 BOE/IDR billion as manifested by a forecast energy saving of 12.7% compared to a business-as-usual (BAU) energy demand scenario in 2014. Moreover, the target electrification ratio by 2019 is 97%.

5. The other notable regulations include: (a) Presidential Regulation No. 61/2011 establishing a National Action Plan to reduce greenhouse gas emissions (RAN-GRK); (b) Local Action Plan to reduce GHG emissions (RAD-GRK, 2012); and, (c) Presidential Regulation No.71/2011 on establishing a National GHG Inventory. All of these regulations support the GHGs emission reduction framework of Indonesia and reflect the Government's voluntary commitment to reduce GHG emissions by 26% by 2020 through national efforts or by 41% with international assistance. National and local actions both on renewable energy (RE) and energy efficiency (EE) have been prioritized to achieve the set GHG emission reduction targets. The energy and transportation sectors are targeted to reduce 38 million tons of CO2 by 2020 (26% target emission reduction).

6. The RAN-GRK and RAD-GRK are working documents with recommended climate change mitigation (CCM) actions covering the agriculture, forestry, industry, and transport and energy

[1] Handbook of Energy and Economic Statistics of Indonesia, Pusdatin MEMR, 2012.
[2] Presentation of Indonesia National Energy Council, 2014
[3] Law No. 30/2007 article 4

sectors of the country. Collectively, they are referred to as the compilation of Indonesia's potential NAMAs, in which some of the actions will be implemented as unilateral NAMAs (targeting 26% emission reduction from the business-as-usual GHG emission trajectory in 2020) and as supported NAMAs (targeting additional 15% emission reduction from BAU emission trajectory in 2020). Nonetheless, the suggested CCM actions in these plans are only indicative, without binding commitment and uncertain budget allocations for their implementation. Despite their inclusion in the RAN-GRK or RAD-GRK, there is no assurance that these CCM actions will be funded and implemented. Indonesia had referenced RAN-GRK as basis for development of Indonesia's Intended National Determined Contributions (INDC) that was summited to the UNFCCC in September 2015.

7. Based on the final draft (2015) of Indonesia's First Biennial Report (BUR) to the UNFCCC, the country's energy sector emits about 512 million tCO2eq or 32.2% of Indonesia's total GHG emission. There is urgency in the energy sector to take action in reducing the level of GHG emissions by involving all the relevant stakeholders. This proposed project, therefore, will contribute to the formulation of the country's Third National Communication (3NC) to the UNFCCC, as well as the Biennial Update Reports particularly for the sections about the energy sector. Furthermore, by supporting market transformation towards the application of renewable energy and energy efficiency technologies, the proposed project is relevant with the aforementioned commitments of the national government.

8. The GoI acknowledges the importance of private sector investments in achieving the above-mentioned emission reduction targets. To attract such investment, the GoI issued Regulation of Minister of Energy No.04/2012 on the electricity purchase price by PLN for small and medium scale RE-based power generation with capacity up to 10 MW. Furthermore, this regulation has been updated for purchasing of RE-based generated electricity up to 10 MW and its excess power by PLN for specific types of RE-based power generation:

    - MEMR Regulation No. 17/2013 on Feed-in Tariff (FiT) for Photovoltaic Solar Power Plant.
    - MEMR Regulation No.22/2014 on Feed-in Tariff for Hydropower generation.
    - MEMR Regulation No.27/2014 on Feed-in Tariff (FiT) for Biomass and Biogas Power Generation.

9. In the area of energy efficiency, the industrial and commercial sectors use 45% of the national total final energy, followed by the residential sector. Both sectors are expected to save energy of respectively 17% and 15% by 2025[4]. The national government has issued Regulation No. 70/2009 that mandates all entities with energy consumption more than 6,000 TOE per year to establish an energy efficiency program, appoint an energy manager within the organization, conduct regular energy audits, and implement and report the audit findings and recommendations. The Asian Development Bank (ADB) in 2009 estimated the amount of investments for energy efficiency improvements in Indonesia at about US$ 4 billion, 25% of which is for EE improvement in commercial buildings, and 75% for retrofitting industrial processes and systems.

10. Specific policies and regulations on energy efficiency have been put in place by the national government to manage energy demand from the industrial, commercial and residential sectors, which are the energy end use sectors in the country. The pertinent regulations are:

    a) National Master Plan of Energy Conservation (RIKEN, 2005) – This targets the reduction of the country's primary energy intensity at least 1% per year until 2025.
    b) Government Regulation No. 70/2009 on Energy efficiency – This refers to the: (a) obligation for large energy consumer to conduct energy audit and designate energy manager, and (b) application of energy efficiency labelling for home appliances.

[4] Draft of Master Plan for Energy Conservation, 2013.

c) Presidential Instruction No. 13/2011 on Water and Energy Saving – This mandates all government-owned building, equipment and activities to save 20% energy consumption compared to the average consumption level in February 2011.
d) Regulation of the Minister of Energy No.13/2012 on Electricity Usage Saving
e) Regulation of the Minister of Energy No.14/2012 on Energy Management
f) Regulation of the Minister of Energy No.01/2013 on Control in consumption of subsidized fuel.
g) Regulation of the Minister of Energy No. 18/2014 on Standard and Labeling for CFL.
h) Regulation of the Minister of Energy No.17/2015 on Minimum Energy Performance Standard and Labeling for Air Conditioner.

11. Furthermore, Indonesia's policy on the provision of fuel and electricity subsidies has created market distortion for RE and EE investments. Since the last decade, the energy subsidy has been increasing annually. In 2014, the energy subsidy for fuel and electricity reached 17% of the total national budget from about 15% in 2013. It is acknowledged that the energy subsidy restricts greater renewable energy and energy efficiency potentials and efforts to displace/substitute oil with indigenous RE resources. The government is working to gradually remove the energy subsidies. But in the meantime, the remaining bulk of subsidies are still substantial.

12. Efforts have been taken by GoI to foster and accelerate the development of renewable energy utilization and energy efficiency technology applications, through the issuance of various enabling policies and regulations. However, the achievements so far are still far from satisfactory. Renewable energy and energy efficiency initiatives still heavily depends on government budget, and are not driven by the market. The National and Local Action Plans (RAN-GRK and RAD-GRK) for the energy sector have not been able to attract private investments. Despite the endowment of abundant RE resources (hydro, biomass, solar) in some provinces, the electrification ratios in these provinces are still low[5]. In general, the private sector is not interested to invest in the provinces, particularly in those where the economic growth is low. As government funding is limited, the sustainability of renewable energy and energy efficiency projects in these areas is a cause of concern with the private sector.

2. Summary of Barriers to the Transforming Market towards Appropriate Renewable Energy and Energy Efficiency Projects

13. Despite the GoI efforts in promoting RE development and utilization and EE technology applications, and the fact that there is significant potential for utilizing RE resources in generating energy with low GHGs emission, the increased share of RE in the national primary energy mix and the improved primary energy consumption index both remain much to be desired. There are clearly significant barriers, some of them persistent, that hinders the widespread application of RE and technologies in the energy generation and energy end use sectors of the country. The energy market in Indonesia shows limited transformation towards the use of more sustainable renewable energy resources for power generation, and towards more energy efficient appliances, equipment and systems. Studies in respect to project developers and financial institutions reveal a number of barriers that hinders the realization of the energy saving and GHG emission reducing potentials of RE and EE technologies. Among others, project developers need improvements in the access to affordable finance, technical capacity, project permitting procedures, and revenue generation stability for RE/EE projects. Tailored government support can address these barriers and provide a much needed boost for the sector. During the consultation meetings with key stakeholders during a Logical Framework Analysis workshop, the following barriers were confirmed as

[5] For example, the provincial electrification ratio (% of households with access to electricity) is 61% (511,233 households) in Jambi; 60.8% (900,679 households) in Riau; 47% (132,556 households) in West Sulawesi; and 48% (522,221 households) in East Nusa Tenggara. Ref: PLN Annual Report 2013.

hindering the energy market transformation towards sustainable RE technologies and energy saving EE projects:

Policy, Institutional and Capacity barriers in the planning, prioritization and implementation of appropriate climate change mitigation actions

- There is low capacity in planning and prioritization of appropriate and cost-effective mitigation actions at the national and sub-national levels in the energy generation and energy end-use sectors. The selection of mitigation actions is not undertaken in an integrated and systematic manner, resulting in fragmented and uncoordinated approaches that struggle to attract private-sector investment. Although the RAN-GRK (National mitigation action plan) and RAD-GRK (Local mitigation action plan) list mitigation actions and establishes emission reduction targets, these were not developed based on accurate data on emission projections and mitigation potential per sector and there is no business-as-usual (BAU) baseline against which to measure the reductions achieved. These factors present a considerable challenge to translating RAN-GRK and RAD-GRK activities into Nationally Appropriate Mitigation Actions (NAMAs), even though the Government considers the NAMA approach to be crucial for the transparent and verifiable implementation of the RAN-GRK/RAD-GRK and for assisting Indonesia to mobilize international climate finance and obtain UNFCCC recognition for its mitigation efforts.
- The RAN-GRK of MEMR for 2010 to 2020 states target of 1.2 GW target for installed RE and 4.5 million tCO_2 reduction. To date, the main funding source for achieving the target was solely from government budget. The low rate of implementation is due to the fact that RAN-GRK is somehow just a list of climate change mitigation activities that did not benefit from a feasibility assessment. Thus, the implementation is still uncertain due to limited government budget and low interest from the private sector in investing in RE/EE projects. Similar to the RAN-GRK implementation, the RAD-GRK implementation report (2010-2014) in targeted pilot provinces (Jambi, West Sulawesi, NTT and West Kalimantan provinces) showed that common contributor to the GHGs emission is from fossil-fueled power generation. However, the climate change mitigation actions were mainly on small size off-grid power generation, such as solar home system, household biogas generation and utilization, and microhydro-based power generation since most of these mitigation actions were funded from the provincial budget. The implemented actions resulted in rather insignificant emission reductions compared to the provincial target for GHG emission reductions by 2020. The provincial governments have limited capacity to attract private investment in the implementation of their respective RAD-GRK and to prioritize the cost-effective mitigation actions. Further, there is no effective coordination among government authorities and institutions working on RE and EE in terms of project planning, policies and monitor progress of RE/EE implementation.
- In meeting emission reduction target through energy efficiency improvement efforts, the implementation of energy audit recommendations is low. Based on the monitoring report of 344 buildings and industries that have been audited during 2011-2012, around 57% implemented no-cost/low-cost recommendations (awareness action, internal management policy, monitoring), medium cost measures - 39%; and, high cost - 28%. Moreover, the GoI has not yet established any green/EE building certification system, while the Green Building Council Indonesia (GBCI), a private institution, has issued voluntary Greenship Certificates for new and existing buildings.
- There is currently no standardized, official approach to establish and to update reference baseline data or assessing individual climate change mitigation actions' contributions to the national GHG emission reduction targets, both in the energy generation and energy end-use sectors. The Ministry of Environment and Forestry (MoEF) through its SIGN programme (National GHGs Inventory System) has launched in 2012 guideline for GHG Inventory for various sub-sectors and has conducted GHG inventory at provincial level in 2014. Such GHGs inventory guidance must be used as reference by the respective line ministries and provincial governments in monitoring, and reporting their contribution to RAN-GRK and

RAD-GRK. However, the adoption of the guidelines by other government agencies is slow due to the lack of coordination, and required capacities to utilize the system for development planning. Moreover, the provincial GHG Inventory works need improvement. The resulting inventories lack detailed information, making them insufficient for emission reference baseline setting and for planning and prioritization of climate change mitigation actions at the provincial level. Also, there is no agreed institutional arrangement between sub-national and national level (SIGN) on updating GHG inventory and in maintaining the GHG inventory database for planning and monitoring purposes of mitigation actions at the provincial level.

- There is no streamlined procedure for getting required permits for projects approvals. This is in terms of the submission of required supporting documents, institutional roles in approving certain documents while issuing permits for the implementation of RE and EE projects. In Indonesia, the permit issuance process for RE is complex and involves many stakeholders and oftentimes take a long time. A lot of effort is required to monitor the progress of permit processing, and that adds cost to project developers. In case if RE projects are to be implemented on forest land, the issuance of permits may take a long time for processing and as a consequence other permits that have already been issued become expired and this situation hinders the implementation of RE. The delays in project implementation will be very costly to project developers and can cause cost overrun that leads to the delay in paying back any debt financing for the project investment cost.

Awareness and Market Barriers

- Insufficient guidelines on RE project development procedure, particularly for small-medium size RE-based power generation projects of up to 10 MW. Ministerial regulation only contains a list of administrative and technical documents required, but lack of procedural guideline for project developers on how to accomplish the required document.
- Limited availability and accessibility of reliable data on RE resource potentials and geo-reference for off-grid and on-grid, as well as limited information about potential of energy saving and business opportunities in the implementation of EE projects to attract investments.
- Inefficient permitting system for RE/EE investment, which involves various institutions from district, province and national levels of government, land status problem and lack of transparency in the permitting process. There is no existing institution or system that particularly guides potential investors and project developers of the possible investments in RE and EE at local and national government level.

Technical Barriers to sustainable RE/EE project implementation

- Limited infrastructure hinders the implementation of RE projects due to inaccessibility of project sites. This is a major barrier particularly for biomass-based energy projects, which oftentimes results in the unreliable supply of biomass fuel inputs. A number of remote locations exist in the country where grid connectivity is not a feasible option.
- Low level of technical capacity and poor quality of services from local energy service providers. Poor quality of feasibility studies are often the case in various renewable energy projects. Also, low quality of technical documents such as geological survey, topography, hydrology, and energy resources supply sustainability and basic engineering, have contributed to the unsustainable RE projects that have been implemented in the past. Particularly for those RE projects that are handed over to sub-national levels of government, the sustainability aspects such as technical capacity for operation and maintenance of energy infrastructure, buy-in from local stakeholders and local budget allocation are commonly overlooked during the project design. The current technical capacity in the identification, implementation, monitoring and evaluation of initiatives that employ RE/EE technologies lies mainly at the national level.
- There are very few national experts that are able to offer energy efficiency solutions in the buildings sector. There is limited guidance available on energy efficiency performance

standards, regulations and certification institutions that can guide in the energy efficient design of buildings or facilities and can create market confidence.

Financial Barriers (Banking and Financial Institutions Perspectives towards mobilizing RE/EE investments)

- Limited source of funds for RE/EE project financing. With the diversification of RE and EE investments, many banks do not have knowledge about the nature of RE/EE business in terms of investment risks and potential rate of returns. Some of the banks and financial institutions are not confident enough in supporting RE/EE projects. In addition, the payback period for some of the RE/EE projects is over 8 years and IRR is less than 16%. This IRR is not comparable with the investments in fossil fuel based power sector.
- Because of poor quality of feasibility studies, the performance of RE/EE projects is not as good as predicted. Cost overrun and low revenue occurred in some projects and these experiences make the banks/financial institutions take extra care in financing new projects on RE and EE. Limited number of RE and EE projects that are backed up with bankable feasibility studies and strong project sponsors (with enough equity, experience and collateral) are available to be financed.
- There is no support for the banking/financial institutions in terms of policies that can motivate them to provide more financing to RE and EE projects with minimum financial risks.
- Refer also to barriers described in Annex D (Sustainable Energy Fund).

Financial Barriers (RE and EE Project Developers Perspectives)

- Majority of Re/EE project developers are new entrepreneurs. Therefore, they are still learning about the nature of business and they are not familiar with proper evaluation of risks and potential returns of these types of projects. Over optimistic revenues and underestimated investments were quite common issues in RE and EE projects due to lack of expertise and underfunded feasibility study. The level of technical capacity of project developers to make bankable project documents on RE and EE is generally low. Because of this, RE and EE projects are considered risky by some banks and financial institutions.
- There is rather limited number of RE and EE project developers or entrepreneurs who are able to meet the requirements of banks/FIs in terms of equity, collateral and quality of RE and EE project feasibility study. Hence, the number of RE and EE project implementations is not growing fast enough to collectively contribute to achieving the national GHG emission reduction targets.
- RE and EE projects are site specific. Site characteristics, infrastructure support, technical suitability and economic viability are by and large affect the viability of such projects. In general, the project preparation cost is relatively high due to the long permitting process, time consuming and costly project preparations.

Barriers in Measurement, Reporting and Verification of RE and EE projects

- There is neither an operational national registry mechanism for climate change mitigation actions nor operational Measurement, Reporting and Verification (MRV) guidelines and standard methodologies for compliance assessment of programs and projects that are supposed to be contributing to the achievement of the national GHG emission reduction targets. The limited understanding of MRV systems and the low level of capacity to implement them are major barriers to the effective implementation of the RAN-GRK and RAD-GRKs. Future energy policy-making and forward-looking development planning in the areas of energy and environment half a decade from now (2020) will not be effective, useful or meaningful if data/information from activities carried out under the RAN-GRK and RAD-GRK are not properly managed and verified, and if the enabling environment for investment is not conducive at all.

3. Baseline Scenario

14. Currently all of the necessary elements for a systematic implementation of climate change mitigation actions under the RAN-GRK and RAD-GRKs are not in place; the planning, funding and implementation of the various climate change mitigation initiatives (RE and EE) are fragmented, heavily reliant on public funds. This situation is unlikely to result in synergies between national and sub-national initiatives. Hence, these will only have limited impacts relating to Indonesia's energy-based emission reduction targets for 2020 and beyond. While the RAN-GRK of the MEMR (2010-2020) states a target of 1.2 GW target for installed RE-based power generation capacity and 4.5 million tCO_2 emission reduction, the RAN-GRK report for 2011-2014 showed that the actual installed RE-based power generation capacity was only 249 MW, equivalent to 2.5 million tCO_2 emission reduction[6]. This means that the annual realization of the planned RE-based power generation capacity as stated in RAN-GRK is only about 5% annually. This is the baseline situation in regards the annual RE-based power generation capacity growth rate (about 1 MW to 3 MW per year for solar PV, biomass and mini hydro power plants).

15. The climate change mitigation projects and programs of the national government, particularly the MEMR as responsible institution in reducing emission in the energy sector based on the RAN-GRK, are implemented through the following baseline programs: (1) Rural Energy Programme (2015-2019, annual budget allocation of about USD 50 million), which aims to increase the country's electrification ratio; and (2) Program Kemitraan Audit Energi or Partnership Programme on Energy Audit (2011-2019, average annual budget allocation of about USD 1 million), which facilitates energy efficiency improvements by providing free of charge energy audit services in the industry and building sectors, and for the certification of energy managers. The investment made by the government to promote RE and EE are still dependent on public budget and have not yet successfully leveraged private sector financing.

16. In 2010, the Indonesia's National Council for Climate Change (DNPI) has published Indonesia's GHG Abatement Cost Curve for several sub-sectors including that for the power sector[7]. The report gives initial information on the marginal abatement cost (MAC) of various mitigation options in the Indonesian power sector based on prediction of current and future available technologies and project cost from several sample RE and EE projects. However, these initial estimations of climate change mitigation project cost and baseline situation did not take into account the influence of various essential factors such as those relating to geography, infrastructure condition and GHG emissions inventory of project locations. In that case, the developed MAC curves were barely used as reference by the national and local governments in the planning of climate change mitigation actions.

17. Furthermore, to promote private investment, the country's new administration has emphasized the implementation of an "Integrated Single-Window Policy" and simplification of permitting procedures for mobilizing investments and to increase Indonesia's global competitiveness. The implementation of the system is both at national and sub-national levels, led by the national/sub-national Investment Agency. This is regarded as a good initiative. However, the coordination mechanism between the relevant agencies and effectiveness of its implementation are still weak particularly at the provincial and district levels.

18. Development partner agencies such as the GIZ, USAID and US-Millennium Challenge Account-Indonesia (MCA-I) have been conducting projects to promote implementation of RE in Indonesia. However, these projects are not always realized or if implemented, are not usually replicable. The non-replication is due to the high grant financing of these projects. The UNDP through the GEF-

[6] Monitoring, Evaluation and Reporting (MER) of MEMR RAN-GRK implementation, 2014.
[7] http://www.mmechanisms.org/document/country/IDN/Indonesia_ghg_cost_curve_english.pdf

funded Wind Hybrid Power Generation Market Development Project (WHyPGen) has established cooperation with PT. Sarana Multi Infrastruktur (PT. SMI), a state-owned financing institution, to develop specific financing mechanism for wind power projects in Indonesia. PT. SMI managed USD 300,000 of the project fund as seed funding that was leveraged by PT. SMI in combination with other financing resources to serve as loan guarantee or to reduce interest or other financial requirements thereby reducing the financial risk of wind power project developer/investors. This financial de-risking instrument is still implemented by PT. SMI for wind power projects. The ongoing RE projects supported by other development partners as well as the financing scheme with PT. SMI are among the baseline activities that will be subsumed in the proposed project. Where applicable, these will be further enhanced to realize more positive global environmental benefits through the facilitation efforts that will be carried out under the MTRE3 project.

19. Furthermore, UNDP through the Low Emission Capacity Building Programme (LECB) has been working with six city governments in Greater Jakarta, including the Jakarta City Government, to exercise prioritization of mitigation actions and to develop a NAMA proposal. As a result, one financed-ready NAMA proposal for Bus Rapid Transit (involving 3 cities) and one financed-ready unilateral NAMA proposal for Green Building to increase energy efficiency in the Jakarta City Hall have been developed. The implementation of the green building NAMA is currently being planned using the local government budget. Lessons from the LECB program in conducting participatory approach in the prioritization of climate change mitigation actions will be considered in the proposed project in facilitating similar process in the pilot provinces.

20. The registry system for climate change mitigation actions and the MRV agency are still under development by the MoEF. Transitional guidelines for Monitoring, Evaluation and Reporting (MER) for RAN-GRK and RAD-GRK have been developed by the RAN-GRK Secretariat with support from the GIZ and JICA. The MER approach is considered as initial step towards the standardized MRV system. Efforts to upgrade the MER to MRV system and to put in place institutional capacity for its implementation are still lacking. So far the Provincial MERs for 2013 and 2014 from key line ministries and provinces are with the RAN-GRK Secretariat. Up to now, the MER system is just the initial step towards the development and implementation of a standardized MRV system, since the "Verification" part is currently missing.

21. The Third National Communication (3NC) project under MoEF is supporting establishment of national registry and MRV system for mitigation actions, and is focused on GHG Inventory in waste and land-based sectors. The MTRE3 project will use the 3NC project activities as baseline and contribute in strengthening GHG Inventory at provincial level for the energy sub-sector, build provincial government technical capacity for its implementation and ensuring linkage between provincial and national systems for RE-based power generation and energy efficiency projects.

22. The MoEF in July 2012 has launched the guidelines for National GHG Inventory for the energy, industry, agriculture, and forestry and waste sectors in response to the Presidential Regulation No.71/2011 on National GHG Inventory and followed by establishment of National GHG Inventory System (SIGN), which has conducted the first provincial GHG Inventory in 2013. Furthermore, led by MoEF, Indonesia's 3NC to the UNFCCC project funded by GEF is supporting development and improving data system of GHG Inventory for agriculture/land-based and waste sectors, but not for energy sector. The SIGN and 3NC works will be the baseline activities of the MTRE3 project, which will further fill in the gaps in supporting establishment of registry and MRV system for energy sub-sector, enhance capacity of national and sub-national stakeholder for its implementation and strengthening coordination mechanism between province and national registry and MRV system.

4. Key Stakeholders

23. The successful implementation of MTRE3 Project will depend on the development of effective partnerships between various agencies at multiple governmental levels. A very important feature of the MTRE3 project is the parallel introduction of the technical interventions needed to improve renewable energy and energy efficiency investment, through implementation of appropriate mitigation actions in energy sector stated in RAN-GRK (National Action Plan for GHG emission reduction), in line with National Energy Policy, and complying to Measurable, Reportable and Verifiable guidelines set out by Ministry of Environment and Forestry. Therefore, the project will be implemented in conjunction with the RAN-GRK, National Energy Policy and national MRV guidelines. This synergistic collaboration is expected to significantly enhance the contributions to the target energy savings for commercial buildings, renewable energy share in the national primary energy mix, as well as the resultant GHG emission reduction as targeted by the projects. Partnerships will also be pursued very actively with project developers in RE/EE investment, financing agencies, and commercial building owners who will stand to benefit from energy efficiency scenario for commercial building and improved renewable energy investment in Indonesia.

24. During the conduct of the project design and preparation activities (i.e., PPG exercise), a number of government institutions, financing agencies, energy experts and private sector energy development companies were consulted about the MTRE3 approach and the possible institutional arrangements. The following lists the stakeholders of the MTRE3 Project and their respective roles in the project implementation:

Role of MTRE3 Stakeholders

Stakeholder	Role
Ministry of Energy and Mineral Resources (MEMR) Directorate General for New and Renewable Energy and Energy Conservation (DG-NREEC) Education and Training Center for Renewable Energy & Energy Conservation (Pusdiklat EBTKE)	As implementing Partner of MTRE3 project in close coordination with the Ministry of Environment and Forestry (MoEF), Ministry of National Development Planning, Ministry of Finance and Ministry of Public Works. MEMR is responsible in enactment of renewable energy and energy efficiency policy and regulation for providing technical assistance in relation to improving energy efficiency and renewable energy measures of energy investment. The training center is a structure within MEMR that is responsible to conduct energy-related education and trainings for government officials in Indonesia.
Ministry of Environment and Forestry (MoEF) Deputy Minister's Office for Climate Change Mitigation and Environmental Damage Control SIGN (*Sistem Inventarisasi GRK Nasional/* National GHG Inventory System) Center	Lead agency in implementation of MRV scheme for RAN-GRK and will be the focal point for coordinating MRV scheme for energy sector in provinces and HPMP beneficiaries with MTRE3 activities. Lead agency in implementation of National GHG Inventory and will be the focal point for coordinating GHG Inventory for energy sector in provinces.
National Planning Agency (BAPPENAS) Directorate for Environment	Focal point for coordinating NAMAs (Nationally Appropriate Mitigation Actions) framework in Indonesia. MTRE3 project will work closely with BAPPENAS in the implementation of proposed project interventions.
Ministry of Public Works (MPW) Directorate General of Cipta Karya, Directorate of Environment and Building Management	Leading agency for regulation on building code – supporting energy efficiency program for commercial buildings in the proposed project.
Ministry of Finance (MoF) Centre for Policy on Climate Change and Multilateral Financing,	Leading agency for provision of policies and regulations for financial packages and incentives in supporting RE investment and EE for commercial buildings. MTRE3 project will work closely with MoF in

Stakeholder	Role
Fiscal Policy Office (FPO/BKF)	the implementation of Sustainable Energy Fund.
Agency for the Assessment and Application of Technology (BPPT) • Center for Energy Conversion and Conservation Technology (PTKKE) • Center for Energy Technology Laboratory (B2TE)	• Focal point for technical support and recommendation of RE and EE technology for MTRE3 project. • Focal point for Laboratory and Test of RE and EE and Energy Auditing for MTRE3 project.
Financial Service Authority (OJK)	Focal points for enabling financial packages in banking sectors and incentives in supporting RE and EE for commercial buildings investments in the proposed project.
Ministry of State-owned Enterprise (BUMN)	Focal points for enabling State-owned Enterprise (BUMN) in banking sectors in supporting RE investment and EE for commercial buildings. MTRE3 project will closely consult with BUMN while mobilizing finance for proposed project interventions.
Local Governments of Pilot Provinces	Local governments will be partner of MTRE3 in implementing renewable energy and energy efficiency related regulations for energy investment, implementation of energy appropriate mitigation actions and establishment of Integrated Market Service Center.
PT Sarana Multi Infrastructure (PT SMI)	Focal partner for Sustainable Energy Fund supporting RE and EE investments. MTRE3 project will work closely with PT SMI in the implementation of proposed project interventions.
BKPM – Indonesia Investment Coordinating Board	Focal point for investment permit in RE and EE. MTRE3 project will work closely with BKPM to mobilize investments for the proposed project interventions.
RE Project Developers in RE/EE: • PT. Pasadena Engineering Indonesia • PT. Multi Fabrindo Gemilang • PT. Daun Biru Engineering	Association or corporation focusing on RE investment, focal point for project development, engineering, procurement and construction service providers and co-financing partners. MTRE3 project work them closely in the design of project interventions and implementation of demonstration projects. These companies have conducted pre-feasibility study for several potential RE-power projects, which implementation can be supported by results from MTRE activities (i.e. access to financing, streamlined permit, improved feasibility, etc.). These companies are negotiating loans for financing their project, which will be subsumed into the MTRE3 project. Their investment is part of the MTRE3 project co-financing.
Building Managers	Individuals or service companies that guide investment decisions on EE and in few cases, focal point for energy efficiency technology implementation in commercial buildings. MTRE3 project work them closely in improving the services they provide.
Green Building Council Indonesia (GBCI)	Association that provides technical assistance, assessment, EE related information, EE standard for commercial buildings. The association will be involved in market development activities under MTRE3 and certification.
Civil Society Organizations (CSOs), Indigenous communities and women groups	CSOs, Indigenous communities and women groups at project locations will be engaged during feasibility assessment and prioritization of RE/EE projects in provinces as part of FPIC process. They will be targeted participants in technical training on RE/EE operation and maintenance. It is expected that employment and local entrepreneurs as service providers can be enhanced from CSOs and local communities.
Development Partners	Development partners are potential to co-finance feasibility study, RE construction/EE instalment and to participate in Sustainable Energy Fund. MTRE3 project work them closely in maximizing the global environmental benefits through accessing additional co-finance.

II. PROJECT STRATEGY

5. Alternative Scenario

25. Taking into account the barriers and the baseline situation in the country's energy generation and energy end use sectors, the proposed MTRE3 project will facilitate the realization of the voluntary emission reduction targets as stated in the RAN-GRK (national) and RAD-GRK (sub-national) for the energy generation and energy end use sectors. This will be achieved by establishing the necessary enabling conditions that would make possible the mobilization of the required investments in RE-based power generation and the application of feasible EE technologies in the energy end-use sub-sectors[8]. The proposed project will bring about an alternative scenario wherein the realization of the RE and EE targets in the provinces (at least in the pilot provinces) will be more enhanced and contribute significantly to the achievement of the RAN-GRK and RAD GRK targets[9]. By addressing the barriers through the implementation of appropriate incremental barrier removal activities, the expected enhanced mobilization of public and private investment for the implementation of RE and EE projects in the alternative scenario will be realized.

26. The approach will be through NAMA implementation and MRV in 4 pilot provinces. Such approach, which is relatively new in Indonesia for realizing verifiable achievements of RAD-GRK and RAN-GRKs. In this case, the incremental activities would include those that will facilitate or enable the design, financing, and sustainable implementation of the RE and EE projects at pilot provincial level, and their MRV. The proposed project will also develop and update MAC curves to the provincial level by closely working with the country's GHG inventory mechanism and the RAN-GRK and RAD-GRK reporting procedures. It should be noted that in case the development of guidelines, methodologies, studies are to be conducted as part of the MTRE3 Project interventions, all available applicable guidelines, methodologies, standards, and studies will be used, and where possible modified to make them tailor-made for the intended beneficiaries.

27. Lessons learned from the completed EE and RE projects in Indonesia[10] by development agencies, as well those under the CDM have been considered in the project design to avoid repeat of shortcomings, adoption of best practices, and enhancement of outcomes. During project implementation, the project team will still keep track of ongoing EE/RE projects to also benefit

[8] The main idea behind the project is to enable the design and implementation of appropriate climate change mitigation actions particularly application of renewable energy (RE) and energy efficiency (EE) technologies, for fossil fuel consumption displacement/or reduction, and ultimately GHG emission reduction. While RET applications can be done in various energy end use sectors, these interventions produce more significant impacts (in terms of fossil fuel consumption reduction, and GHG emission reduction) in the energy sector (particularly power generation). Hence, the decision to consider the energy sector. Energy efficiency technology applications can also be carried out in the energy generation sector (particularly in energy guzzling fossil fuel-fired power generation facilities). However, considering the fact that this particular segment of the energy generation sector is not an area supported by GEF, the decision of the project proponents was to work on the energy end-use sector that would appreciate better the application and benefits of EE technologies, particularly for EE promotion and advocacy purposes. Hence, the focus on commercial buildings for showcasing the design, planning, engineering, installation and operation of systems that improve the energy utilization of commercial building facilities/services.

[9] The target settings that were made were based on the MEMR and PLN, and supplemented by information derived from the various relevant study reports produced by other energy projects. Actions that have been identified by the MEMR, PLN and/or the baseline studies to achieve the set targets were further analyzed by the project development team (PDT) to identify: (a) aspects that are not fully addressed by the proposed actions; (b) additional interventions, or enhancements to the proposed actions, that can be done; and, (c) follow-up interventions to enhance the realization of the set targets. From the analyses of the proposed climate change mitigation actions that were done, the potential RE & EE projects (as climate change mitigation actions) were identified.

[10] This include the regional ALGAS project (Indonesia is one of the countries involved). Lessons learned from this were also taken into account in the design of the MTRE3 Project. The way the least cost abatement strategies were arrived at in the ALGAS project considering the minimal GHG emission data available in the early 2000s can be applied in this new project particularly when doing the provincial GHG inventories, where provincial energy and GHG emission data are lacking.

not only from their results but also from lessons learned and best practices from them and where feasible, apply them (if necessary) in any adjustments that maybe made in the scheduling, or implementation strategy of the project activities.

28. To bring about the alternative scenario, the project will address and remove barriers that are mentioned above particularly in RE-based power generation and energy efficiency technology applications in commercial buildings in Indonesia. The proposed project will make use, and promote best practices in the design and implementation of feasible and cost-effective RE and EE projects at the provincial level (i.e., provincial level NAMAs) and put in place enabling environment for transforming market towards RE and EE investments, which will simultaneously support implementation of mitigation actions covered in the RAN-GRK and RAD-GRK and in so doing adequately meet the country's voluntary climate change mitigation targets of either 26% (national efforts) or 41% (with external assistance) by 2020. Without these incremental activities, the implementation of RE and EE programs in Indonesia will remain fragmented, ad-hoc, highly dependent on limited public budget, and their impacts (in terms of GHG emission reductions) are unverifiable or at best just best estimates.

29. The baseline situation in RE project development and implementation realizes only 5% of RAN-GRK target. The proposed MTRE3 project will facilitate RAN-GRK achievement through the implementation of a total of 15 MW RE-based power generation capacity as provincial NAMA projects (7 MW mini-hydro, 6 MW biomass and 2 MW solar PV projects) during 5 year project implementation, and an additional 85 MW (39 MW mini-hydro, 40 biomass and 6 MW solar PV) as direct post-project impact. In the area of energy efficiency in commercial buildings, the MEMR audit results showed that the baseline or BAU Specific Energy Consumption (SEC) is 285 kWh/m^2, which is still the current level. According to benchmark from Green Building Council Indonesia only about 10% reduction from BAU that has been voluntarily implemented by building owners through low-medium cost investments. Only few of building owners implement high-cost investment that resulted in about 20% reduction from BAU SEC. The alternative scenario offered by MTRE3 project during its 5 years of implementation is the realization of an average of 20% reduction[11] in the baseline SEC to 228 kWh/m^2 in commercial buildings with minimum floor area of 500 m^2, through the implementation of EE NAMAs in the 4 pilot provinces. The EE projects that will be implemented will be for commercial buildings (total collective floor area = 50,000 m^2). Energy savings from EE project implementations in additional commercial buildings (total collective floor area = 50,000 m^2) are expected to be realized as direct post-project impact[12].

6. MTRE3 Project Objective, Outcomes and Outputs

30. The objective of the proposed MTRE3 project is to support the design and implementation of appropriate climate change mitigation actions in the energy generation and energy end use sectors. The project will tackle the barriers to the widespread application of RE and EE in the country's energy generation and energy end use sectors through the strategically defined activities that are grouped into the following project components:

- **Component 1:** Climate change mitigation options for the RE based energy generation and energy efficiency

[11] Based on MEMR energy audit results and energy saving benchmark set by the Indonesia Green Building Council for the highest voluntary EE measures implemented in commercial buildings. There are currently no regulations for EE in buildings.
[12] For Energy Efficiency interventions, the MTRE3 project focuses on energy efficiency in commercial building. This is based on the stakeholder consultations, government priority and high potential for energy saving in the commercial buildings sector. However, other high impact energy efficiency projects besides commercial building will be supported, wherever feasible. The demonstration sites for EE projects in commercial buildings will not be limited only to the capital cities of the 4 pilot provinces. EE technology application demonstration will also be implemented in big cities such as Jakarta, Bandung, Surabaya and Medan, taking into account the rapid development of commercial building in these cities.

- **Component 2:** Market transformation through implementation of appropriate mitigation actions in the RE based energy generation and energy efficiency
- **Component 3:** MRV system and national registry for mitigation actions in the RE based energy generation and energy efficiency.

COMPONENT 1: CLIMATE CHANGE MITIGATION OPTIONS FOR THE RE-BASED ENERGY GENERATION AND ENERGY EFFICIENCY

Component 1 is comprised of activities that will address barriers in the planning, prioritization and implementation of appropriate climate change mitigation actions. The major activities summarized in Table 1 below will be conducted to achieve the required outputs that will contribute to the realization of prioritized appropriate mitigation actions in the RE-based energy generation and energy efficiency.

Table 1: Major Activities under Component 1

COMPONENT 1: CLIMATE CHANGE MITIGATION OPTIONS FOR THE RE-BASED ENERGY GENERATION AND ENERGY EFFICIENCY	
Outcome 1: Prioritized appropriate mitigation actions in the RE-based energy generation and energy efficiency.	
Output	**Activities**
1.1. Defined and established sectoral and sub-national reference baselines for the RE-based energy generation and energy efficiency in commercial building sectors in pilot provinces.	1.1.1. Development of data inventory for energy generation, its use, available renewable resource and GIS mapping of potential available resources of RE and its value chain at provincial level
	1.1.2. Development of reference baseline and GHG inventory of the energy sector in four pilot provinces (Jambi[13], West Sulawesi, East Nusa Tenggara, and Riau pilot provinces)
	1.1.3. Review and improvement of the Local Energy Planning Document (RUED) including RAD-GRK activities in 4 pilot Provinces
	1.1.4. Design and conduct of a capacity building to strengthen Energy Working Group on RAD GRK in 4 pilot Provinces
	1.1.5. Design and conduct of a benchmarking program for Specific Energy Consumption (SEC) or Energy Consumption Intensity (IKE) or Energy Performance Index (EPI) in the commercial buildings sector
1.2. Developed and published detailed marginal GHG abatement cost curves for renewable energy and energy efficiency options in the selected provinces.	1.2.1. Development of Marginal Abatement Cost Curves (MACCs) for applicable climate change mitigation options in 4 pilot provinces
	1.2.2. Conduct of a revamp program for the existing energy demand and consumption data system to support the development of GHG MACCs
	1.2.3. Design, conduct and evaluation of capacity building and awareness activities for the utilization of MACCs
1.3. Selected appropriate and prioritized mitigation options that are integrated into national and provincial development plan	1.3.1. Review of RAD-GRK activities using the MACC results and formulation of recommendations for enhancing the RAD-GRKs
	1.3.2. Mainstreaming RUED into the provincial development plan (RPJMD) of the 4 pilot provinces including updating of the RAD-GRKs and building synergy with RAN-GRK of MEMR
1.4. At least two projects designed,	1.4.1. Participatory selection and prioritization of climate change

[13] The MEMR changed the pilot province from West Nusa Tenggara to Jambi considering more potential private investments in mini-hydro and biomass-based power generation after the assessment that was carried out during the project preparation stage (PPG exercise). However, results and lessons learned from the "Mitigation Momentum" have been taken into account in the design of the project activities. The project team (under the MEMR) will also coordinate with the "Mitigation Momentum" project team for exchange/sharing of data/information that can be useful during project implementation.

Output	Activities
each for the implementation of selected prioritized mitigation actions in RE-based energy generation and energy efficiency in commercial building sectors.	mitigation actions and identification of pilot sites
	1.4.2. Development of 2 NAMA proposals for the selected prioritized RE projects (resulted from Output1.3) as pilot/demonstration climate change mitigation projects.
	1.4.3. Development of 2 NAMA proposals for the selected prioritized EE projects (resulted from Output1.3) as pilot/demonstration climate change mitigation projects.

OUTPUT 1.1: Defined and established sectoral and sub-national reference baselines for the RE-based energy generation and energy efficiency in commercial building sectors in pilot provinces.

31. This output will be achieved through the implementation of activities that will develop energy sector reference baseline data at the provincial level as a way to improve the quality of RAD GRK planning, implementation and reporting. The development of reference baseline in the provincial energy sector will be supported by data inventory and Global Information System (GIS) mapping that contains information on benchmark of Specific Energy Consumption (SEC) in commercial building sector, potential of RE projects, related-value chain activities and transmission and distribution lines. The baseline and data inventory will be used in the development of Local Energy Planning Document (RUED) that contains RAD-GRK as part of its action plan. Scoping and rapid assessment will be conducted in each pilot province to collect lesson learned from previous similar initiatives and to avoid duplications.

    GEF funding will be used to provide the technical assistance (TA), survey and stakeholder consultations at pilot provinces for data collection, developing reference baseline, GHG inventory, benchmarking exercise, developing GIS interactive mapping and improvement of RUED. For this Output, amount of funding that will be allocated from GEF is USD 500,000 and USD 450,000 from MEMR.

    - *Activity 1.1.1: Development of data inventory for energy generation, its use, available renewable resource and GIS mapping of potential available resources of RE and its value chain at provincial level.* This activity involves the collection of relevant the information from the 4 pilot provinces on energy generation, energy consumption, RE resources potentials and RE-based energy generation projects (e.g., hydro, biomass, solar PV), value chain activities, land and infrastructures availability, etc. Appropriate GIS tools will be used to generate interactive maps and linked with the collected data including RE inventory that includes relevant information.

    - *Activity 1.1.2: Development of reference baseline and GHG inventory of energy sector in four pilot provinces (Jambi, West Sulawesi, East Nusa Tenggara, and Riau pilot provinces)*. This activity entails the conduct of the review of the energy chapter in the RAD GRK document and GHG inventory in energy sector in Jambi, Riau, West Sulawesi, and East Nusa Tenggara provinces. The GHG inventory results will be used to develop reference baseline (under a Business as Usual (BAU) case) for updating the provincial RAD-GRKs.

    - *Activity 1.1.3: Improvement of Local Energy Planning Document (RUED) including RAD-GRK activities in 4 pilot Provinces.* This activity is comprise of actions that will improve the existing provincial RUED, among which is the integration of the updated RAD-GRK. The improvement of the RUED is to, among others, ensure that it is in line with the National Energy Planning Document (RUEN). The Provincial Regulation (PERDA) will be released as the legal basis for RUED implementation following a series of public consultation with relevant stakeholders.

- *Activity 1.1.4: Design and conduct of a capacity building to strengthen Energy Working Group on RAD GRK in 4 pilot Provinces.* Since 2011, every province has established its Energy working group that is responsible in developing the RAD-GRK for the energy sector of the province. However, the technical capacity of the working group is considered low and the institutional arrangement for the operation of the working group is unclear. The activity involves the strengthening the institutional capacity, and increase the technical capacity of the working group members. The involvement of the local university personnel will be advocated. Related activities like training on *Long-range Energy Alternative Planning system* (LEAP) and GHG Inventory tools will be conducted. The coordination between MEMR and Energy Working Group in pilot provinces will be strengthened. Part of the training will be on the application of the *LEAP model and the GHG inventory methodology.*

- *Activity 1.1.5: Design and conduct of a benchmarking program on Specific Energy Consumption (SEC) or Energy Consumption Intensity (IKE) or Energy Performance Index (EPI) in the commercial buildings sector (link to Activity 2.2.2).* Currently there is no national standard for SEC/IKE/EPI for commercial buildings. This activity involves the conduct of a survey and data collection from significant sample of commercial buildings in selected cities in Indonesia referring to the initial results of energy audit data from MEMR. The review and verification will be conducted on the proposed standard BAU for SEC/IKE/EPI of 285 kWh/m^2/year for office buildings based on limited benchmark data from Green Building Council Indonesia. MEMR suggested reasonable target for the SEC by end of project is 20% lower from the baseline, which is 228 kWh/m^2/year. The survey result from this activity will be used as input to develop improved new SEC/IKE/EPI standard for commercial building to be proposed to MEMR.

OUTPUT 1.2: Developed and published detailed marginal GHG abatement cost curves (MACC) for renewable energy and energy efficiency options in the selected provinces.

32. This output will be achieved through the implementation of activities that will help inform decisions made by public or private investors through the use of the marginal abatement cost approach as a tool to assess the cost-effectiveness of a RE or EE project. Particularly for investment by local government on mitigation actions under RAD-GRK, selection of most feasible and cost-effective mitigation actions is crucial taking into account the country's development priorities. The development of such abatement cost curve depends on available resources and infrastructure in a particular area as well as the choice of technology. Providing a detailed marginal abatement cost curve (MACC) for each province can: (a) inform policy making at the national level; (b) screening potential renewable energy and energy efficiency projects in targeted provinces; (c) establishing the cost-effectiveness and GHG abatement potentials of climate change mitigation actions; and, (d) trigger public-private partnership in RE and EE projects investment. Lessons learned from the development and use of the Indonesia Marginal Abatement Cost Curve (DNPI, 2010) will be considered in development of provincial level MACCs.

GEF funding is required to provide technical assistance to conduct the review, to develop specific abatement cost curve for 4 pilot provinces, to collect required data and to revitalize data system for development of MACCs and to conduct awareness activities and publications. To deliver this output, the estimated amount of GEF funding that will be allocated is USD 500,000 and USD 350,000 from MEMR.

- *Activity 1.2.1: Development of Marginal Abatement Cost Curve (MACCx) in 4 Pilot Provinces.* This activity involves the conduct of a review of the Abatement Cost Curve report for Indonesia (previously developed by the National Climate Change Council) to learn how the model was developed. This will include the step-by-step study approach in the calculation of a range of GHG emission abatement potentials of each RE and EE technology and

associated costs, in each province. Relevant stakeholders will be consulted in this process to review the cost assumptions and finalize the document.

- *Activity 1.2.2: Conduct of a revamp program for the existing energy demand and consumption data system to support the development of GHG MACCs*. This activity involves the design of a program to revamp the existing energy data system in each pilot province to develop MACCs, in particular for RE and EE mitigation measures in regards to data updates and data management. Various discussions will be conducted involving key stakeholders, such as: electricity provider (PLN), policymakers (Ministry of Energy and Mineral Resources, Ministry of Finance, Ministry of Public Works, sub-national Government Officials/Agencies, etc.), energy project investors/developers, energy technology providers, energy experts (academics, researchers, development partners), NGOs and CSOs.

- *Activity 1.2.3: Design, conduct and evaluation of a capacity building and awareness enhancement program for the utilization of MACCs.* This activity includes the organization and conduct of workshops and awareness raising activities to enhance the use of provincial MACCs and the revamped/updated energy data system. Dissemination of information materials and guidelines for application on certain GHG mitigation actions in energy generation and energy end-use sectors will be conducted. Advocacy campaigns to the MEMR to adopt the updated data and provincial MACC will be pursued. The final version of the provincial level MACCs for selected RE/EE technologies will be published and disseminated as knowledge product.

OUTPUT 1.3: Selected appropriate and prioritized mitigation options that are integrated into national and provincial development plan.

33. This output will be achieved through activities that facilitate the integration of RUED and the RAD-GRK into the Provincial Development Planning (RPJMD) process, thereby enabling the development and budgeting of RE and EE projects funded through the local government budget. Furthermore, synergy between updated RAD-GRK with RAN-GRK of the MEMR will be developed thereby ensuring effective implementation between national and provincial climate change mitigation actions and avoiding double counting of GHG emission reductions.

    GEF funding will be used to provide technical assistance in reviewing RAD-GRK, developing policy recommendations, advocacy and stakeholder consultations. For the delivery of this o*utput the amount of funding that will be allocated from GEF is USD 316,296 and from MEMR will be USD 300,000.*

    - *Activity 1.3.1: Review of RAD-GRK activities by using the MACC results and formulation of recommendations for enhancing the RAD-GRKs.* This activity is a participatory review and updating of RAD-GRK for energy sector in pilot provinces based on the results of the development of reference baseline and provincial level MACCs. It is expected that RAD-GRK will reflect the cost-effective mitigation actions in energy sector at provincial level. Thereafter, policy recommendations will be formulated concerning the need and guidance in enhancing the RAD-GRKs using reference baselines and MACCs.

    - *Activity 1.3.2: Mainstreaming RUED into Provincial development plan (RPJMD) and building synergy with RAN-GRK of MEMR*. This activity advocates integration of the RUED that includes the updated RAD-GRK into the RPJMD. Policy recommendation will be developed and a series of Focus Group Discussion and Workshop involving key stakeholders; BAPPEDA (Local Development Planning Agency) and Local Parliament (DPRD) will be conducted.

OUTPUT 1.4: At least two projects designed, each for the implementation of selected prioritized mitigation actions in RE-based energy generation and energy efficiency in commercial building sectors.

34. This output will be achieved through activities that support the implementation of selected climate change mitigation actions utilizing renewable energy for power generation and energy efficiency in improving the energy performance of commercial buildings. The activities will address existing problems related to the poor technical assessments done in energy planning and the lack of measurement tools to gauge the success of planned, prioritized and implemented climate change mitigation actions. This output will help develop the criteria/indicator for the selection and prioritization of climate change mitigation actions, support the technical assistance to conduct quality of feasibility study for selected RE and EE projects and to develop NAMAs proposals. Synergies will be explored and utilized as well as lessons learned from previous and ongoing initiatives on RE/EE, such as Mitigation Momentum (ECN), USAID-ICED projects and UNDP-BRESL/Japan Partnership Fund.

GEF funding will be used to provide technical assistance in developing selection criteria, development of NAMAs proposals, stakeholder consultation process, and publication of guidelines. For the delivery of this output, the amount of GEF funding that will be allocated is USD 500,000 and from co-financiers will be USD 400,000 (MEMR), USD 250,000 (Pasadena Engineering) and USD 350,000 (PT. Daun Biru).

- *Activity 1.4.1: Participatory selection and prioritization of climate change mitigation actions and identification of pilot sites.* The activity entails the development of a set of selection criteria for appropriate climate change mitigation actions for the pilot provinces[14]. The criteria will be developed and discussed with MEMR, Local Governments and other key stakeholders (e.g., private sector, CSOs, energy-technology industries). The selection criteria shall reflect the priority for the climate change mitigation actions of the local government, readiness of supporting stakeholders, available resources, local government capacity, MRV measure, potential for scale-up and/or replication, potential negative impacts to the natural environment and surrounding communities, potential socio-cultural impacts, etc. A series of meetings and consultations will be conducted. Information from the developed MACCs will be used as reference in the selection process. Guidelines for criteria development and prioritization process will be developed, each for RE and EE project respectively. It is expected that at least two cost-effective RE projects and two EE projects and the corresponding pilot project sites will be selected.

- *Activity 1.4.2: Development of 2 NAMA proposals of the selected prioritized RE projects (resulted from Output1.3) as pilot/demonstration mitigation projects.* This will focus on the development of financed-ready NAMA proposals for the two selected RE projects that will be identified and prioritized (Output 1.3) as demonstration climate change mitigation actions at the provincial level. It is expected that the implementation of these RE NAMA demos that will be facilitated by investment-enabling environment will have transformational impact for promoting investment in RE-based energy generation and utilization projects in the energy and energy end use sectors. The draft RE NAMA proposals will be communicated to stakeholders to get feedback for refinement and to get buy-in for its implementation. The key stakeholders include electricity utility (PLN), national and sub-national policymakers

[14] The selection of the projects will be based on a set selection criteria that will include, among others, the replicability potential of the projects in other provinces. It must be reiterated that the demonstration or showcasing of these projects (conceptualization, design, planning, engineering, financing, installation, commercial operation, and maintenance) is meant to encourage the scale-up and/or replication of these demos, and the adoption of the approaches/strategies that were applied in the entire process of RE/EE project evolution, to other stakeholders (e.g.., local governments, energy project developers/investors, financial institutions, engineering firms, etc.) in other provinces of Indonesia. This way, although the project interventions can be considered relatively small, the target impact/influenced sectors cover a wide swath of energy producers and energy end users of the country.

(Bappenas/RAN-GRK Secretariat, MEMR, Ministry of Finance, Ministry of Public Works, sub-national Government Offices/Agencies, etc.), energy project investors and/or developers, technology providers (ESCO), energy experts (academics, researchers, development partners), and NGOs and CSOs involved in RE and energy access activities. The final proposals will be registered by provincial government or by MEMR in the NAMA Registry and depending on the financing sources and registry system to be developed, the NAMA proposals can be submitted to the UNFCCC through the RAN-GRK Secretariat for international support (i.e., supported NAMAs) or financed by domestic sources (i.e., unilateral NAMAs) through public-private partnerships. The RE NAMA proposals will also be reported in Indonesia's 3NC to the UNFCCC as part of mitigation measures in the country's energy generation sector. During the RE NAMA development, synergies and lessons will be taken from previous and ongoing initiatives on RE development and utilization in the country, such as the Mitigation Momentum (ECN) on RE-NAMA, and the USAID-ICED project that has assessed potential RE projects in Indonesia. The results of the assessments done by other agencies (e.g., WB, ADB, and JICA) of other potential RE-based energy generation and utilization options, as well as lessons learned from developed and implemented RE CDM projects, particularly those implemented in the 4 pilot provinces, will be taken into account to avoid duplications, and repeating of not-so-good results and experiences. The initial identification of potential RE NAMA projects has been done in the 4 pilot provinces.

- *Activity 1.4.3: Development of 2 NAMA proposals of the selected prioritized EE projects (resulted from Output1.3) as pilot/demonstration mitigation projects.* This will focus on the development of financed-ready NAMA proposals for the two selected EE projects that will be identified and prioritized (Output 1.3) as demonstration climate change mitigation actions at the provincial level. It is expected that the implementation of these EE NAMA demos that will be facilitated by investment-enabling environment will have transformational impact for promoting investment in the application of EE technologies in the country's buildings sector. As with the RE NAMA demos, the draft EE NAMA proposals will be communicated to stakeholders to get feedback for refinement and to get buy-in for its implementation. The key stakeholders are practically the same as those for the RE NAMA demos, non-government stakeholders are those involved mainly in EE and "green building" initiatives. The final proposals will also be registered in the NAMA Registry and depending on the financing sources and registry system to be developed, the NAMA proposals can be submitted to the UNFCCC through the RAN-GRK Secretariat for international support (i.e., supported NAMAs) or financed by domestic sources (i.e., unilateral NAMAs) through public-private partnerships. The EE NAMA proposals will also be reported in Indonesia's 3NC to the UNFCCC as part of mitigation measures in the country's energy generation and energy end use sectors. During NAMA development, synergies and lessons will be taken from previous and ongoing initiatives on EE technology and techniques/measures promotion and implementation in the country, such as the UNDP-BRESL/Japan Partnership Fund project on energy efficiency standard of appliances and capacity of testing laboratories to measure energy performance as part of green building interventions. The results of the assessments done by other agencies (e.g., WB, ADB, and JICA) of other potential EE technology options, as well as lessons learned from developed and implemented EE CDM projects, particularly those implemented in the 4 pilot provinces, will be taken into account to avoid duplications, and repeating of not-so-good results and experiences. The initial identification of potential EE NAMA projects has been done in the 4 pilot provinces.

COMPONENT 2: MARKET TRANSFORMATION THROUGH IMPLEMENTATION OF APPROPRIATE MITIGATION ACTIONS IN THE RE-BASED ENERGY GENERATION AND ENERGY EFFICIENCY

35. Component 2 is comprised of activities that will address barriers to the sustainable market-based investment of RE and EE projects, particularly barriers related to permit issuance, policy, technical, access to information and financing. The following activities will be conducted to deliver the pertinent outputs that will collectively bring about enhanced and sustainable market diffusion of renewable energy and energy efficiency technologies:

Table 2: Major Activities under Component 2

COMPONENT 2: MARKET TRANSFORMATION THROUGH IMPLEMENTATION OF APPROPRIATE MITIGATION ACTIONS IN THE RE-BASED ENERGY GENERATION AND ENERGY EFFICIENCY
Outcome 2: Enhanced and sustainable market diffusion of renewable energy and energy efficiency technologies

Output	Activities
2.1. Operational Integrated Market Service Center (IMSC) in the pilot provinces	2.1.1. Development of enhanced "Single-Window service" institutional guidelines for streamlining RE and EE investment permitting system at the provincial level.
	2.1.2. Operationalization of the IMSC to support sustainable investment in RE and EE.
	2.1.3. Development of an effective networking and knowledge sharing system with other provinces and with the national government to promote replication of successful pilot projects.
2.2. Established technical support system to provide training for operation and maintenance of RE & EE technologies including MRV aspects of projects to local service companies.	2.2.1. Design and conduct of a technical capacity building program for key stakeholders on RE/EE systems operation, maintenance, and monitoring.
	2.2.2 Development of standards, certification and accreditation of RE/EE technology and service providers.
2.3. Implemented financing mechanism, and supporting activities, to accelerate domestic financial sector investment in climate change mitigation activities	2.3.1. Design and conduct of a capacity building program for domestic financial institutions on risk and investment appraisal for RE/EE financing opportunities.
	2.3.2. Review and recommend broader fiscal and financial sector reforms to promote domestic investment in RE/EE activities.
	2.3.3. Establishment of the ***Sustainable Energy Fund*** for financing appropriate RE/EE projects.
2.4. Implemented and operational two RE and two EE demonstration NAMA projects (Output 1.3) through public-private partnership modality and supported by conducive environment for sustainable investment.	2.4.1. Evaluation and selection of investment proposals (inclusive of design) for the implementation of RE and EE pilot projects.
	2.4.2. Implementation and operationalization of two RE projects and two EE pilot projects in the 4 pilot provinces.

OUTPUT 2.1: Established integrated Market Service Center in the pilot provinces

36. This output will be delivered through activities that will enhance the existing facility referred to as Integrated Single Window Service (ISWS)[15] at the provincial level into an Integrated Market Service Center (IMSC) with additional functions that focus on the enablement of RE and EE projects. This will be done in each of the pilot provinces. Such expanded service, which will become a full-pledged provincial government unit, will be hosted by the provincial government, specifically by the regional investment agency. This unit will still carry out the same service but under the MTRE3 project, it will focus on developing networking hub for RE/EE businesses and supply chain. The

[15] This is based on Presidential Regulation 27/2009, which is now being enforced by the current government administration.

Center will build RE/EE data system, introduce a Single-Window investment permit system and provide information to government/project developers/investors on technologies, policies/guidelines, and feasible investments in RE and EE projects, public-private partnership facilitation, and conflict resolution mechanism. It is also expected to provide technical assistance services that include, among others, feasibility studies, guidance on MRV compliance and coordination with the national level agencies and national registry.

The GEF funding that is required is for the development of IMSC for RE/EE through technical assistance, stakeholder convening process, establishment of data management system and operational cost for the Center in first two years. For the delivery of this output, the amount of funding that will be allocated from GEF is USD 1,000,000 and from co-financiers will be USD 750,000 (MEMR) and USD 150,000 (PT. Daun Biru).

- *Activity 2.1.1: Development of enhanced "Single-Window service" institutional guidelines for streamlining RE and EE investment permitting system at the provincial level.* The activity involves the design of the institutional arrangement for the enhanced Single-Window facility, including the roles and mandates and operational procedures. It will include review and effectiveness analysis of existing investment permitting regulations and procedures, to come up with the necessary modifications or supplementary processes that have to be incorporated to accommodate the intended functions of the IMSC. Moreover, the activity will also include the training of the existing staff of the ISWS, and newly recruited personnel that will carry out the services of the IMSC.

- *Activity 2.1.2: Operationalization of the IMSC to support sustainable investment in RE and EE.* This activity involves the actual operation of the IMSC in each of the pilot provinces carrying out the designed functions to serve as professional RE/EE business center and investment permit issuance unit of the provincial government. It will be staffed by adequately trained and qualified personnel, and will be equipped with necessary systems to deliver its intended services. Each IMSC will also provide offline and online information including; (1)Technical Assistance (technologies, technical guidelines, regulations, standards, certification, list of service providers for feasibility studies, operation & maintenance of RE and EE systems, project monitoring and evaluation, etc.); (2) Financing Assistance (FAQ for project financing, project financing guidelines, links to websites of financing institution, etc.); (3) Investment Permit Assistance (FAQ for project permit procedure, links to websites of related institutions, links to online forum related to project permit, etc.); and (4) online registry for RE and EE project developers/investors.

- *Activity 2.1.3. Development of an effective networking and knowledge sharing system with other provinces and with the national government to promote replication of successful pilot projects.* This activity aims to revitalize and support the existing fora/dialogues/ workshops in respect to renewable energy and energy efficiency investment. It involves key stakeholders in the areas of RE and EE including other provincial governments and related line ministries. At least two national fora will be conducted yearly as media for knowledge and experience sharing and in addressing issues for the successful implementation of RE/EE investments.

OUTPUT 2.2: Established and operational technical support system to provide training for operation and maintenance of RE & EE technologies including Monitoring & Evaluation aspects of projects to local service companies.

37. This output will be delivered through the activities that will design, develop and prepare the necessary tools, instruments, institutional and support frameworks that will enable the IMSCs to provide the technical support system that it will put into work in each pilot province. The IMSC will build local capacity through long term technical trainings and knowledge about RE/EE technology, operation, maintenance and monitoring will be pursued by targeting key stakeholders in RE/EE investment such as project developers, Independent Power Producers, government

officials, ESCO companies and other service providers along the project value chain. Development of technical guidelines, standard, certification and accreditation of RE/EE technology or related services will be facilitated in coordination with MEMR, PLN and national certification/standardization agency. It is expected that local RE/EE service providers will be established at the pilot provinces.

GEF funding is required for conducting long-term trainings, development of guidelines, facilitation of technology certification and personal/organizational technical accreditation. For the delivery of this output, the amount of funding that will be allocated from GEF is USD 852,900 and from co-financiers will be USD 1,500,000 (MEMR), USD 250,000 (Pasadena Engineering), USD 500,000 (PT. Daun Biru) and USD 150,000 (Multifab).

- *Activity 2.2.1. Design and conduct of a technical capacity building program for key stakeholders on RE/EE technology, operation, maintenance, and monitoring.* The activity involves the conduct long-term training and Training of Trainers on RE/EE technology, project development, operation, maintenance, monitoring, certification and accreditation. Such capacity building program is expected to be provided by local training institutions in the future. Part of the activities include the production and dissemination of technical materials to improve skills related to RE/EE project development, feasibility studies development, and effective method to access financing scheme, project permit process, grid interconnection study, and transmission and distribution services and also on MRV activities. The capacity building activities will be implemented in close coordination with local universities, RE/EE associations, Green Building Council Indonesia, Education and Training center of MEMR and PT. PLN and management officers of the IMSC.

- *Activity 2.2.2. Development of standards, certification and accreditation of RE/EE technology and service providers.* This activity entails the conduct of research and development of Standards on RE/EE technologies, improvement of existing accreditation and certification procedures and support individual and corporate accreditation to support development of accredited RE/EE local service providers. This activity will be carried out closely with existing institutions such as the Agency for Professional Certification (LSP), National Standardization Agency (BSN), National Professional Certification Agency (BNSP), and National Accreditation Committee (KAN).

OUTPUT 2.3: Implemented financing mechanism, and supporting activities, to accelerate domestic financial sector investment in climate change mitigation activities.

38. To achieve this output, capacity building activities will be conducted for domestic banks in the financial appraisal of RE/EE investments; and, the formulation of broader fiscal and financial sector reforms to promote domestic investment in RE/EE activities. One of the important element in the delivery of this output is the development and operationalization of a financial support scheme that can address the current types of financial barriers in regards RE and EE project financing. As part of the barrier removal activities, the project will facilitate the establishment and operation of a financing scheme called the Sustainable Energy Fund (SEF) to be administered by a local financial institution, PT.SMI. The SEF will disburse grants and offer performance guarantees and commercial loan guarantees for RE/EE investments. Refer to Annex D for the rationale, description and implementation arrangement for the SEF.

GEF funding is required for the conduct of capacity building activities for targeted stakeholders in the energy generation and energy end use sectors, to provide technical assistance for RE/EE projects appraisal, to conduct policy, regulatory and institutional advocacy to establish Sustainable Energy Fund and to provide initial seed funding to operationalize the SEF in providing financial services for de-risking RE/EE investment, particularly for small-medium size RE and EE projects. For the delivery of this output, the amount of GEF funding that will be allocated is USD 2,680,000 and USD 1,000,000 from MEMR.

- *Activity 2.3.1: Capacity building for financing institutions, project developers, consulting companies and policy makers on RE/EE project and investment risk analysis*. This activity entails the conduct of capacity building for RE/EE multi stakeholders to establish convergent understanding related to RE/EE project, associated technology and investment risks. This activity also involves engaging the fiscal policy agency under the Ministry of Finance, which has the authority to issue policies on incentives/disincentives for RE/EE projects. This activity will be closely coordinated with the activities under Output 1.3 and 1.4. This activity is intended to address technical barriers related to the limited experience and track record of the domestic financial industry in funding RE/EE investments. Capacity building will be provided to domestic equity and debt investors to raise awareness or RE/EE technologies, and to develop concrete analytical and investment appraisal skills. UNDP's highly-regarded de-risking methodology will be used to identify and illustrate the range of investment risk categories for RE/EE. Financing and contractual structures, such as project finance, aggregative models, and energy performance contracting, will be introduced to domestic investors. Capacity building activities will be closely coordinated with domestic financial actors accessing SEF. A learning-by-doing approach will be prioritized throughout.

- *Activity 2.3.2: Review of existing financing policies, incentive and disincentive mechanism and development of policy recommendation.* This activity is intended to address upstream fiscal and financial sector regulatory barriers which are limiting the availability and flow of low-cost, domestic financing to RE/EE activities. With regard to tax policy, the project will review current national regulations which disfavor and penalize RE/EE investment, for example PMK-21-2010 (Regulation of the Minister of Finance on import duty exemption). With regard to financial sector regulations, the project will review the current banking capital and liquidity requirements by OJK (Financial Services Authority), and visit domestic financial sector targets and indicators for RE/EE investments. Reforms will be suggested to create, at minimum, a level fiscal playing field, and to free up balance sheets for RE/EE investing.

- *Activity 2.3.3: Establishment of the **Sustainable Energy Fund** for financially supporting appropriate RE/EE projects.* This activity involves the detailed design of the financing schemes that will make up the Sustainable Energy Fund (SEF) based on the detailed concept presented in Annex D. It will also involve the work leading to the SEF's official establishment and operationalization. It is expected that the financial instruments that comprise the SEF will address the financing needs of off-grid RE-based energy generation projects and small-to-medium size RE/EE projects. In this scheme, SEF will co-invest with the commercial financial sector/lenders, which is expected to ensure the SEF's sustainability over the medium/long term. The MTRE3 project team together with PT.SMI will conduct project appraisal and selection of RE & EE projects in cooperation with the IMSCs in pilot provinces. The RE/EE projects that are assisted and facilitated by the IMSCs will be prioritized for financing by the SEF.

OUTPUT 2.4: Implemented and operational two RE and two EE demonstration NAMA projects (Output 1.3) through public-private partnership modality and supported by conducive environment for sustainable investment.

39. The activities that will deliver this output include: (a) demonstration of operational RE and EE pilot projects that are supported by enabling environment for sustainability; (b) selection of potential RE/EE public and private project proponents; (c) development of bankable RE projects and investment grade audit for each EE project; (d) assessment and development of supply chain enhancement interventions; (e) facilitation of the securing of land for RE-based energy generation project; and, (f) technical assistance to government in pursuing special power tariff policy for on-grid and off-grid RE-based power generation facilities. Finally, the operationalization and monitoring and evaluation of the operational performance of two RE and two EE pilot projects

that will be used to demonstrate the design, development, planning, engineering, construction, commercial operation and maintenance of RE-based power generation facilities and energy efficient systems that can be replicated or scaled up in other provinces in Indonesia. Partnership with Millennium Challenge Account- Indonesia (MCA-I) under Green Prosperity Programme will be pursued to mobilize grant, particularly to support RE-infrastructure development in isolated grid areas and leverage public-private partnership for supplementary investments inasmuch as it is typical that such projects are not economically feasible but are necessary for social development purposes.

GEF funding is required to conduct activities and provision of technical assistance for conducting assessment, monitoring and policy advocacy. It includes selection of RE/EE project proponent and due diligence, developing high quality feasibility study for RE and investment grade audit (IGA) for EE project, to conduct electricity tariff assessment for RE-based isolated grid power generation and to operationalize and maintain the pilot projects. For the delivery of this Output, the amount of GEF funding that will be allocated is USD 400,000 and from financiers will be USD 1,000,000 (MEMR), USD 9,250,000 (Pasadena Engineering), USD 38,500,000 (PT. Daun Biru) and USD 1,700,000 (Multifab).

- *Activity 2.4.1:* Evaluation and selection of investment proposals (inclusive of design) for the implementation of RE and EE pilot projects - This activity involves the conduct of technical and financial investment-grade feasibility studies, assessment and development of supply chain activities including business model for operation and maintenance of the potential pilot EE and RE projects. Project proponents will be selected through call for proposal and a due diligence process involving local government, MEMR and financial partners. It will also involve the assessment of the potential projects' compliance with "clear and clean" land use policy for RE investment, and the regulations concerning special electricity tariff for RE-based power generation that supply isolated-grid/captive market. The assessment of load factors in isolated off-grid areas, economic activity of the communities and infrastructure conditions of the project areas will also be assessed.

- *Activity 2.4.2: Implementation and operationalization of two RE projects and two EE pilot projects in the 4 pilot provinces.* This activity involves the provision of technical assistance and implementation support to ensure the trouble-free engineering, procurement and construction, commissioning, operation and maintenance of the 2 RE and 2 EE pilot projects including the application of the designed MRV system that will be used during the project implementation. Monitoring and Evaluation activities will be conducted to evaluate the impacts of supporting enabling instruments and institutions (e.g., Integrated Market Service Center, Sustainable Energy Fund, permit system, tariff policy for RE-based isolated-grid project, clean and clear land use for RE investment, development of supply chain activities).

COMPONENT 3: MRV SYSTEM AND NATIONAL REGISTRY FOR MITIGATION ACTIONS IN THE RE-BASED ENERGY GENERATION AND ENERGY EFFICIENCY.

40. Component 3 is comprised of activities for the improvement and operationalization of registry system and MRV for climate change mitigation projects in RE-based power generation and energy efficiency commercial buildings. The following activities will be conducted to deliver relevant outputs that will altogether realize the accurate measurement and accounting of GHG emission reductions from mitigation actions in the RE-based energy generation and energy efficiency technology applications:

Table 3: Major Activities under Component 3

COMPONENT 3: MRV SYSTEM AND NATIONAL REGISTRY FOR MITIGATION ACTIONS IN RE-BASED ENERGY GENERATION AND ENERGY EFFICIENCY.
Outcome 3: Accurate measurement and accounting of GHG emission reductions from mitigation actions in the RE-based energy generation and energy efficiency applications.

Output	Activities
3.1. Improved and operational registry mechanism for mitigation actions in energy sector.	3.1.1. Development of a "sub-registry" of climate change mitigation actions for energy and energy end use sectors.
	3.1.2. Capacity building for the implementation, monitoring and evaluation of GHG registry mechanism in energy and energy end use sectors.
3.2. Developed Measurement, Reporting and Verification (MRV) guidelines and standard methodologies for RE-based energy generation and energy efficiency in commercial building sectors.	3.2.1: Development of project-level MRV methodology and guidelines for the selected RE/EE pilot projects.
	3.2.2: Development of procedures for GHG Audits that are conducted by third party entities.
3.3 Implemented MRV system for the selected appropriate mitigation actions in RE-based energy generation and energy efficiency in commercial building sectors.	3.3.1. Capacity building on the implementation of the MRV system and certification of GHG Auditors.
	3.3.2. Evaluation of MRV system for the selected RE and EE pilot projects.

OUTPUT 3.1: Improved and operational registry mechanism for mitigation actions in RE-based energy generation and energy efficiency in commercial building sectors.

41. The activities that will deliver this Output include the facilitation of the establishment and operationalization of a sub-registry for climate change mitigation actions in energy and energy end use sectors that will be integrated into the national registry system. A capacity building activity for RE/EE project developers and government officials will also be designed and conducted on the application of the procedures, as well as the monitoring and evaluation of the implementation of the registry system is included, with the aim of setting up a reliable and effective system.

    GEF funds are required for providing technical assistance in the development of registry system for energy sector and for building capacity of stakeholders in its implementation and M&E. For the delivery of this output, *the amount of GEF funding needed is USD 250,000 and USD 500,000 from MEMR.*

    - *Activity 3.1.1: Development of a "sub-registry" of climate change mitigation actions for energy and energy end use sectors.* This activity involves the provision of technical support in drafting the framework and mechanism for energy sector sub-registry system and its integration to the National Registry System. This activity will be conducted in close coordination with local government and with the MoEF, which is responsible for development and implementation of National Registry System for climate change mitigation actions. The draft will include important clauses such as the Proof of Title (ownership of reduced GHG emission), classification of projects to be registered, and database-sharing mechanism with the National Registry (under MoEF), Secretariat of RAN-GRK and RAD-GRK and the sustainability measures for the registry system over the years.

    - *Activity 3.1.2: Capacity building for implementation, monitoring and evaluation of GHG registry mechanism in energy and energy end use sectors.* This activity involves the development of training materials and conduct of training courses for various stakeholder groups in close coordination with the MoEF, Secretariat RAN-GRK and RAD-GRK at the national level. Furthermore, a Monitoring and Evaluation (M&E) system will be developed to strengthen the coordination mechanism and to ensure achievement of GHG Sub-Registry for

the energy and energy end use sectors. Institutionalization of the registry system and its M&E mechanism will be facilitated to ensure integration of the system as the government programme.

OUTPUT 3.2: Developed Measurement, Reporting and Verification (MRV) guidelines and standard methodologies for RE-based energy generation and energy efficiency in commercial building sectors.

42. The activities that will deliver this Output includes the development of standard methods and guidelines for the implementation of the MRV schemes for pilot projects on RE-based power generation and energy efficiency in commercial building sectors. Compliance to the procedure will allow project owners to report verified GHG emission reduction to the authorized MRV entity at the national level, which is the MoEF (according to the Presidential Decree No.71/2011). The activities will be aligned with the existing and planned national MRV system as in the country's BUR/3NC, particularly for the energy and energy end use sectors. The development of procedures for third party entities, i.e., auditors, to conduct validation and verification of GHG reductions from mitigation actions in RE/EE will also be facilitated in coordination with MoEF. Stakeholder consultations will be conducted to get input on the draft of standard method and procedure.

GEF support is required for providing technical assistance for developing MRV standard method, guideline and GHG reduction audit procedure by third party. For the delivery of this output, the amount of GEF funding needed is USD 250,000 and USD 750,000 from MEMR.

- *Activity 3.2.1: Development of project-level MRV methodology and guidelines for the selected RE/EE pilot projects.* This activity entails the development of MRV guidelines for RE-based power generation project and energy efficiency project in commercial building sectors. The standard method will refer to the UNFCCC method in coordination with MoEF. The RE MRV guidelines will be applicable for RE project connected to a PLN grid (*on-grid* application), or to a commercial isolated-grid. GHG emission projections and the monitoring system for tracking such projections will be properly designed, including the required communication procedure and organizational structure for the field operation team.

- *Activity 3.2.2: Development of procedures for GHG Audits that are conducted by third party entities.* This activity involves the development of a GHG Project's Validation and Verification Standard and the procedures for the MRV activities that will be implemented by a third-party GHG auditor. It will also include the development of Best Practice Guidance for issuing the Validation Report and Verification Report (including an Auditor Statement / Deed of Representative). This activity requires close coordination between the MoEF (MRV agency) and the potential GHG Auditors during development of the standard and procedure.

OUTPUT 3.3 Implemented MRV system for the selected appropriate mitigation actions in RE-based energy generation and energy efficiency in commercial building sectors.

43. This Output is a showcase of an operational MRV system for RE and EE pilot projects, specifically for the RE & EE demos of the MTRE3 project. The activities to deliver this output include capacity building of project owners, government officials and third party GHG Auditors on the implementation of the MRV procedures that will support certification of verified emission reductions from GHG Auditors. The testing of the usefulness of MRV-required equipment, the reporting system and verification ability of the system will also be assessed. The result of the assessment will be used to evaluate the designed MRV system for any necessary further improvement.

The GEF support that is required is for technical assistance to build capacity in implementation of MRV system and certification of GHG Auditor; procurement of measurement equipment

required for the implementation of the MRV system, and for demonstrate implementation of MRV system. For the delivery of this output, the required amount of GEF funding is USD 373,700 and from financiers will be USD 1,000,000 (MEMR), USD 250,000 (Pasadena Engineering), USD 500,000 (PT. Daun Biru) and USD 150,000 (Multifab).

- *Activity 3.3.1: Capacity building on the implementation of MRV system and certification of GHG Auditors.* This activity will involve the design and conduct of capacity building courses for project developers/owners, government officials and third party GHG Auditors on the MRV and GHG Audit standard and procedures for the selected pilot projects in RE and EE. It will include transfer of knowledge in the areas of measurement system design, development of equipment installation/testing, operation and maintenance. It involves in-country and if necessary international trainings to enable local key stakeholders to meet the international standards on the design of GHG Project Monitoring Report by implementing the MRV system procedures.

- *Activity 3.3.2: Evaluation of MRV system for the selected RE and EE pilot projects.* This activity involves the monitoring and evaluation of the implementation of the MRV system for the RE/EE pilot projects, including data analysis, stakeholder discussions and dissemination of updated MRV system. The establishment of coordination platform between project owners with MEMR, RAN-GRK Secretariat and MoEF (MRV National Agency) will also be facilitated by strengthening the existing coordination arrangements. In pace with the technology development and measurement system installation for the RE and EE projects, this activity would ensure quality testing and certification process that is constantly monitored and evaluated, as well as ensuring that the institutions are obliged to continuously improve capability based on the evaluation results and recommendations.

7. Key indicators, Risks and assumptions

44. The key performance indicators of the project are the following:

Indicator	EOP Target
• Cumulative direct CO2 emissions reduction, tons CO2 eq	27,019
• Cumulative energy produced from RE systems facilitated by the project, MWh	79,190
• Cumulative energy saved from EE in commercial buildings facilitated by the project, MWh	8,550
• Cumulative volume of public and private investment mobilized for SEF, US$ million	25
• Cumulative number of additional households (from baseline) having access to electricity in pilot provinces	80,000
• No. of small-to-medium scale RE/EE projects that were financially supported by the Sustainable Energy Fund	10
• Cumulative amount of funds from the SEF used in financially supporting small-to-medium scale RE/EE projects, US$ million	25

Risks and Assumptions

45. During the project implementation, there are some anticipated risks (e.g., internal or external factors beyond the direct control of the project management and implementation) that might prevent the project objectives from being achieved. These risks are listed and analyzed in terms of the risk type, level of risk, and the proposed risk mitigation actions, in the table below. These risks are to be viewed in the context of Indonesia in both national and local levels of government. Most of these risks are mainly due to the various levels of understanding of, and attitude towards,

the application of RE and EE technologies in the energy generation, and energy end-use sectors of the country.

#	Description	Date Identified	Risk Type	Risk Level (Impact & Probability)	Risk Management
1	Lack of coordinating authority for implementation of mitigation actions in the government	February 2015	Political	• Shifting of government energy program priorities leads to reduced technical and budgetary support to ES&L program • Poor coordination among line ministries and RE/EE industry leads to slow policy execution and poor implementation of the program. *P=3* *I= 2* *Risk Level: Low*	• Government commitment to the project will be clearly established and confirmed through annual budget allocations. PD and PMO will ensure adequate liaison and discussion with MEMR regarding this. • Government (national and local) commitment on EE and RE priorities are ensured through the RAN-GRK and RAD-GRKs. Facilitation will be provided to the MEMR to lead the coordination of activities on climate change mitigation actions in the energy and energy end-use sectors with other relevant line ministries and sub-national level government agencies.
2	Lack of successful involvement of private sector, Public-Private Partnership	February 2015	Regulatory, Institutional	• Local government and private sector not participating adequately in the project, due to lack of awareness, interest, disruption to operation and business priorities. • Financing of investments for engaging in RE and EE business are not forthcoming or not available. P=3 I=4 *Risk Level: High*	• Continuous updating (through the provision of expert advice) of the policies and action plans of the energy sector to ensure sustained promotion of RE and EE initiatives. • Continuous review and adjustment of the institutional framework for the implementation of RE and EE projects. • Regular capacity enhancement for local governments in the areas of low carbon development and energy-integrated development planning • Private sector, professional organizations will be consulted and involved in the annual project work planning; for further enhancement of working relationships to ensure cooperation. • Regular policy dialogue between governments (national/local) and private sector in regards RE & EE project development, financing and implementation though private-public partnerships (PPP). BAPPENAS will be involved in the PPP policy

#	Description	Date Identified	Risk Type	Risk Level (Impact & Probability)	Risk Management
					dialogues at provincial level. • Regular dissemination of information on successful PPP initiatives and mechanisms on RE/EE project development, financing and implementation. • Close collaboration with the GBCI.
3	Technology maturity and performance are not meeting expectation.	February 2015	Technology	• Failure of RE and EE products to perform as claimed by manufacturers resulting to customer dissatisfaction. • Government is not able to implement and enforce testing procedures and standards on RE/EE equipment production and application P=3 I=3 *Risk Level: Moderate*	• Thorough evaluation of the technical and economic performance of RE/EE technologies (and the associated hardware) that will be showcased under the project • Strict implementation and enforcement of set performance standards in the country, as well as in the country of origin of the relevant equipment and instruments that will be used in the RE/EE technology application demos. • The project includes interventions on strengthening the capacities of government agencies that are mandated to implement and enforce product testing and certification. Government to plan (and implement plan) on the provision, and training on the use, of testing equipment for specific RE/EE appliances/equipment.
4	Social and climate-related risks impacts the sustainability of renewable energy projects implementation	February 2015	External	• Climate change hinders full performance of RE technologies due to disturbance to supply of renewable energy resources and impacts of climate events like flood/drought/landslide. • Low level of social acceptance by local communities of renewable energy projects due to benefit-sharing issues. P=3 I=4 *Risk Level: High*	• Climate factors and climate scenario will be taken into account in the feasibility studies that will be conducted in the potential RE/EE demonstration projects, as well as in the design and engineering of the selected RE/EE technology application demos. • The design of the demonstration projects will be as such that climate-related (direct and indirect) risks, including insurance coverage. • Free, Prior Informed Consent (FPIC) principle will be implemented for RE projects as part of social and environmental safeguard measure.
5	Lack of financial institutions sustained commitment for	February 2015	Institutional	• Unwillingness of private sector to participate in RE and EE investments due to lack	• The capacities of financial institutions will be strengthened, particularly in the assessment of renewable energy

#	Description	Date Identified	Risk Type	Risk Level (Impact & Probability)	Risk Management
	sustainable energy investments.			of financial support and high initial investment cost leading to failure of the project to induce an increase in market driven RE and EE initiatives. The recent decision of the state-owned utility company (PLN) not to issue Power Purchase Agreement (PPA) for renewable-based power generation projects will further increase the uncertainty among the private sector entities in investing in RE-based power generation projects. PLN requested the government to guarantee provision of state budget for the feed-in tariff payments if PLN it is mandated and required to purchase RE-based generated electricity from IPPs. P=3 I=3 *Risk level: High*	and energy efficiency project proposals. • Close collaboration with the banking/financial sector in the design and development of appropriate financing schemes that are mutually beneficial for the potential clients (e.g., RE/EE project developers/investors) and the bank/financial institution. • Close monitoring and evaluation of the implementation of whatever suitable financing instruments that will be developed and promoted by the project. • Provision of up-to-date technology and market information on the economic and environmental aspects of RE/EE technologies • Assist the government MEMR) in its continuing dialog with all pertinent parties (particularly PLN) in establishing the funding for the feed-in tariff payments, including how this will be made sustainable through the succeeding years.
6	Lack of manpower and institutional capacity at MEMR and local provincial government to support project implementation and programme continuity.	February 2015	Institutional	• Capacity (human and institutional) at national and sub-national level is insufficient to make breakthrough for promoting investments in integrated RE/EE projects that need adequate stakeholder coordination. P=3 I=2 *Risk level: Moderate*	• The establishment of Integrated Market Service Center at provincial level in the selected provinces will address the institutional capacities at the local provincial government. • Adoption of an integrated approach in the strengthening of institutional capacities of designated agencies for the promotion of RE and EE including MEMR personnel. For this purpose, the commitment of the relevant personnel to allocate adequate time and efforts for such capacity building shall be ensured.

46. The following are the critical assumptions for the successful implementation of the project, and realization of the project outcomes and objective:

- GOI's commitment to climate change mitigation remains unchanged.
- Continued commitment of local government officials in supporting IMSCs in their regions (Presidential Regulation No.27/2009).

- Continues support of GOI agencies and partner financing institutions to SEF
- RAN-GRK and RAD-GRK are still mandatory based on Presidential Regulation No.61/2011.
- Local government continue to consider climate change mitigation as part of local development agenda.
- Continuous cooperation and coordination between provincial and national government agencies implementation of national and local action plans on climate change mitigation actions.
- Data availability at local level to support MRV

8. Expected Global, National and Local Benefits

47. The project is designed to support both national development objectives and contributing to the achievement of global environmental benefits. In the baseline scenario, the level of awareness of decision-makers about the economic and social benefits for promoting RE & EE is not sufficient to lead to substantial RE/EE investments in the country. The MTRE3 project consists of incremental activities that are mainly for removing barriers and creating enabling environments for transforming market toward RE/EE investments, and realize more global benefits from GHG emissions reduction that can be derived from the displacement of fossil fuels for power generation in rural Indonesia, and the widespread utilization of RE and implementation of EE. At the same time the project assists Indonesia in meeting its voluntary emission reduction commitment by 2020.

48. The design of the incremental activities took into account all the relevant baseline activities that are currently being done and those that will be carried out in the country even without the GEF assistance. It also took into consideration the national priorities particularly the increased contribution of RE in the primary energy mix, increase the electrification ratio in rural Indonesia and reducing energy consumption in the country's urban areas. The project involves innovative approaches to de-risking public-private investments such as improved prioritization approach for cost-effective mitigation actions, creation of integrated market service center, streamlining permit system, establishment and operationalization of financing mechanism, and supporting activities, to accelerate domestic financial sector investment in climate change mitigation activities,; and the deployment of a registry and MRV mechanism to better track and confirm the achievement of GHG emission reduction targets.

49. The proposed project is expected to bring about the following socio-economic benefits to Indonesia at the national and local levels.

- Clear policy, regulation and institutional framework promoting decentralized, market-based RE and EE in public-private partnership manner, leading to increased number of properly vetted and approved RE/EE projects implemented by the private sector or community-based RE-based power projects through private-public partnerships.
- Increased technical capacity of national and local government officials in planning and prioritization of cost-effective and appropriate RE and EE projects supported by reliable data, leading to more effective planning, budgeting and implementation of public RE/EE projects.
- Increased support for RE and EE project financing that can encourage national project developers/investors to participate in RE/EE investments. More RE/EE project investments and RE/EE projects implemented.
- Increased job opportunities from the implementation of various RE/EE projects.
- Reduced burden of public funding due to more private sector investments in RE/EE projects.
- Easier national government reporting of the verifiable achievement of mitigation commitment to international for a, with the operationalization of registry and MRV system linking sub-national and national level.

- Sustainable rural energy from renewable resources that can stimulate rural socio-economic development.
- Rural households gain access to electricity from RE-based energy systems.
- Enhanced growth of local economy and increase employment from operation and maintenance services business as well as other productive economy benefitted from access to electricity.
- Women gains access to technical knowledge of RE/EE and about operation/maintenance of RE systems. Reduced gender inequality and enhance women's empowerment in rural socio-economic development activities.

50. Following GEF-prescribed methodology for estimating GHGs emissions reduction, Table 5 below summarizes the estimated global environmental benefits from the implementation of MTRE3 Project. These global benefits (tons CO_2 emission reduction) consist of Direct Project CO_2 emission reduction (from implemented demonstrations during project implementation period and post project 10-year influence period) and Consequential CO_2 emission reduction (based on the bottom-up and top-down approaches).

Table 4: Summary of Cumulative Direct and Consequential GHG Emission Reduction Resulting from the MTRE3 Project

Summary Project Contribution to GHG Emission Reduction (tCO_2 eq)	
Direct Project Emissions Reduction, tCO_2	1,289,846
Direct Post Project Emissions Reduction, tCO_2	3,864,586
Total Direct Project and Post Project (over their useful lifetime), tCO_2	**5,154,432**
Consequential Emission Reductions (BU Approach), tCO_2	15,417,696
Consequential Emission Reduction (TD Approach), tCO_2	1,440,000
GEF Finance	**8,025,000**
Unit Abatement Cost Reduction, US$/$tCO_2$	**1.56**

Detailed calculation of CO_2 emission reduction is in Annex B.

9. Sustainability and Replicability

51. The sustainability of the project impact will be ensured through the leadership of the MEMR and the firm and unwavering commitments of the provincial governments of at least the pilot provinces. The climate change mitigation programs and administrative functions that will be developed under the project are geared towards promoting and supporting renewable energy and energy efficiency project implementation as a key elements of the country's overall environmental and development agenda. The serious enforcement of clear policies, regulations, fiscal/financial instruments, through a supportive regulatory and institutional system, for RE/EE investment, and engagement of private sector in RE/EE investment, will ensure the sustainability of the project impacts. Moreover, the availability of local capacity to operate and maintain the installed RE/EE infrastructures and operational RE/EE project financing mechanism are also essential elements for ensuring sustainability of project impact.

52. The project activities are designed to promote and facilitate scale up and replication. Firstly, the project will support the Government of Indonesia in integrating the climate change mitigation program as part of development agenda through improved RAN-GRK/RAD-GRK as appropriate climate change mitigation action plan and development of Local Energy Planning (RUED). Thus, annually the prioritize climate change mitigation actions will be planned and budgeted. Secondly, the scale up is also expected from creation of enabling environment for RE/EE investments, such as streamlined permit system; establishment and operationalization of Integrated Market Service Centers (IMSCs) in the provinces; and, the establishment of Sustainable Energy Fund (SEF) that will provide financing assistance for RE/EE project developers/investors. Finally, the scale up is

also expected to result from the replication of similar approach in other provinces under the leadership of the MEMR. Since the IMSC in each province will be integral part of the existing Integrated Single-Window Service (PTSP) facility in each province, the sustainability of its operation is ensured. The SEF will be operated by the MEMR after completion of the MTRE3 Project. In that case, the continuous capitalization of the SEF after the MTRE3 Project completion will be ensured through the annual budget allocations of the MEMR[16].

53. Among the criteria in selecting the RE and EE projects that will be implemented and showcased in the pilot provinces is their potential for replication. Hence, such projects are deemed replicable and the project identification process will ensure that reliable data for assessing their investment potentials and viability are available. The selection process will ensure that the final line up of projects consists of those that can be replicated in other sectors and regions in Indonesia. The data and best practices information will be codified as knowledge materials that are accessible for wide stakeholders in the energy and energy end use sectors. The main vehicle for knowledge management intervention of the project is through the establishment of the IMSC in each pilot province. In the IMSC, data on potential renewable resources, spatial map of clean and clear areas for RE investment, verified feasibility study of RE/EE projects, regulation, standards, guideline, technology and service providers will be managed and made available and accessible for interested developers/investors. The Center will serve as host for Single-Window Permit Issuance for investment as well as knowledge hub for the national RAN-GRK Secretariat, GHGs inventory, registry, MRV systems and training centers hosted by MoEF, BAPPENAS and the MEMR. The Center can serve as market place for project owners/developers, government and investors. The best practices from the MTRE3 Project will be compiled, analyzed and communicated to wider stakeholders, particularly in other provinces, via workshop, learning exchange, trainee opportunity and regular publications in printed and social media.

[16] The initiative to establish the Sustainable Energy Fund (SEF) was announced by the Minister of MEMR in December 2015. The MTRE3 Project facilitates the establishment of this fund through the provision of "seed funding", mobilize other resources from development partners, and institutional strengthening.

10. Project Results Framework

Project Title	**Market Transformation through Design and Implementation of Appropriate Mitigation Actions in the Energy Sector (MTRE3)**
Project Objective:	To support the design and implementation of appropriate climate change mitigation actions in the energy generation and energy end use sectors
UNDP Integrated Results and Resources Framework 2014-2017 Outputs:	Output 1.5. Inclusive and sustainable solutions adopted to achieve increased energy efficiency and universal modern energy access (especially off-grid sources of renewable energy).
Expected CPAP 2010-2015 Outputs	2.2.1. National energy policies and guideline developed and integrated into sub-national development plan. 2.2.2. Sub-national authorities and key partners are able to implement programmes, mobilize resources and develop public-private partnership for RE/EE, which will contribute to the reduction of national greenhouse gases emission.
Applicable GEF-5 Strategic Objectives:	Climate Change Mitigation Objective-2: Promote Market Transformation for Energy Efficiency in Industry and the Building Sector. Climate Change Mitigation Objective-3: Promote Investment in Renewable Energy Technologies
Applicable GEF-5 Outcomes:	Outcome 2.2: Sustainable financing and delivery mechanisms established and operational. Outcome 3.1: Favorable policy and regulatory environment created for renewable energy investments. Outcome 3.2. Investment in renewable energy technologies increased Outcome 3.3. GHG emissions avoided.

Project Outcomes	Indicators	Baseline	Targets (End of the Project)	Source of Verifications	Critical Assumptions
Objective: To support the design and implementation of appropriate climate change mitigation actions in the energy generation and energy end use sectors	• Cumulative CO_2 emissions reduction, tons CO_2 eq • Cumulative energy produced from RE systems facilitated by the project, MWh • Cumulative energy saved from EE in commercial buildings facilitated by the project, MWh • Cumulative volume of public and private investment mobilized for SEF, US$ million • Cumulative number of additional households (from baseline) having access	• 0 • 0 • 0 • 0 • 0[17]	• 27,019 • 79,190 • 8,550 • 25 • 80,000	• Report of RAN/ RAD-GRK; Report of Registry and MRV Agency • Annual government expenditure report. • PLN Annual Report, MEMR & ESCOs report. • Project monitoring report, MRV report.	GOI's commitment to climate change mitigation remains unchanged.

[17] The baseline value is 2,066,689 households (HHs). This comprise: 511,233 HHs (@ 61% ratio electrification) in Jambi Province; 900,679 HHs (@ 60.8%) in Riau; 132,556 HHs (@ 47%) in West Sulawesi; and, 522,221 HHs (@ 48%) in NTT. Source: PLN Annual Report 2013.

Project Outcomes	Indicators	Baseline	Targets (End of the Project)	Source of Verifications	Critical Assumptions
	to electricity in pilot provinces				
Component 1: Climate Change Mitigation Options for the RE-based Energy Generation and Energy Efficiency.					
Outcome 1: Prioritized appropriate mitigation actions in the RE-based energy generation and energy efficiency.	Number of provinces with updated sub-national GHG Inventory and GHG Marginal Abatement Cost Curve (MACC) for energy sector	0[18]	4	GHG Inventory Report Publication of provincial MACC.	
Component 2: Market Transformation through Implementation of Appropriate Mitigation Actions in the RE-based Energy Generation and Energy Efficiency.					
Outcome 2: Enhanced and sustainable market diffusion of renewable energy and energy efficiency technologies.	Total number of provinces with operational "Integrated Market Service Center" (IMSC) to support sustainable RE & EE investments.	0	4	Annual report of Provincial Investment Agency. Reports from the IMSCs on RE/RE projects that were assisted in development and implementation	Continued commitment of local government officials in supporting IMSCs in their regions (Presidential Regulation No.27/2009).
	No. of small-to-medium scale RE/EE projects that were financially supported by the Sustainable Energy Fund Cumulative amount of funds from the SEF used in financially supporting small-to-medium scale RE/EE projects , US$ million	0[19] 0	10[20] 25[21]	Reports on SEF-financed RE/EE projects Financing agreements for SEF-financed RE/EE projects	
	Cumulative number of NAMAs proposals developed for RE and EE projects in pilot provinces, based on the identified and	1[22]	4 (2 RE and 2 EE)	Registry system database/Secretariat of RAN-GRK for	Continues support of GOI agencies and partner financing

[18] Data in Provincial GHG inventory 2012 are available with MoEF for all 34 provinces in Indonesia; but no sub-national MACC available.
[19] A Letter of Agreement between UNDP/WHyPGen and PT.SMI on financing support for wind power projects was signed in 2013.
[20] The average size of the identified demo RE projects for demonstration is below 2 MW.
[21] The SEF is expected to mobilize investments of US$ 25 million, targeting the MTRE3 demonstration of 15 MW RE-based power generation and energy efficiency improvement projects in commercial buildings with floor area of 50,000m^2.
[22] This is a financed-ready NAMA on energy efficiency in buildings developed for the Jakarta City Hall.

Project Outcomes	Indicators	Baseline	Targets (End of the Project)	Source of Verifications	Critical Assumptions
	prioritized RE/EE projects.			submission of NAMAs proposals.	institutions to SEF
	Cumulative capacity of RE investment projects implemented, MW	0	15	Reports on approved, financed and implemented RE projects.	Local government continue to consider climate change mitigation as part of local development agenda.
	Cumulative floor area of buildings that were made energy efficient, m^2.	0	50,000	Reports on approved, financed and implemented EE projects.	
Component 3: MRV System and National Registry for Mitigation Actions in the RE-based Energy Generation and Energy Efficiency.					
Outcome 3: Accurate measurement and accounting of actual GHG emission reductions from mitigation actions in the RE-based energy generation and energy efficiency.	No. of registered mitigation actions in energy sector that are endorsed by the MEMR and MoEF.	0	14[23]	Documents of registered projects Website of Registry system of MoEF.	Continuous cooperation and coordination between provincial and national government agencies.
	Total number of MRV reports submitted to MoEF following nationally agreed standard method and guideline.	0	4[24]	Submitted MRV reports.	Data availability at local level to support MRV process.

[23] At least 10 small-medium size RE/EE demonstration projects, 2 RE and 2 EE NAMAs
[24] MRV reports for implemented RE and EE NAMAs projects.

11. Total Budget and Work Plan

Award ID:	**00086173**
Project ID:	**00093506**
Award Title:	**Market Transformation through Design and Implementation of Appropriate Mitigation Actions in the Energy Sector (MTRE3)**
Business Unit:	**IDN 10**
Project Title:	**Market Transformation through Design and Implementation of Appropriate Mitigation Actions in the Energy Sector (MTRE3)**
PIMS No.	**4673**
Implementing Partner:	**Ministry of Energy and Mineral Resources**

GEF Outcome/Atlas Activity	Responsible Party/Implementing Agency	Fund ID	Donor	Atlas Budgetary Account Code	ATLAS Budget Description	Amount Year 1 (USD)	Amount Year 2 (USD)	Amount Year 3 (USD)	Amount Year 4 (USD)	Amount Year 5 (USD)	Total (USD)	Budget Note
Outcome 1: Prioritized appropriate mitigation actions in the energy generation and energy end-use sectors are designed	MEMR	62000	GEF	71200	International consultants	14,396	14,000	0	-	11,500	39,896	1
	MEMR	62000	GEF	71300	Local Consultant	120,000	140,000	120,000	110,000	100,000	590,000	2
	MEMR	62000	GEF	71400	Contractual services-Individual	20,000	20,000	15,000	20,000	-	75,000	3
	MEMR	62000	GEF	72100	Contractual services-Companies	100,000	150,000	150,000	-	-	400,000	4
	MEMR	62000	GEF	71600	Travel	30,000	42,000	82,400	82,000	72,000	308,400	5
	MEMR	62000	GEF	75700	Training, Workshops	60,000	70,000	70,000	70,000	80,000	350,000	6
	MEMR	62000	GEF	72500	Supplies	6,000	10,000	13,000	12,000	12,000	53,000	7
					Sub-Total Component 1:	**350,396**	**446,000**	**450,400**	**294,000**	**275,500**	**1,816,296**	
Outcome 2: Enhanced and sustainable market diffusion of renewable energy and energy efficiency	MEMR	62000	GEF	71200	International consultants	-	-	23,000	-	11,500	34,500	8
	MEMR	62000	GEF	71300	Local Consultant	100,000	150,000	180,000	160,000	160,000	750,000	9
	MEMR	62000	GEF	71400	Contractual services-Ind	-	15,000	17,500	37,500	-	70,000	10
	MEMR	62000	GEF	72100	Contractual		100,000		100,000		200,000	11

					services-Com							
	MEMR	62000	GEF	72630	Capital Grant – Gov't	-	-	2,680,000	-	-	2,680,000	12
	MEMR	62000	GEF	71600	Travel	40,600	80,000	80,000	86,400	86,400	373,400	13
	MEMR	62000	GEF	75700	Training, Workshops	50,000	80,000	85,000	85,000	92,000	392,000	14
technologies.	MEMR	62000	GEF	72500	Supplies (publication, printing stationaries)	6,000	10,000	13,000	12,000	12,000	53,000	7
	MEMR	62000	GEF	72400	Comm. & Audio visual	-	200,000	200,000	-	-	400,000	15
					Sub-Total Component 2:	**196,600**	**635,000**	**3,278,500**	**480,900**	**361,900**	**4,952,900**	
Outcome 3: Accurate measurement and accounting of actual GHG emission reductions from mitigation actions in the RE-based energy generation and energy efficiency	MEMR	62000	GEF	71200	International consultants	-	-	-	30,000	11,500	41,500	16
	MEMR	62000	GEF	71300	Local Consultant	120,000	120,000	130,000	120,000	110,000	600,000	17
	MEMR	62000	GEF	71400	Contractual services-Individuals	-	-	-	25,000		25,000	18
	MEMR	62000	GEF	71600	Travel	5,000	10,000	17,000	17,000	17,000	66,000	19
	MEMR	62000	GEF	75700	Training, Workshops	5,000	13,000	23,000	29,300	39,600	109,900	20
	MEMR	62000	GEF	72500	Supplies	5,300	6,000	6,000	7,000	7,000	31,300	21
					Sub-Total Component 3:	**135,300**	**149,000**	**176,000**	**228,300**	**185,100**	**873,700**	
Project Management	MEMR	62000	GEF	71200	International consultants	0		14000	0	14,000	28,000	22
	MEMR	62000	GEF	71300	Local Consultant	26,000	25,000	25,500	31,500	34,000	142,000	23
	MEMR	62000	GEF	71600	Travel	5,000	5,000	5,000	5,300	7,500	27,800	24
	MEMR	62000	GEF	75700	Training, Workshops	5,000	5,000	5,000	5,000	5,000	25,000	25
	MEMR	62000	GEF	72500	Supplies	3,000	3,000	4,000	4,000	5,000	19,000	26
	MEMR	62000	GEF	72400	Communication & Audio visual	22,000	2,000	2,000	1,000	3,000	30,000	27
	MEMR	62000	GEF	74100	Professional Services -	3,000	3,000	3,000	3,000	3,000	15,000	31

					Audits							
	UNDP	62000	GEF	74500	Direct Project Costing (DPC)	20,000	17,450	10,826	23,202	23,826	95,304	28
					Sub-Total Project Management	**84,000**	**60,450**	**69,326**	**73,002**	**95,326**	**382,104**	
TOTAL GEF BUDGET						**766,296**	**1,290,450**	**3,974,226**	**1,076,202**	**917,826**	**8,025,000**	
Project Management	UNDP	04000	UNDP	75700	Training, Workshops	5,000	7,500	7,500	10,000	10,000	40,000	29
	UNDP	04000	UNDP	71600	Travel	5,000	10,000	15,000	15,000	15,000	60,000	30
TOTAL UNDP Co-Finance						**10,000**	**17,500**	**22,500**	**25,000**	**25,000**	**100,000**	

Allocation of the GEF Budget for CCM-2 (EE) and CCM-3 (RE) Activities

Project Component	**Expected Outputs**	**GEF Funding (US$)**	
		CCM-2	**CCM-3**
Component 1: Climate Change Mitigation Options for the RE-based Energy Generation and Energy Efficiency.	Output 1.1 Defined and established sectoral and sub-national reference baselines	50,000	450,000
	Output 1.2 Developed and published detailed marginal GHGs abatement cost curves	50,000	450,000
	Output 1.3 Selected appropriate and prioritized mitigation options that are integrated into national and provincial development plan	31,630	284,666
	1.4. At least two projects designed, each for the implementation of selected prioritized mitigation actions in RE-based energy generation and energy efficiency in commercial building sectors.	50,000	450,000
	Total Outcome 1	**181,630**	**1,634,666**
2. Market Transformation through Implementation of Appropriate Mitigation Actions in the RE-based Energy Generation and Energy Efficiency.	Output 2.1 Established Integrated Market Service Center in the pilot provinces	100,000	900,000
	Output 2.2 Established technical support system to provide training for operation and maintenance of RE & EE technologies including MRV	85,290	767,610
	Output 2.3 Implemented improved financing mechanisms for investments in the climate change mitigation projects	270,000	2,430,000
	Output 2.4 Implemented and operational pilot testing of two RE and two EE investments	40,000	360,000
	Total Outcome 2	**495,290**	**4,457,610**
3. MRV System and National Registry for Mitigation Actions in the RE-based Energy Generation and Energy Efficiency.	Output 3.1 Improved and operational registry mechanism for mitigation actions in energy sector.	25,000	225,000
	Output 3.2 Developed Measurement, Reporting and Verification (MRV) guidelines and standard	25,000	225,000
	Output 3.3 Implemented MRV system for the selected appropriate mitigation actions	37,370	336,330
	Total Outcome 3	**87,370**	**786,330**
Project Management Cost		38,210	343,894
TOTAL		**802,500**	**7,222,500**

Budget Notes

BUDGET NOTES	Total (US$)	Description
1	39,896	Short term International expert consultant to review development of GHG Inventory and MACC (60 days/year * US$ 350/day + travel cost)
2	590,000	Technical RE/EE Experts of 4 Provincial liaisons (4 months/y), 1 TWG Officer (12 month/y) and CTA (3 months/y) for 5 years
3	75,000	Short term national expert on public policy in energy sector to assist development of RUED and NAMAs proposal (in average 120 days/y * USD 250/day)
4	400,000	Lump sum contract to 2 companies to conduct in 4 pilot provinces: Lot 1 (GHG Inventory for energy sector, energy audit in commercial building for benchmarking SEC) and Lot 2 (GIS Mapping for RE potential and development of provincial MACC) 2 Lot * USD 200,000= US$ 400,000
5	308,400	Average annual travel and allowance cost for 4 government officials and 1 PMU staff to attend consultative meeting with stakeholders in sub-district level in 4 provinces (6 times/y* 5 persons* 3days*4 provinces; allowance US$100/day and travel US$ 150/time)
6	350,000	Average annual meeting cost for public consultation and training workshops on MACC, RUED and NAMAs in 4 provinces (6 times/year*30 participants*3 days*4 provinces; meeting package and local transport US$ 50/person)
7	106,000	Lump sum annual cost for publication and dissemination of results, average 3 times/year for 5 years in 4 provinces.
8	34,500	Short term International expert consultant to support development of RE technology standard, Green building standard, accreditation mechanism (60 days*US$ 350 + travel cost)
9	750,000	Technical RE/EE Experts of 4 provincial liaison officers (4 months/y), TWG2 officer (12 months) and CTA (6 months/y) for 5 years.
10	70,000	Short term national experts to support assessment of RE/EE financing mechanism/business model; training materials; and RE/EE standard and accreditation (3 experts for 100 days* USD 250/day)
11	200,000	Lump sum contract to 2 companies to conduct 4 financed-ready RE/EE Feasibility Study: Lot 1 (RE-based power generation) and Lot 2 (energy efficiency in commercial building) 2 Lot * US$ 100,000 = US$ 200,000
12	2,680,000	Loan Guarantee for Sustainable Energy Fund to be managed by accredited financing institution. Sufficient as start-up fund to guarantee loan for 1 MW biomass power plant from Palm Oil Mill Effluent (POME).
13	373,400	Average annual travel and allowance cost for government officials, PMU staff and local service providers to visit potential RE/EE sites in 4 provinces (6 times/y* 8 persons* 3days*5 years* 4 provinces; allowance US$100/day and travel US$ 150/time)
14	392,000	Average annual meeting cost for public consultation and training workshops on RE/EE operation and maintenance, technology standard and accreditation, data collection (7 times/year*30 participants*3 days*4 provinces; meeting package and local transport US$ 50/person)
15	400,000	Purchasing of hardware for data management and remote monitoring system in 4 provinces to support Integrated Market Service Centre, lump sum US$ 100,000/province
16	41,500	Short term International expert consultant to develop RE/EE MRV method and GHGs audit procedure (100 days/year * US$ 350/day)
17	600,000	Technical RE/EE Experts of 4 Provincial liaisons (4 months/y), 1 TWG Officer (12 month/y) and CTA (3 months/y) for 5 years
18	25,000	Short term national experts to develop RE/EE MRV method and GHG audit procedures in collaboration with International Expert (1 expert for 100 days* USD 250/day)
19	66,000	Average annual travel and allowance cost for government officials, PMU staff and project developers to attend public consultation for development of registry system and MRV method in 4 provinces (3 times/y* 5 persons* 3days*4 provinces; allowance US$100/day and travel US$ 150/time)
20	109,900	Average annual meeting cost for public consultation and training workshops on RE/EE operation and maintenance, technology standard and accreditation, data collection (3 times/year*30 participants*3 days*4 provinces; meeting package and local transport US$ 50/person)
21	31,300	Lump sum annual cost for publication and dissemination of results, average 3 times/year.

22	28,000	Short term International expert consultant to conduct mid-term and terminal evaluation of project (40 days/year * US$ 350/day)
23	142,000	Project management unit: project manager SC9, 2 admin/finance personnel SC6 (12 months/y for 5 years) including national expert consultant to conduct mid-term and terminal evaluation of project
24	27,800	Local travel cost for project management purposes, monitoring, coordination with other agencies (US$ 50/day in average 4 times/month)
25	25,000	Internal coordination meeting (in average 4 times/y, 2 days, including daily allowance for provincial officers)
26	19,000	Lump sum annual cost for printing and stationaries for project management US$ 600/month.
27	30,000	Lump sum annual cost for communication, audio visual and equipment for project management US$ 500/month and US$ 21,000 for laptop/PCs/LCDs, etc.
28	95,304	Indicative service cost provided by UNDP for the project, based on Universal Price List 2015.
29	40,000	Project management learning sessions on result-based management, gender responsive actions, theory of change, monitoring and knowledge management (4 times/y during 5 years, external resource persons fee US$ 2000/session)
30	60,000	Average travel and daily allowance cost for UNDP programme staffs to oversee the project implementation, compliance to admin/finance for 5 years
31	15,000	Budget set up for audit costs - towards professional audit services

Summary of annual financing distribution (US$)

	Year 1	Year 2	Year 3	Year 4	Year 5	Total
GEF	766,296	1,290,450	3,974,226	1,076,202	917,826	8,025,000
MEMR	2,000,000	2,000,000	1,500,000	1,500,000	1,000,000	8,000,000
UNDP	10,000	17,500	22,500	25,000	25,000	100,000
PT. Pasadena Engineering Indonesia	1,000,000	1,000,000	6,000,000	1,000,000	1,000,000	10,000,000
PT. Daun Biru	2,000,000	2,000,000	30,000,000	3,000,000	3,000,000	40,000,000
PT. Multi Fabrindo Gemilang	200,000	300,000	1,000,000	300,000	200,000	2,000,000
Total	**5,976,296**	**6,607,950**	**42,496,726**	**6,901,202**	**6,142,826**	**68,125,000**

12. Management Arrangement

54. The project will apply adaptive management approach in its implementation. Based on the partnerships defined and firmed up during the project development, the management arrangements have always been anchored on co-operation and mutual sharing of benefits where accountability and responsibility for implementing the project and achieving the project outputs. Such arrangement should also be based on collective decision making through a Project Steering Committee (PSC) composed of the project key stakeholders such as the MEMR/DGNREEC, MOEF, MOF, BAPPENAS, UNDP and government representative from pilot provinces. The MTRE3 Project Organizational Structure is seen in **Figure 1** below:

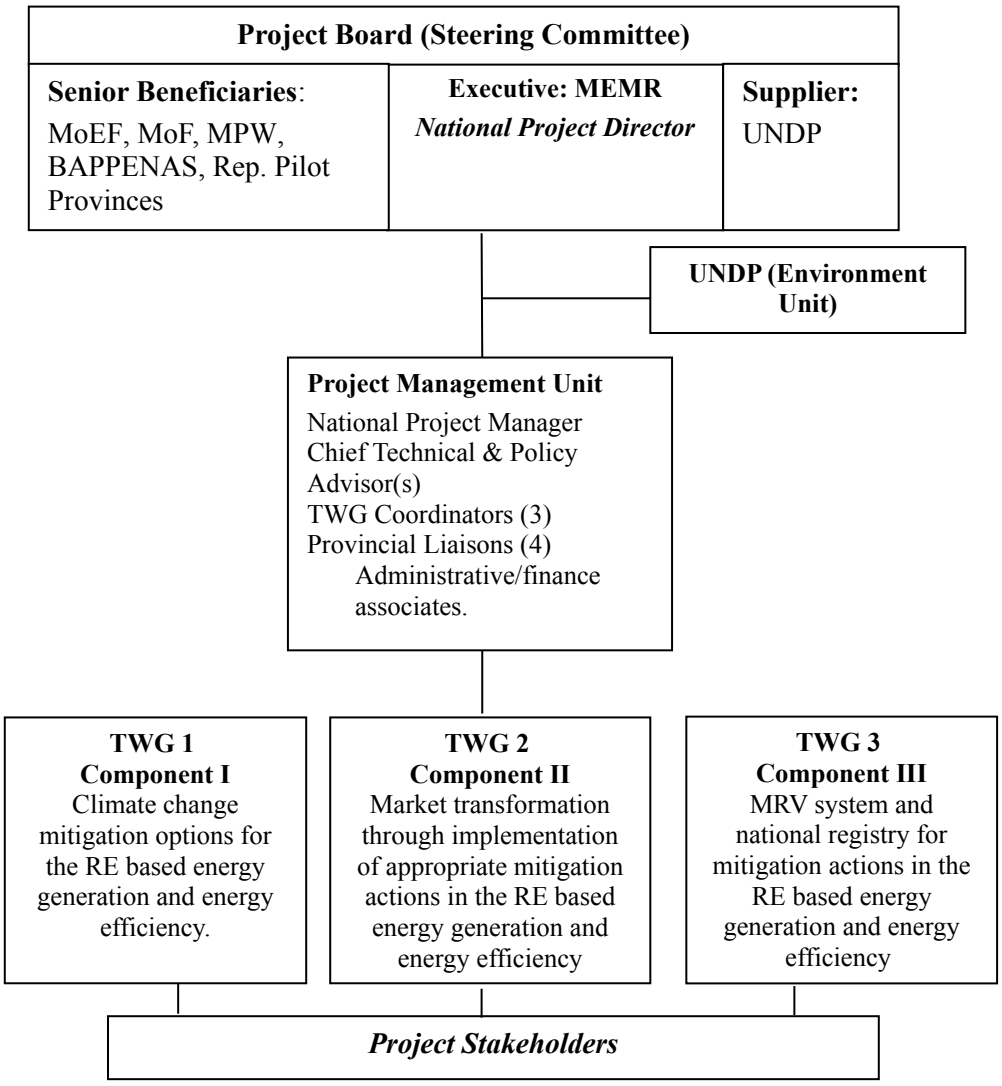


Figure 1: MTRE3 Project Organizational Structure

55. The Technical Working Groups (TWGs) represent the coordination among the Stakeholders for RE and EE in carrying out the project activities according to an agreed plan and approach. Table 6 describes the scope of cooperation in the work areas.

56. UNDP's Country Office in Indonesia will be responsible for ensuring transparency, appropriate conduct and financial responsibility. This office will oversee annual financial audits, as well as the execution of independent Project mid-term review and terminal evaluation. All financial

transactions and agreements, including contracts with staff and consultants if provided by UNDP, will follow the rules and regulations of United Nations. The UNDP Bangkok Regional Hub (BRH) will provide regular programmatic and administrative oversight as well.

Table 5: Scope of cooperation for each Technical Working Group (TWG)

TWGs	Scope of Cooperation
TWG 1: Climate change mitigation options for the RE based energy generation and energy efficiency.	• Policy, Regulation and Institutional Setup for Climate Change Mitigation Action in RE and EE
TWG 2: Market transformation through implementation of appropriate mitigation actions in the RE based energy generation and energy efficiency.	• Training, public awareness and policy review on promoting RE and EE investments • Institutional development: Integrated Market Service Center for RE/EE investments, revitalizing and strengthening the existing institutional setup • Database and network enhancement for RE and EE stakeholders • RE and EE project pilot testing (TA, FS, and engineering assessment)
TWG 3: MRV system and national registry for mitigation actions in the RE based energy generation and energy efficiency	• MRV system in place and investment projects of MTRE3 start following the standards and guidelines • Functional national Registry for mitigation actions in RE and EE sectors • Capacity development and Institutional Arrangement to apply MRV scheme for RE and EE project investment

National Coordination and Implementation Arrangements

57. The MTRE3 Project will be managed and coordinated at two levels: (a.) policy and institutional level and (b.) operational level. The first level will primarily focus on institutional coordination and policy advocacy to maintain the project's relevance to national priorities, while the second level will focus on implementation of project activities. A Project Steering Committee (PSC) will be established and will comprise of the representatives of the UNDP-Indonesia, and the participating government agencies: MEMR/DGNREEC, MOEF, MOF, MPW and BAPPENAS, and the Director of the Project Management Unit (PMU). The National Project Director (NPD) will come from the Implementing Partner, which is the MEMR/DGNREEC and appointed by the Government of Indonesia through the MEMR/DGNREEC.

58. The PSC will play the role of an advisory committee. The PSC members will also be invited to participate in the annual project review meetings. The Chairperson of the PSC is the designated NPD. The NPM provides the management oversight to the whole project and assists the NPD in the operational aspects of the project implementation. The PMU will be established to be responsible for day-to-day coordinating and implementing the activities of the project and carrying out the prescribed monitoring system. The PMU Director will serve as the Secretary of the PSC.

59. Implementation of the MTRE3 project will be through the UNDP's National Implementation Modality (NIM) on behalf of the Government of Indonesia through the MEMR / DGNREEC as the Implementing Partner. The Implementing Partner will assume the overall responsibility of ensuring that all activities are executed accordingly and as per the approved Project Document. The NPD will be responsible for the achievement of the project objectives through institutional

coordination with the key stakeholder members of the PSC and overall alignment of the MTRE3 Project with the relevant national RE and EE programs of Indonesia. He/she takes overall responsibility for all projects' timely reporting, including the submission of Annual Work Plans (AWP), Annual Project Report/Project Implementation Review (APR/PIR) and financial reports. He/She will ensure the delivery of the project outputs and the judicious use of the project resources. This will ensure that expected outputs are delivered using the most efficient and cost effective implementation strategies and procedures. The NPM shall also see to it that the Implementing Partner (DGNREEC) supports the project in the planning, coordination, secretariat, administration and financial management of the project in coordination with UNDP-Indonesia. As the project's Main Implementing Partner, the MEMR will also provide in-kind contribution to implement the MTRE3 Project.

60. The PSC will be responsible for the following: (a). Reviewing of annual progress reports for necessary guidance; (b) Reviewing and approving the annual work plans and budgets; (c) Providing guidance on the effectiveness of MTRE3 project implementation, and its linkages to corporate UNDP policy decisions, and other UNDP initiatives; and, (d) Monitoring and evaluating the implementation of MTRE3 towards the intended outputs, after a year and half year of project execution. As a minimum, the PSC will meet at least once a year, allowing for the stakeholders to review the progress with the project implementation and to agree on a coordinated annual project implementation strategy and plan. The first PSC meeting should be held within the first 12 months following the inception workshop.

61. UNDP-Indonesia, in close coordination with the UNDP-GEF Regional Technical Advisor (RTA) for Climate Change in the Asia-Pacific Region will carry out the GEF oversight and ensure that expected project performance standards are met. RTA will be also responsible for monitoring and evaluation (M&E), including organizing project reviews, approving annual implementation work plans and budget revisions, monitoring progress, identifying problems, suggesting actions to improve project performance, facilitating timely delivery of project inputs, and provide linkages to other sub-regional, Asia-Pacific regional and global initiatives. All M&E functions will be carried out in line with standard UNDP and UNDP-GEF procedures. UNDP Indonesia will also provide country office support for all the activities of the project as coordinated with the Implementing Partner.

62. A Project Management Unit (PMU) will be established and is responsible for the day-to-day management of all the project activities including those on capacity building, demonstration sub-projects and dissemination activities at the national level. The PMU will be managed by a National Project Manager (NPM), who will be supported by Chief Technical and/or Policy Advisor (CTA), administrative staff and Coordinators corresponding to the 3 TWG groups. A CTA provides strategic technical/policy support and advice to ensure that the project is implemented according to the agreements in the Project Document and the standards of UNDP and GEF in project implementation, as well as to provide high-level and strategic advisory to enhance impact of the project. Apart from coordinator for each TWG, there will be provincial liaison officer for each pilot sites. The liaison officer will be responsible to liaise government and stakeholders in terms of technical and administration at local level.

63. The MTRE3 PMU will prepare its overall project 5-year work plan at the inception stage of the project based on the project activities that are described in this Project Document, specifying the level of activities that will be carried out for the year, the targets to be achieved, and the corresponding inputs (in terms of manpower and budget). During the inception stage, the Implementing Partner with assistance of the PMU will prepare its first year work plan and submit this to the UNDP Indonesia to be approved for the allocation of funds for the implementation of the initial year activities. Succeeding annual work plans, based on the results of the previous year and the planned activities for the current year, will be prepared and submitted for approval and budget allocations at the start of each year.

64. Also during the inception stage, the MTRE3 PMU will prepare its overall 5–year M&E plan including Annual Targets based on the Project Planning Matrix. The M&E plan will consist of success indicators with realistic targets and time lines, and backed up with clear means of verification, and assumptions. Each activity/task that will be carried out and monitored in terms of the appropriate output indicators (for the activity deliverables) and the impact indicators (for the impacts). The targets will be reviewed each year and any necessary revision or adjustment of these, as well as the assumptions will be done on a continuous basis during the life of the project as part of adaptive management. Consultation and synergy with other relevant stakeholders as well as with other GEF climate change mitigation projects and other similar initiatives will be conducted to enhance impact.

13. Monitoring Framework and Evaluation

65. The project will be monitored through the following M&E activities.

Project start:

66. A Project Inception Workshop will be held within the first two months of project start with those with assigned roles in the project organization structure, UNDP country office and where appropriate/feasible regional technical policy and program advisors as well as other stakeholders. The Inception Workshop is crucial to building ownership for the project results and to plan the first year annual work plan.

67. The Inception Workshop should address a number of key issues including:

    a) Assist all partners to fully understand and take ownership of the project. Detail the roles, support services and complementary responsibilities of UNDP CO and BRH staff vis-à-vis the project team. Discuss the roles, functions, and responsibilities within the project's decision-making structures, including reporting and communication lines, and conflict resolution mechanisms. The Terms of Reference for project staff will be discussed again as needed.
    b) Based on the project results framework and the relevant GEF Tracking Tool if appropriate, finalize the first annual work plan. Review and agree on the indicators, targets and their means of verification, and recheck assumptions and risks.
    c) Provide a detailed overview of reporting, monitoring and evaluation (M&E) requirements. The Monitoring and Evaluation work plan and budget should be agreed and scheduled.
    d) Discuss financial reporting procedures and obligations, and arrangements for annual audit.
    e) Plan and schedule Project Steering Committee meetings. Roles and responsibilities of all project organization structures should be clarified and meetings planned. The first Project Steering Committee meeting should be held within the first 12 months following the inception workshop.

68. An Inception Workshop report is a key reference document and must be prepared and shared with participants to formalize various agreements and plans decided during the meeting.

Quarterly:

69. The quarterly monitoring will involve:

- Project progress shall be monitored in the UNDP Enhanced Results Based Management Platform.
- Based on the initial risk analysis submitted, the risk log shall be regularly updated in ATLAS. Risks become critical when the impact and probability are high.

- Based on the information recorded in Atlas, Project Progress Reports (PPR) can be generated in the Executive Snapshot.
- Other ATLAS logs can be used to monitor issues, lessons learned etc. The use of these functions is a key indicator in the UNDP Executive Balanced Scorecard.

Annually:

70. Annual Project Review/Project Implementation Reports (APR/PIR): This key report is prepared to monitor progress made since project start and in particular for the previous reporting period (30 June to 1 July). The APR/PIR combines both UNDP and GEF reporting requirements.

    The APR/PIR includes, but is not limited to, reporting on the following:

    - Progress made toward project objective and project outcomes - each with indicators, baseline data and end-of-project targets (cumulative)
    - Project outputs delivered per project outcome (annual).
    - Lesson learned/good practice.
    - AWP and other expenditure reports
    - Risk and adaptive management
    - ATLAS QPR
    - Portfolio level indicators (i.e. GEF focal area tracking tools) are used by most focal areas on an annual basis as well.

Periodic Monitoring through site visits:

71. UNDP CO and the UNDP BRH will conduct visits to project sites based on the agreed schedule in the project's Inception Report/Annual Work Plan to assess first hand project progress. Other members of the Project Board may also join these visits. A Field Visit Report/BTOR will be prepared by the CO and UNDP BRH and will be circulated no less than one month after the visit to the project team and Project Board members.

Mid-term of project cycle:

72. The project will undergo an independent Mid-Term Evaluation at the mid-point of project implementation (approximately end of 2018). The Mid-Term Evaluation will determine progress being made toward the achievement of outcomes and will identify course correction if needed. It will focus on the effectiveness, efficiency and timeliness of project implementation; will highlight issues requiring decisions and actions; and will present initial lessons learned about project design, implementation and management. Findings of this review will be incorporated as recommendations for enhanced implementation during the final half of the project's term. The organization, terms of reference and timing of the mid-term evaluation will be decided after consultation between the parties to the project document. The Terms of Reference for this Mid-term evaluation will be prepared by the UNDP CO based on guidance from the BRH and UNDP-GEF. The management response and the evaluation will be uploaded to UNDP corporate systems, in particular the UNDP Evaluation Office Evaluation Resource Center (ERC). The relevant GEF Focal Area Tracking Tools will also be completed during the mid-term evaluation cycle.

End of Project:

73. An independent Terminal Evaluation will take place three months prior to the final Project Board meeting and will be undertaken in accordance with UNDP and GEF guidance. The terminal evaluation will focus on the delivery of the project's results as initially planned (and as corrected after the mid-term evaluation, if any such correction took place). The terminal

evaluation will look at impact and sustainability of results, including the contribution to capacity development and the achievement of global environmental benefits/goals. The Terms of Reference for this evaluation will be prepared by the UNDP CO based on guidance from the BRH and UNDP-GEF.

74. The Terminal Evaluation should also provide recommendations for follow-up activities and requires a management response, which should be uploaded to PIMS and to the UNDP Evaluation Office Evaluation Resource Center (ERC).

75. The relevant GEF Focal Area Tracking Tools will also be completed during the final evaluation.

76. During the last three months, the project team will prepare the Project Terminal Report. This comprehensive report will summarize the results achieved (objectives, outcomes, outputs), lessons learned, problems met and areas where results may not have been achieved. It will also lay out recommendations for any further steps that may need to be taken to ensure sustainability and replicability of the project's results.

Monitoring and Evaluation Budget

77. The indicative project monitoring and evaluation budget is presented in the table below.

Table 6: Monitoring Work Plan and Indicative Budget

Type of M&E activity	***Responsible Parties***	***Budget*** *(US$) excluding project staff time; all figures are indicative*	***Time frame***
Inception Workshop (IW) & associated arrangements	▪ Project Manager (PM) ▪ UNDP CO	8,000	Within first two months of project start up
Inception Report	▪ Project Team ▪ UNDP CO		Immediately following IW
Measurement of Means of Verification of project results (baseline and end-of-project impact study)	• Project Manager /Implementing Partner ▪ Project team members	Included in Project Management	Start, mid and end of project (during evaluation cycle) and annually when required
Measurement of Means of Verification for Project Progress and Performance (measured on an annual basis)	• Oversight by UNDP-GEF BRH Technical Advisor and PM • Measurements by regional field officers and local IAs	Included in Project Management	Annually prior to APR/PIR and to the definition of annual work plans
APR/PIR	▪ PMU - PM ▪ M&E team ▪ UNDP CO ▪ UNDP-GEF BRH	0 (included in routine project staff activity)	Annually
Meetings of Steering Committee and relevant meeting proceedings (minutes)	▪ PM ▪ UNDP CO ▪ Regional advisory boards ▪ National implementing agency	5,000	PSC at least once a year, ideally immediately following Regional Advisory Board meetings
Quarterly Operational status reports	▪ M&E team	0 (included in routine project staff activity)	To be determined by Project team and UNDP CO
Technical monitoring, evaluation, and reporting	▪ Project team ▪ National and international	0 (included in	Continuous, starting from project inception

Type of M&E activity	*Responsible Parties*	***Budget** (US$) excluding project staff time; all figures are indicative*	*Time frame*
within project components.	consultants as needed	routine project staff and counterpart activity)	
Midterm Evaluation (external)	▪ Project team ▪ UNDP CO ▪ UNDP/GEF BRH ▪ External Consultants (i.e. evaluation team)	35,000	At the midpoint of project implementation.
Final Evaluation (external)	▪ External Consultants (i.e. evaluation team) ▪ Project team ▪ UNDP CO ▪ UNDP/GEF BRH	35,000	At the end of project implementation
Final Report	▪ External Consultant ▪ Project team ▪ UNDP CO	(costs included in Terminal Evaluation, above)	At least one month before the end of the project
Compilation of lessons learned	▪ M&E team ▪ Project team ▪ UNDP CO ▪ UNDP/GEF BRH	16,000	Annually
Financial audit	▪ UNDP CO ▪ Project team ▪ External auditors	Indicative cost per year: 3,000	Yearly @5 years
Achievements and project performance monitoring to field sites	▪ Project Team ▪ GoI ▪ Media ▪ Donors ▪ UNDP CO* ▪ UNDP/GEF BRH* * *covered by IA fees*	44,000	Annually or more frequently
TOTAL INDICATIVE COST	(Excluding project team staff time and UNDP staff and travel expenses)	158,000	

78. Project monitoring and evaluation will be conducted in accordance with established UNDP and GEF procedures and will be provided by the project team and the UNDP Country Office (UNDP-CO) with support from UNDP-GEF. The Project Planning Matrix provides *performance* and *impact* indicators for project implementation along with their corresponding *means of verification*. These will form the basis on which the project's Monitoring and Evaluation system will be built.

Learning and knowledge sharing:

79. Results from the project will be disseminated within and beyond the project intervention zone through existing information sharing networks and forums.

80. The project will identify and participate, as relevant and appropriate, in scientific, policy-based and/or any other networks, which may be of benefit to project implementation though lessons learned. The project will identify, analyze, and share lessons learned that might be beneficial in the design and implementation of similar future projects.

81. Finally, there will be a two-way flow of information between this project and other projects of a similar focus.

Communications and visibility requirements:

82. Full compliance is required with UNDP's Branding Guidelines. These can be accessed at http://intra.undp.org/coa/branding.shtml, and specific guidelines on UNDP logo use can be accessed at: http://intra.undp.org/branding/useOfLogo.html. Amongst other things, these guidelines describe when and how the UNDP logo needs to be used, as well as how the logos of donors to UNDP projects needs to be used. For the avoidance of any doubt, when logo use is required, the UNDP logo needs to be used alongside the GEF logo. The GEF logo can be accessed at: http://www.thegef.org/gef/GEF_logo. The UNDP logo can be accessed at http://intra.undp.org/coa/branding.shtml.

83. Full compliance is required with the GEF's Communication and Visibility Guidelines ("GEF Guidelines"). The GEF Guidelines, which shall be adhered to can be accessed at: http://www.thegef.org/gef/sites/thegef.org/files/documents/C.40.08_Branding_the_GEF%20final_0.pdf. Amongst other things, the GEF Guidelines describe when and how the GEF logo needs to be used in project publications, vehicles, supplies and other project equipment. The GEF Guidelines also describe other GEF promotional requirements regarding press releases, press conferences, press visits, visits by Government officials, productions and other promotional items.

84. Where other agencies and project partners have provided support through co-financing, their branding policies and requirements should be similarly applied.

General

UNDP support services

85. MEMR has entered into an agreement with UNDP for direct project support services in the form of procurement of goods and services during the project implementation process (see Annex A). In such a case, appropriate cost recovery will be charged as per UNDP rules and regulations. The support services will be outlined in the form of Letter of Agreement signed between MEMR and UNDP.

Prior obligations and prerequisites

86. No prior obligations or prerequisites have been identified.

Audit Clause

87. Audit will be conducted according to UNDP Financial Regulations and Rules and applicable Audit policies. The Government of will provide the UNDP Resident Representative with certified periodic financial statements, and with an annual audit of the financial statements relating to the status of UNDP (including GEF) funds expended on the project according to the established procedures set out in the appropriate UNDP programming and finance manuals. The audit will be conducted by the legally recognized auditor of the Government of Indonesia, or by a commercial auditor engaged by the Government.

Agreement on the intellectual property rights and use of logo on the project's deliverables

88. In order to accord proper acknowledgement to GEF for providing funding, a GEF logo will appear on all relevant GEF-supported project publications, including among others, project hardware, if any, purchased with GEF funds. Any citation on publications regarding projects

funded by GEF should also accord proper acknowledgement to GEF. The UNDP logo should be prominent – and separated from the GEF logo. Alongside GEF and UNDP logo, the MERE logo may also feature as the Implementing Partner of the project.

Assets

89. The ownership of the assets procured under the project from GEF grant money lies with the UNDP Resident Representative until the end of the project. At the end of the project, the assets would be transferred to the implementing Ministry of the Government of Indonesia (MEMR) following UNDP applicable rules and regulations.

14. Legal Context

90. This document together with the CPAP signed by the Government and UNDP which is incorporated by reference constitute together a Project Document as referred to in the SBAA and all CPAP provisions apply to this document.

91. Consistent with the Article III of the Standard Basic Assistance Agreement, the responsibility for the safety and security of the implementing partner and its personnel and property, and of UNDP's property in the implementing partner's custody, rests with the implementing partner.

92. The implementing partner shall:
    a) Put in place an appropriate security plan and maintain the security plan, taking into account the security situation in the country where the project is being carried;
    b) Assume all risks and liabilities related to the implementing partner's security, and the full implementation of the security plan.

93. UNDP reserves the right to verify whether such a plan is in place, and to suggest modifications to the plan when necessary. Failure to maintain and implement an appropriate security plan as required hereunder shall be deemed a breach of this agreement.

94. The implementing partner agrees to undertake all reasonable efforts to ensure that none of the UNDP funds received pursuant to the Project Document are used to provide support to individuals or entities associated with terrorism and that the recipients of any amounts provided by UNDP hereunder do not appear on the list maintained by the Security Council Committee established pursuant to resolution 1267 (1999). The list can be accessed via http://www.un.org/Docs/sc/committees/1267/1267ListEng.htm. This provision must be included in all sub-contracts or sub-agreements entered into under this Project Document.

95. The UNDP Resident Representative in Indonesia is authorized to effect in writing the following types of revision to this Project Document, provided that he/she has verified the agreement thereto by the UNDP Regional Coordination Unit and is assured that the other signatories to the Project Document have no objection to the proposed changes:

- Revision of, or addition to, any of the annexes to the Project Document;
- Revisions which do not involve significant changes in the immediate objectives, outputs or activities of the project, but are caused by the rearrangement of the inputs already agreed to or by cost increases due to inflation;
- Mandatory annual revisions which re-phase the delivery of agreed project inputs or increased expert or other costs due to inflation or take into account agency expenditure flexibility; and
- Inclusion of additional annexes and attachments only as set out here in this Project Document

96. Standard text has been inserted in the template. It should be noted that although there is no specific statement on the responsibility for the safety and security of the Implementing Partner in the SBAA and the supplemental provisions, the second paragraph of the inserted text should read in line with the statement as specified in SBAA and the supplemental provision, i.e. “the Parties may agree that an Implementing Partner shall assume primary responsibility for execution of a project”.

III. ANNEXES

Annex A: Country Office Support Service Agreement

COUNTRY OFFICE SUPPORT SERVICE (COSS) AGREEMENT

AGREEMENT BETWEEN UNDP AND THE GOVERNMENT FOR THE PROVISION OF SUPPORT SERVICES

1. Reference is made to consultations between officials of the Government of Indonesia Ministry of Energy and Mineral Resources (hereinafter referred to as "the Government") and officials of UNDP with respect to the provision of support services by the UNDP country office for nationally managed programmes and projects. UNDP and the Government hereby agree that the UNDP country office may provide such support services at the request of the Government through its institution designated in the relevant programme support document, as described below.
2. The UNDP country office may provide support services for assistance with reporting requirements and direct payment. In providing such support services, the UNDP country office shall ensure that the capacity of the Government-designated institution is strengthened to enable it to carry out such activities directly. The costs incurred by the UNDP country office in providing such support services shall be recovered from the administrative budget of the office.
3. The UNDP country office may provide, at the request of the designated institution, the following support services for the activities of the programme:
   (a) Identification and/or recruitment of project and programme personnel and technical expertise;
   (b) Procurement of goods and services to undertake agreed activities;
   (c) Administration of the donor contribution;
   (d) Management of grant agreements and related disbursements for project-related activities. (To be specified in the project details)
4. The procurement of goods and services and the recruitment of programme personnel by the UNDP country office shall be in accordance with the UNDP regulations, rules, policies and procedures.
5. The relevant provisions of the Revised Basic Agreement for Technical Assistance signed 29 October 1954 between the United Nations, the International Labour Organization, the Food and Agriculture Organization of the United Nations, the United Nations Educational, Scientific and Cultural Organization, the International Civil Aviation Organization, and the World Health Organization and the Government of the Republic of Indonesia, the Standard Agreement on Operational Assistance signed 12 June 1969 between the United Nations, the International Labour Organization, the Food and Agriculture Organization of the United Nations, the United Nations Educational, Scientific and Cultural Organization, the International Civil Aviation Organization, the World Health Organization, the International Telecommunication Union, the World Meteorological Organization, the International Atomic Energy Agency, the Universal Postal Union, the Inter-Governmental Maritime Consultative Organization and the United Nations Industrial Development Organization and the Government of the Republic of Indonesia and the Agreement signed 7 October 1960 between the United Nations Special Fund and the Government of the Republic of Indonesia including the provisions on liability and privileges and immunities, shall apply to the provision of such support services. The Government shall retain overall responsibility for the nationally managed programme through its designated institution. The responsibility of the UNDP country office for the provision of the support services described herein shall be limited to the provision of such support services detailed in the annex to the programme support document.
6. Any claim or dispute arising under or in connection with the provision of support services by the UNDP country office in accordance with this letter shall be handled pursuant to the relevant provisions of the above mentioned agreements.
7. The manner and method of cost-recovery by the UNDP country office in providing the support services described in paragraph 3 above shall refer to the enhanced UNDP UPL (Universal Price List) effective date 1 January 2015.
8. The UNDP country office shall submit progress reports on the support services provided and shall report on the costs reimbursed in providing such services, as may be required.
9. Any modification of the present arrangements shall be effected by mutual written agreement of the parties hereto.

If you are in agreement with the provisions set forth above, please sign and return to this office two signed copies of this letter. Upon your signature, this letter shall constitute an agreement between your Government and UNDP on the terms and conditions for the provision of support services by the UNDP country office for nationally managed programmes and projects.

Indicative COSS for the MTRE3 Project has been calculated as follows:

No	Service Name	Unit	Volume	Unit Cost (US$)*	Total (US$)
1	Payment of DSA advance and its settlement	Financial Resource Management Unit	4	67.06	804.72
2	Payment process	Financial Resource Management Unit	32	33.53	63,304.64
3	Payment Process for Cash Distribution via Service Provider	Financial Resource Management Unit	4	60.57	726.84
4	Review and approval of APJV and GLJE (adjustment/correction/petty cash report)	Financial Resource Management Unit	4	22.37	1,073.76
5	Procurement Process < USD 5,000	Procurement	4	196.44	4,714.56
6	Procurement Process Without ACP Approval (below USD 50,000)	Procurement	10	196.44	1,964.40
7	Individual Consultant without ACP Approval (below USD 100,000)	Procurement	2	217.25	434.50
8	Individual Consultant < USD 5,000	Procurement	10	86.90	869.00
9	Vendor registration for supplier	Procurement	2	18.11	72.44
10	Request for PO creation only	Procurement	9	49.11	3,093.93
11	RECRUITMENT : Service Contract (SC 3 – 7)	Human Resource	8	779.12	6,232.96
12	RECRUITMENT : Service Contract (SC 8 – 11)	Human Resource	7	810.12	5,670.84
13	Vendor registration for staff and SC	Human Resource	15	18.11	271.65
14	Extension of Service Contract	Human Resource	15	86.00	5,160.00
15	PO Ticket/Travel Creation (For Travel with Ticket Value USD 2.500 and Above Only)	Administration Unit	2	45.96	91.92
16	Courier services (City Courier, Domestic, International)	Administration Unit	4	25.78	618.72
17	Vendor Registration (for Workshop/Meeting Participant Only)	Administration Unit	2	18.11	36.22
18	Creation UNDP's email address for new recruit under UNDP' contract management	Information Communication Technology	15	10.88	163.20
	TOTAL				95,304.30

*Based on UPL 2015

Yours sincerely,

_____
Signed on behalf of UNDP
Christophe Bahuet
Country Director
[*Date*]

_____
For the Government
Rida Mulyana
DG for New and Renewable Energy & Energy Conservation
Ministry of Energy and Mineral Resources
[*Date*]

Annex B: Calculation of CO_2 Emission Reduction Attributable to MTRE3 Project

Energy Efficiency (EE) in Commercial Building Sectors

Direct Project and Direct Post Project Emission Reduction

Business as Usual Scenario (Baseline)

The baseline of Specific Energy Consumption (SEC) for commercial building in Indonesia according to Indonesia Green Building Council Indonesia (GBCI) and the MEMR energy audit data is 285 kWh/m^2/year. This is the baseline average SEC of commercial buildings in the country. The average reduction in specific energy consumption in new and existing commercial building sectors from EE improvement projects that were voluntarily implemented by the building owners is about 10% from the baseline. Only very few commercial buildings could reduce 20% from baseline SEC. The average grid emission factor for Indonesia in 2013-2014 according to State-owned Electricity Company (PLN) data was 0.8 tCO_2/MWh.

Alternative scenario

Depart from the BAU condition, the MTRE3 project will demonstrate during its 5 years duration a reduction of 20% SEC or down to 228 kW/m^2/year in commercial buildings, whose collective floor areas total 50,000 m^2(as Direct Emission Reduction). Considering successful implementation of the demonstrations (which can be replicated) during the MTRE3 project period, and the effective enabling conditions that will be facilitated by the project, it is expected that more commercial building EE improvement projects will be implemented in other commercial buildings during the projects influence period. This assumptions are in-line with the MEMR's target in RAN-GRK to build energy conservation partnership with 300 building objects during the period 2015-2020. Based on the average size of commercial building in the cities, the MTRE3 project will target buildings with minimum area of 500 m^2. After the completion of the project, a potential 20% SEC reduction in at least 100 buildings (totaling 50,000 m^2 floor area) in other cities in Indonesia can be realized. This is mainly from EE projects whose design will be assisted by the project but will be implemented by their developers/owners after the completion of the GEF assistance. These are considered as Direct Post Project Emission Reduction. Including the demonstrations, the project will directly impact 20% SEC reduction in commercial buildings whose collective floor areas total 100,000 m^2 in 10 years.

Following the GEF's prescribed methodology for estimating GHGs emission reduction from energy efficiency projects, the **Direct emission reduction (direct project and direct post project) attributable to the project** is calculated using the:

$$\mathbf{CO_2 \text{ direct (DER)} = E * c}$$

CO_2 direct =	Direct GHG emission savings of successful project implementation, tons CO_2 eq.
E =	Cumulative energy saved or substituted, e.g., in kWh, across all technologies that are affected by the intervention, and cumulated over the lifetime of the respective investments
c =	CO_2 emission factor of the national power generation portfolio, tCO_2/MWh.

The Direct **CO_2 direct (project and post project) (DPPER) impacted by the MTRE3 energy efficiency activities**

= (Direct Project Energy Saving + Direct Post Project Energy Saving) * c*10 years

= {((285–228kWh/m^2/y)*50,000m^2/1000 + (285–228kWh/m2/y)*50,000m^2/1000)}*0.8 tCO_2/MWh *10 years
= (2,280 tCO_2 eq + 2,280 tCO_2 eq)*10
= **45,600 tCO_2 eq.**

Consequential CO_2 Emission Reduction from EE activities in commercial building sector

Following the expected success of energy saving in 100,000 m^2 of commercial building and the establishment of Sustainable Energy Fund by the MTRE3 project to support EE project developers/investors, further replication of the energy efficiency efforts in the country is expected to take place and result in indirect CO_2 emission attributable to MTRE3 project. In accordance to the GEF method, the Consequential CO_2 emission reduction can be calculated using the Bottom-Up (BU) and Top-down (TD) approaches as follows:

Consequential CO_2 Emission Reduction (CER_{BUA}) = CO_2 direct* RF [Bottom-up Approach]

CO_2 indirect BU =	Emissions saved with investments after the project, as estimated using the bottom- up approach, tons CO_2 eq
RF =	Replication factor, i.e., how often will the project's investments be repeated during the 10 years after project implementation; assume a RF of 2.
CO_2 direct =	Estimate for direct and direct post-project emission reductions, tons CO_2 eq

Consequential CO_2 Emission Reduction (CER_{TDA}) = P10 * CF [Top-Down Approach]

CO2 indirect TD =	GHG emission savings in CO_2 eq as assessed by the top-down methodology;
P10 =	Technical and economic potential for GHG savings with the respective application within 10 years after the project; and
CF =	Causality factor.

Assumptions:

- For the bottom-up approach, since the MTRE3 project will make possible supporting EE project developers/investors and ESCOs through the establishment and operationalization of the SEF, the probable replication of the project investment during 10 years after project implementation is assumed twice, thus the replication factor (RF) is 2.
- For the top-down approach, taking into account the MEMR's planned target of 0.49 million tons CO_2 emission reduction from energy conservation in the buildings sector until 2020 as stated in RAN-GRK, the P10 is 0.49 million tons CO_2. This target is within 10 years of project's completion. In this regard, the expected impact of the MTRE3 Project would be "modest and substantial". The corresponding GEF causality factor is 0.40.
- The average grid emission factor for Indonesia in 2013-2014 according to State-owned Electricity Company (PLN) data still applies (0.8 tCO_2/MWh).

Therefore, the consequential bottom-up and top-down CO_2 emission reduction that can be attributed to the MTRE3 project are as follows:

Consequential CO_2 emission reduction from the MTRE3 Project's energy efficiency activities:
= 45,600 tons CO_2 * 2 = 91,200 tCO_2 (BUA)
= 490,000 tons CO_2 * 40% = 196,000 tCO_2 (TDA)

Based on the above, the resulting conservative estimate of the unit abatement cost (UAC) is about 17.6 GEF US$/ton CO2 considering only the direct and direct post project CO2 emission reductions. However, considering the anticipated consequential CO2 emission reductions, the UAC can range between 3.3 to 5.9 GEF US$/ton CO2.

Renewable Energy (RE) Electricity Generation

Direct Project and Direct Post Project Emission Reduction

During the project duration, the MTRE3 project is expected to demonstrate operationalization of 15 MW RE-based electricity generation consist of 7 MW mini-hydro (capacity factor 50%, useful lifetime 20 years), 6 MW biomass (capacity factor 85%, useful lifetime 20 years) and 2 MW on-grid solar PV (capacity factor 22%, useful lifetime 20 years). *Cumulative generated electricity from 15 MW renewable during 5 years project duration is estimated to be 79,190 MWh (See Table B1 below)*. The average grid emission factor for Indonesia in 2013-2014 according to State-owned Electricity Company (PLN) data is 0.8 tCO_2/MWh.

Table B1: Electricity Generation from 15 MW of RE-based Power Generation Capacity during project duration (2016-2020) and 85 MW in 5 years after MTRE3 Project Completion (2021-2025)

	Annual Power Generation (MWh) and Installed Capacity (MW)									
	2016	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**	**2025**
Mini Hydro	0	4,380/ (1MW)	8,760/ (2MW)	8,760/ (2MW)	8,760/ (2MW)	35,040/ (8 MW)	61,320/ (14MW)	26,280/ (6MW)	28,032/ (6.4MW)	20,148/ (4.6MW)
Solar PV	0	964/ (0.5MW)	964/ (0.5MW)	964/ (0.5MW)	964/ (0.5MW)	1,927/ (1MW)	1,927/ (1MW)	1,927/ (1MW)	3,854/ (2MW)	1,927/ (1MW)
Biomass	0	7,446/ (1MW)	7,446/ (1MW)	14,892/ (2MW)	14,892/ (2MW)	1,117/ (3MW)	2,606/ (7MW)	3,723/ (10MW)	3,723/ (10MW)	3,723/ (10MW)
Annual electricity generation (MWh)	0	12,790	17,170	24,616	24,616	38,084	65,853	31,930	35,609	25,798
Cumulative RE-electricity generated (MWh)	0	12,790	29,959	54,575	**79,190**	117,275	183,128	215,058	250,667	**276,466**

Furthermore, in addition to the 15 MW demonstration RE-projects, the MTRE3 project will establish Sustainable Energy Fund, which will provide loan guarantee scheme for RE project investment and assumed will trigger RE investment within 10 years since the start of project. Based on the potential RE investments listed in commercial banks pipeline database, MEMR's strategic plan 2015-2020 and USAID-ICED project assessment result, an additional 85 MW RE-based power generation investments can be realized by 2025 (See Table B2). The 85 MW RE-based power generation investments consist of 39 MW mini-hydro, 40 MW biomass and 6 MW on-grid solar PV. *The expected cumulative generated electricity from the total installation of 100 MW of RE during project duration as well as 5 years from end of the project is 276,466 MWh (See Table B1)*.

The identified 85 MW potential RE-based power generation investment that are expected to be influenced by the results of the MTRE3 project (and manifested in terms of Direct Post Project GHG Emission Reductions) are summarized in Table B2 below.

Table B2: Potential RE-based Power Generation Projects Influenced by the MTRE3 Project

Project Name	Project Type	Capacity (MW)	Project Status	Province	Source
KERPAP	Minihydro	2	Proposal	Aceh	ICED-USAID RE project database
SIMONGGO	Minihydro	6	Proposal	North Sumatera	ICED-USAID RE project database
AEK PULI 1	Minihydro	6	Proposal	North Sumatera	ICED-USAID RE project database
SILINDA	Minihydro	6	Proposal	North Sumatera	ICED-USAID RE project database
KETAHUN TENGAH 1	Minihydro	6.4	Proposal	Bengkulu	ICED-USAID RE project database
KATAHUN ILIR 1	Minihydro	8	Proposal	Bengkulu	ICED-USAID RE project database
TONGAR	Minihydro	4.6	Proposal	West Sumatera	ICED-USAID RE project database
Total Minihydro		**39**			
SAM SAM PALM OIL MILLS	Biomass	3	Proposal	Riau	ICED-USAID RE project database
KETAHUN	Biomass	7	Proposal	Bengkulu	ICED-USAID RE project database
PONTIANAK	Biomass	30	Proposal	West Kalimantan	Bank BCA pipeline
Total Biomass		**40**			
Rote	Solar PV	1	Planned	NTT	MEMR / DG SK 979 K/29/DJE/2014
Sumba Timur	Solar PV	1	Planned	NTT	MEMR / DG SK 979 K/29/DJE/2017
Larantuka	Solar PV	1	Planned	NTT	MEMR / DG SK 979 K/29/DJE/2018
Maumere-Rope-Ende	Solar PV	2	Planned	NTT	MEMR / DG SK 979 K/29/DJE/2019
Bajawa-Ruteng-Labuan Bajo	Solar PV	1	Planned	NTT	MEMR / DG SK 979 K/29/DJE/2020
Total Solar PV		**6**			
Total RE potential replication projects		**85**			

Direct Project and Direct Post Project CO_2 emission reduction can be calculated by using following equation:

CO_2 direct (DER) = E * c = e * l * c; with

CO_2 direct = Direct GHG emission savings of successful project implementation in tons of CO_2 eq
E = Cumulative energy produced by renewable energy, e.g., in MWh; $E = \Sigma\ l*e$
c = CO_2 emission factor of the national power generation portfolio, e.g., in t CO_2/MWh
e = Annual energy replaced, e.g., in MWh
l = Average useful lifetime of equipment in years, assumed 20 years for hydro, biomass and solar PV power generation.

Direct project and direct post project CO_2 emission reduction (DPPER) impacted by MTRE3 project RE activities:
= CO_2 direct project from 15 MW RE + CO_2 direct post project from 85 MW RE
= (1,583,808 MWh*0.8 tCO_2/MWh) + (4,802,232 MWh*0.8 tCO_2/MWh)
= 1,267,046 tCO_2eq + 3,841,786 tCO_2 eq
= 5,108,832 tCO_2 eq

Consequential CO_2 Emission Reduction from project's RE activities

Following the expected success of installing and operating a total cumulative capacity of 100 MW RE-based power generation plants and the operationalization of the SEF by the MTRE3 project in 10 years from project start, further replication of the RE-based power generation investments in the country is expected to take place and will result in consequential CO_2 emission reductions that can be attributed to the MTRE3 project. In accordance with the GEF's prescribed methodology, the consequential CO_2 emission reduction can be calculated using the Bottom-Up (BU) and Top-down (TD) approaches as follows:

Consequential CO_2 Emission Reduction (CER_{BUA}) = CO_2 direct* RF [Bottom-up Approach]

CO_2 indirect BU = Emissions saved with investments after the project, as estimated using the bottom- up approach, tons CO_2 eq
RF = Replication factor, i.e., how often will the project's investments be repeated during the 10 years after project implementation; assume a RF of 3.
CO_2 direct = Estimate for direct and direct post-project emission reductions, tons CO_2 eq

Consequential CO_2 Emission Reduction (CER_{TDA}) = P10 * CF [Top-Down Approach]

CO2 indirect TD =	GHG emission savings in CO_2 eq as assessed by the top-down methodology;
P10 =	Technical and economic potential for GHG savings with the respective application within 10 years after the project; and
CF =	Causality factor.

Assumptions:

- For the BU approach, since the project will facilitate loan guarantees through the SEF, the probable replication of the RE investment after project implementation is assumed three times, thus the replication factor (RF) is 3.
- For the TD approach, taking into account the MEMR's planned target of 3.11 million tons CO_2 emission reduction from RE-based power generation until 2020 as stated in RAN-GRK, the P10 is 3.11 million tons CO_2 and the MTRE3 Project impact is conservatively assumed to "modest and substantial" contributions, with the corresponding causality factor = 0.40.
- The average grid emission factor for Indonesia in 2013-2014 according to State-owned Electricity Company (PLN) data at 0.8 tCO_2/MWh is still valid.

Therefore, the consequential bottom-up and top-down CO_2 emission reduction from future RE-based power generation investments that can be attributed to the MTRE3 project are as follows:

Consequential CO_2 emission reduction (bottom-up and top-down) from the MTRE3 Project's RE activities:
= 5,108,832 tCO_2 eq * 3 = 15,326,496 tCO_2 (BU Approach)
= 3,110,000 t CO_2 * 40% = 1,244,000 tCO_2 (TD Approach)

Summary of CO_2 Emission Reductions

CO_2 Emission Reduction Type	**Quantity, tCO_2**
Lifetime Direct CO_2 Emission Reduction (DER_{TOTAL})	1,289,846
Lifetime Direct Post Project CO_2 Emission Reduction ($DPPER_{TOTAL}$)	3,864,586
Lifetime Consequential CO_2 Emission Reduction - BU Approach (CER_{BUA})	15,417,696
Lifetime Consequential CO_2 Emission Reduction - TD Approach (CER_{TDA})	1,440,000

Range of Consequential CO_2 Emission Reduction: CER_{BUA}& CER_{TDA}: 1.4 – 15.4 million tons

Annex C: Project Annual Targets

Outcomes	Indicators	Baseline	Annual Targets				
			Year 1	Year 2	Year 3	Year 4	Year 5
Objective: To support the design and implementation of appropriate climate change mitigation actions in the energy generation and energy end use sectors	• Cumulative CO_2 emissions reduction, tons CO_2 eq • Cumulative energy produced from RE systems facilitated by the project, MWh • Cumulative energy saved from EE in commercial buildings facilitated by the project, MWh • Cumulative volume of public and private investment mobilized for SEF, US$ million • Cumulative number of additional households (from baseline) having access to electricity in pilot provinces	• 0 • 0 • 0 • 0 • 0	• 2,700 • 7920 • 855 • 2.5 • 8,000	• 5,400 • 15,840 • 1710 • 5.0 • 16,000	• 10,810 • 31,680 • 3,420 • 10.0 • 32,000	• 18,910 • 55,430 • 5,990 • 17.5 • 56,000	• 27,019 • 79,190 • 8,550 • 25.0 • 80,000
Outcome 1: Prioritized appropriate mitigation actions in the RE-based energy generation and energy efficiency.	Number of provinces with updated sub-national GHG Inventory and GHG Marginal Abatement Cost Curve (MACC) for energy sector	0	0	1	2	3	4
Outcome 2: Enhanced and sustainable market diffusion of renewable energy and energy efficiency technologies.	Total number of provinces with operational "Integrated Market Service Center" (IMSC) to support sustainable RE & EE investments.	0	0	1	2	3	4
	No. of small-to-medium scale RE/EE projects that were financially supported by the Sustainable Energy Fund Cumulative amount of funds from the SEF used in financially supporting small-to-medium scale RE/EE projects , US$ million	0 0	1 2.5	2 5.0	4 10.0	7 17.5	10 25.0
	Cumulative number of NAMAs proposals developed for RE and EE projects in pilot provinces, based on the identified and prioritized RE/EE projects.	1	0	1 (EE)	2 (EE)	3 (EE & RE)	4 (EE & RE)
	• Cumulative capacity of RE investment projects implemented, MW • Cumulative floor area of buildings that were made energy efficient, m2.	0 0	1.5 5,000	3.0 10,000	6.0 20,000	11.0 35,000	15.0 50,000
Outcome 3: Accurate measurement and	• No. of registered mitigation actions in energy sector that are endorsed by the MEMR and MoEF.	0	2	3	6	10	14

Outcomes	Indicators	Baseline	Annual Targets				
			Year 1	Year 2	Year 3	Year 4	Year 5
accounting of actual GHG emission reductions from mitigation actions in the RE-based energy generation and energy efficiency.	• Total number of MRV reports submitted to MoEF following nationally agreed standard method and guideline.	0	0	1	2	3	4

Annex D: Sustainable Energy Fund

Among the interventions that will be carried out under the UNDP-GEF **Market Transformation through Design and Implementation of Appropriate Mitigation Actions in the Energy Sector (MTRE3) Project** is the development, establishment and operationalization of an enhanced financing scheme for supporting the implementation of renewable energy (RE) and energy efficiency (EE) projects in Indonesia. The envisioned Sustainable Energy Fund (SEF) for Indonesia is a financing mechanism that is meant to support the funding of small-to-medium size RE-based energy generation projects, particularly in off-grid areas; as well as EE technology application projects in buildings. The financing support is either through the: (1) Guarantees (through a loan guarantee fund); (2) Project development financing, for early stage costs (through a project development fund); and, (3) Viability gap financing (through a viability gap fund).

Rationale:

The Sustainable Energy Fund (SEF) is mean to address major gaps in existing financing services for renewable energy and energy efficiency projects in Indonesia. These are:

(a) Small-to-medium size RE-based power generation projects (< 10 MW, off-grid or on-grid) are generally not economically viable for commercial banks to financially support. For these type of projects, private companies (e.g., banks/financial institutions) are only interested to invest in if there is some sort of financial support or subsidies that will be provided. Currently, there are some sort of a project viability gap funding that is provided by the GOI (Ministry of Public Works) for infrastructure projects such as for building roads and bridges. There is also such kind of financial assistance provided to banks issuing agricultural credit for agricultural projects in the form of interest subsidy. Such forms of financial assistance are not available in the energy sector. Current practice shows that the government will utilize 100% public funding for development of non-economically viable energy infrastructure projects if such projects are for rural development or if under the country's electrification program. The GOI (even with the current donor assistance) has limited financing capacity to solely achieve the 2025 target of 23% RE share in the national primary energy mix, a target which requires cumulative investment of about USD 120 billion to achieve. The GOI cannot also, on its own, achieve the country's 99% electrification goal by 2019. Hence, for achieving these targets funding will have to come from both the private and public sectors, including both domestic and international sources. However, the private sector entities that can assist the GOI in achieving the 2019 and 2025 energy targets by implementing RE-based power generation projects would also need assistance from the government for financial support as part of the creation of the conducive enabling environment that will encourage them to invest in the development and implementation such projects In the provision of financing assistance to potential RE-based power generation project developers in the form of a viability gap fund (VGF), the proposed SEF will co-invest with the commercial financial sector. In this way, the effectiveness of public funds in financing RE/EE projects will be enhanced.

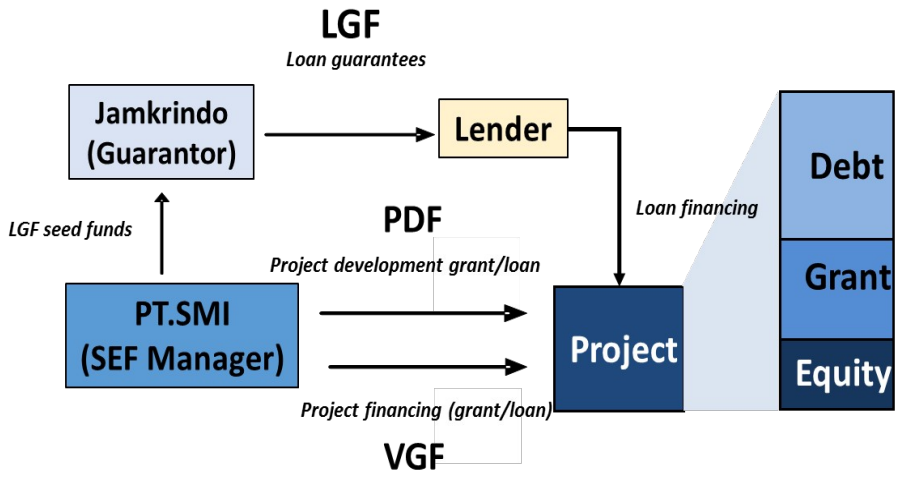
(b) Financial institutions are not supporting RE/EE projects. RE/EE projects are seen high risk by financiers. A review by the Bank of Indonesia on green lending by banks found that the share "green financing" was negligible, with only 1.4% of total lending, or IDR 10.2 trillion (USD 1 billion), belonging to this category. Currently, Jamkrindo, a state-owned credit insurance company, is providing loan guarantee for infrastructure projects but not yet for RE/EE projects. The USAID and ADB have ongoing loan guarantee schemes for RE/EE projects in Indonesia. However, the high final interest rate, and the applicability of loan guarantee for certain type of projects have made the schemes not attractive. The proposed SEF will be in collaboration with Jamkrindo in the establishment and operationalization of a

loan guarantee scheme for lenders (banks/financial institutions) that will be providing loans for financing RE/EE projects. The project activity on the establishment of the SEF shall include also capacity development for the LGF manager/administrator which is Jamkrindo.

(c) There are limited funds for proper and comprehensive project design and development. The lack of good quality feasibility studies and/or detailed design engineering for RE/EE projects often results in these projects being non-bankable. The SEF will provide project development funding for high impact RE/EE projects to enhance project economic feasibility.

Financing Schemes and Implementation Arrangement

The proposed implementation arrangement of the SEF will involve PT. Sarana Multi Infrastruktur (PT.SMI), a state-owned financing institution, as the fund manager. PT.SMI has a specific mandate from the GOI to promote RE/EE initiatives in Indonesia. Where necessary, it will take affirmative action whenever commercial banks are unable to support the RE/EE projects financing. Based on discussions with the project proponents and stakeholders, the general implementation management structure is as shown below.



Sources of Funds:
- **Domestic Public Funds** *(including counterpart cash funds from PT.SMI and Jamkrindo)*
- **Domestic Private Sector** *(e.g., mobilized CSR funds)*
- **International Public Fund**
  - **GEF:** *cash transfer from UNDP to PT. SMI of GEF budget for SEF seed funding*

LGF: Loan Guarantee Fund
PDF: Project Development Fund
VGF: Viability Gap Fund

Fig. 1: Sustainable Energy Fund Framework

The detailed design of the SEF framework, and the terms and conditions, mechanics and implementation arrangements of each SEF financing scheme will be carried out in Activity 2.3.3 of the proposed MTRE3 Project. The general concept of the SEF financing schemes (based on Fig. 1) is as follows:

Financial resources that will be mobilized or secured for the SEF will be utilized in 3 ways: (1) loan guarantees; (2) project development grants; and, (3) viability gap funding (grant/loan). For the provision of loan guarantees, a **loan guarantee fund (LGF)** will be set-up with Jamkrindo, which as its main mandate, will serve as the loan guarantor. It will not necessarily be a new fund, but can build on any suitable existing loan guarantee schemes of Jamkrindo. Seed funds will be provided by the SEF and counterpart funding will be provided

by Jamkrindo. The beneficiaries of the LGF are banks/financial institutions (FIs) that will be willing to provide loan financing to RE/EE projects. The LGF will be a de-risking instrument for these banks/FIs when there are defaults in the loan repayments of the RE/EE project owners. It will help overcome financing barriers in remote off-grid locations where projects lack a track record. The loan guarantees are expected to spur lending to RE/EE project proponents by spreading risks among the guarantor (Jamkrindo), the lender and the borrower (project proponent or owner). Jamkrindo will establish strong linkages with participating banks/FIs towards the promotion of the LGF. The underlying premise is that ultimately the loan guarantee will not be needed because the lender will be convinced that the risks and transaction costs are reasonable and manageable.

For the provision of financial assistance in project development, a **project development fund (PDF)** will be set up by PT. SMI that is intended to support the cost for RE/EE project preparation and packaging (including arrangement of main project financing). The financial assistance can be a grant or a contingent grant (converting to a loan). The detailed design of the scheme will evaluate the right mix of grant provision and the provision of this financing assistance as a conditional grant of up to a certain percentage of the project preparation cost. The loan repayment mechanism can be such that a percentage of the loan (e.g., 25%) would be repaid to PT. SMI if the project preparatory activities came up with a non-viable project result, or the project preparatory activities resulted in a viable project but the project could not be implemented within an agreed period because of other reasons beyond the assistance that can be provided under SEF. If the project preparatory activities resulted in a viable project, a higher percentage (e.g., 75%) of the loan would have to be repaid. The PDF will be held and administered by PT. SMI. This financing scheme shall allow preparation (including arrangement of financing and PPAs) for eligible RE/EE projects. Eligible RE/EE projects will be selected on the basis of their use of financing and their proven commitment. For RE projects, the financing can be utilized for project preparatory activities like wind data analysis, wind turbine generator micro-siting, hydrologic investigation and analysis, detailed topographic surveys, hydraulic study, etc.

For the provision of financial assistance to enhance the viability of RE/EE projects, a **viability gap fund (VGF)** will be developed either as a stand-alone fund or as an extension of any suitable existing financing windows in PT. SMI. The financial assistance can be a grant or a concessional loan. The detailed design of the scheme will evaluate the right mix of grant provision and concessional loan financing. Depending on a set eligibility criteria that will defined as part of Activity 2.3.3 of the MTRE3 project, part of the VGF can be grant assistance to fill the financing gap of specific RE/EE project (e.g., public sector projects). The concessional loan financing scheme is intended for financing small-to-medium scale RE-based energy generation projects such as those that will support rural development with relaxed terms for long-term borrowing.

The process for project selection will be through open call for investment proposal with VGF support. The projects can be those (a) previously determined pipelines that have passed the evaluation by the SEF Technical Team (includes PT.SMI staff, technical experts, Government and UNDP); (b) new project proposals. Then PT.SMI and the SEF Technical Team will evaluate project proposals, based on a set of eligibility criteria that will be developed during the design of the VGF. As part of the MTRE3 project activities, the PT. SMI personnel (i.e., SEF technical team) will be trained in the evaluation of RE/EE project proposals based on the set of eligibility criteria. The provision of assistance to project developers and other local organizations that wish to avail of the financing will also be part of the MTRE3 project activities. Part of the seed money contribution from the GEF will be used for RE/EE project viability gap financing.

The UNDP-GEF MTRE3 project will also provide (as per Activity 2.3.3): (i) complementary technical assistance to the SEF investments (e.g., investment appraisal for the domestic

financial sector), (ii) technical support in the design and establishment of the fund at PT. SMI; (iii) channel GEF funds as seed capital; and, (iv) facilitate additional grant mobilization for SEF from other donors and private sector Corporate Social Responsibility fund resources.

The initiative to establish the Sustainable Energy Fund (SEF) was announced by the Minister of MEMR in December 2015. In line with this, the MTRE3 project will support the SEF establishment (its objective, governance, capitalization, fiduciary management, implementation arrangements and MRV), with the aim of making the SEF a stand-alone financially sustainable fund. Although the SEF design will be finalized during implementation, it is envisioned to promote and engage the domestic financial sector in financing investments in small-medium renewable energy (below 10MW, rural electrification purposes) and energy efficiency activities, particularly ESCOs. The long term objective is to transition the domestic financial sector to providing abundant access to low-cost capital for RE/EE investments.

PT Sarana Multi Infrastruktur (PT SMI), a state-owned financing company, will administer the SEF. PT SMI has a specific mandate to promote RE/EE financing and to support the development of domestic financial sector involvement in RE/EE activities. UNDP has previously worked with PT.SMI for the GEF-funded wind hybrid power generation project, WhyPGen. UNDP is not directly involved in the operation of the SEF. Under the MTRE3 project, UNDP will channel the seed funding from the GEF for the SEF to PT. SMI. It will however be overseeing the activity on the operation of the SEF (as a demonstration of the design, funding and resource mobilization, and operationalization of an effective improved financing mechanism for investments on climate change mitigation projects).

The SEF will receive seed funding of USD 2,680,000 from the GEF through UNDP. Thereafter a variety of domestic sources of funding are envisaged, including provincial budgets and fossil fuel extraction royalties. International/bilateral sources of funding will also be pursued. The SEF's steering committee will be established. GoI membership will include MEMR, MoF, BAPPENAS (Ministry of National Planning). UNDP will also be a steering committee member, for a minimum of the duration of the project.

The SEF will seek to develop a robust pipeline of investment opportunities. Sourcing for the SEF will be linked to RE/EE projects arising from the Integrated Market Service Centres (IMSCs) in pilot provinces. Partnerships will additionally be developed with various entities/initiatives, including but not limited to, the Millennium Challenge Account–Indonesia's Green Prosperity Programme, GIZ, Danida, and USAID-ICED.

Conclusion

It should be noted that as part of the MTRE3 Project, the SEF establishment will by and large demonstrate the design, funding and resource mobilization, and operationalization of an effective improved financing mechanism for investments on climate change mitigation projects, as well build national institutional capacity for the operationalization of such financing mechanisms. The sustainability of the SEF can be ensured by designing it not as a sinking fund that has to be regularly replenished. The right mix of grant provision and loan financing has to be made considering the extent of the financing barriers for small-to-medium scale RE/EE projects. Sustainability will also be guaranteed once it has transitioned into a regular government-supported financial support mechanism under the aegis of a reliable national institution that can further implement the SEF beyond the GEF project by mobilizing resources such as from state budget, petroleum levy or private sector CSR resources. Ultimately, when the local RE/EE market is already well developed, grant or loan guarantee may no longer be required. At that time, the successful SEF may consider supporting the development and application of new advances in RE/EE technologies.

Annex E: Terms of Reference of Key Project Personnel

Position: National Project Manager (NPM)

Level: SC-9, 5 years duration

Scope of Work:

MTRE3 Project is a 4-year project designed to support the design and implementation of appropriate climate change mitigation actions in the energy generation and energy end use sectors. The day-to-day operational management of the Project will be the responsibility of the PMU based within the designated Implementing Partner Ministry of Energy and Mineral Resources (MEMR) for the project. A Project Steering Committee (PSC) will be established to act as a Project Board for overall guidance and approval of key project activities including fund commitments and co-financing arrangements. It shall be the role of the PMU to undertake the following activities under the PMU Manager's direct responsibility:

- Coordination with the PMUs of other related on-going projects for the purpose of facilitating the implementation of top management's decision on delineation of project thrusts and direction
- Preparation of work plans, budget, and TORs of consultants, trainers, and subcontractors;
- Assist MEMR in the selection and hiring of other national professionals, administrative staff and subcontractors needed in the Project;
- Monitoring and evaluation of progress of project activities;
- Arranging of regular review meetings and ensuring effective coordination of project activities;
- Preparation and dissemination of project reports and other information materials. The primary goal of this activity will be to enhance the transparency of project implementation.
- Oversee the financial record-keeping and internal control management of the PMU and the Project as a whole;
- Submission of timely and accurate financial reports and progress reports to UNDP
- Provide technical and administrative support for the initial operation of the PSC.

The Project Manager will take responsibility for the management and administration of the MTRE3 Project. He/she will report directly to the MEMR and UNDP in the overall operation and management of the Project with the following functions:

- Be ultimately responsible for the fiscal management of the project
- Manage the PMU staff and be responsible for ensuring high staff performance and motivation
- Directly and indirectly manage and coordinate multi-faceted local and international consultant contracts to ensure their timely completion and high quality
- Manage the monitoring and evaluation of both the overall project and its components
- Be responsible for ensuring that project objectives are being met in terms of budget and project outputs and must be prepared to recommend changes to the PSC to ensure that all outputs are realized over the project life
- Be the primary project advocate and representative and meet regularly with stakeholders from the public, the energy industry, relevant agencies of Government of Indonesia and international donors to further project objectives.

Specific responsibilities will include:

- Regular reporting to MEMR or its designated agency and PSC on the status of the project activities;
- Synthesis of the works and outputs of consultants and subcontractors to identify potential issues and problems
- Chairing of the quarterly Project Monitoring Committee meetings;
- Preparation of quarterly financial and project progress reports;
- Preparation of annual work plans (AWP), annual project reports (APR), Project Implementation Review (PIR) and others as required by the UNDP.

Qualifications and Experience:

- Master's Degree or higher in a related field;
- At least 10 years of proven track record of project management experience preferably in the area of energy;
- Good writing, training, presentation and reporting skills;
- Proven skills and experience in GHG data collection, data set development and data analysis is an added advantage;
- Substantial involvement in the preparation of GHG emission scenarios; knowledge and experience in modeling tools and applied methodologies of GHG emissions in renewable energy is an added advantage;
- Experience in managing teams with a mix of international and local expertise; and demonstrated strong leadership, effective management skills, good coordination ability, and team working spirit;
- Good interpersonal/communication skills;
- Good oral and written communication skills in English.
- Proven track record of experience in successful contract management;
- An understanding of and links with the Indonesian RE and EE investment is preferable.

Position: Chief Technical Advisor

Level: SC-11 for 4 years duration

Scope of Work:

To provide top-quality advisory services to the Government of Indonesia, particularly the Ministry of Energy and Mineral Resources. The CTA will:

- Provide strategic advisory to support GoI in meeting the targets of RPJMN 2015-2019 to achieve 16% to 20% of RE contribution in primary energy mix and 96.61% electrification ratio.
- Identify sources of information related to policy-driven issues. Identification and synthesis of best practices and lessons learned to be directly fed back to relevant national agencies.
- Provide high quality information and analysis to the government counterparts to inform the strategic direction of national and local development priorities.
- Ensure that required expertise and technical assistance is made available.
- Coordinate the development of policies and institutions involved in renewable energy sector with national agencies and other strategic partners.
- Coordinate and organize relevant studies to collate research-based information, evidences and facts to enrich the achievement of development results.
- Contribute to advocacy networks at national level, knowledge networks and communities of practice.
- Contribute to high-impact advocacy campaigns implemented with key partners.

To provide technical oversight and assistance to the Project Management Unit (PMU) through the Project Manager. The CTA will:

- Advise the PMU on the implementation of the project so as to ensure that it satisfies its objectives and targets and meets the requirement of the MEMR as Implementing Partner and UNDP and GEF;
- Provide advice and technical inputs in the review of outputs by consultants hired under the component activities;
- Actively participate in Technical Working Groups (TWGs), various teams and other ad-hoc task forces for the conduct of specific activities and tasks under the Project;
- Provide technical inputs in the preparation of annual work plan (AWP), annual project reports (APR), project implementation review (PIR), and others as required by the UNDP and Project Manager;
- Provide technical inputs to the conduct of the project Mid-Term Review and Terminal Evaluation and drafting and implementation of the Action Plan to address the Terminal Evaluation Recommendations;
- Act as PMU deputy in various meetings, workshops and other activities, as may be instructed by Project Manager;

- Provide suggestions for resolving problems and barriers as they emerge in the process of implementation, especially in external communications, and in meeting various aspects of UNDP working processes and practice;
- Provide other support services as may be required by Project Manager.
- Conduct synthesis of the works, reports and other outputs of consultants and subcontractors involved in the Project;
- Assist PMU to organize and arrange international workshops, conferences and surveys.

Qualifications and Proven Experience:

- Having a Master degree in engineering, economics or other scientific/technical field with relevance to energy sector/climate change mitigation;
- More than 15 years working experience in energy/climate change mitigation area;
- Having extensive experience in energy/climate change mitigation program development and implementation;
- At least 7 years of international professional experience in international negotiations, GHG inventory, CDM and new market mechanisms is an added advantage;
- Demonstrated experience in developing and analyzing MACC curves;
- Demonstrated experience in economic analysis of GHG abatement options;
- Experience in GHG inventory preparation and with more than 3 years in management of projects related to mitigating or adapting to climate change, and / or project formulation on NAMAS, LEDS, CDM, and / or sequestration projects and carbon emissions;
- Experience in developing methodologies for GHG accounting, monitoring protocols, and reporting process including (i) GHG emissions parameters, and their measurement, frequency, data sources and monitoring criteria, and (ii) Methodology to determine uncertainties, is an added advantage;
- Skills demonstrated in capacity building and mentoring in developing countries, notably at the systemic, institutional and technical levels;
- Strong knowledge of government policies and strategies;
- Understanding well the project participating countries' economic, energy and environment situation;
- Strong existing relationships with energy/climate change mitigation institutions as well as a ready-network of international experts is an added advantage;
- Strong coordinative capability with international and domestic experts;
- Excellent spoken and written English capability;
- Familiarity with projects supported by UNDP/GEF would be an advantage;

Position: TWG Coordinator

(3 national posts)

Level: SC-8 for 5 years duration

Scope of Work:

To provide technical expertise on **NAMAs development/Investment Promotion/ Renewable Energy and Energy Efficiency MRVs** to Working Group stakeholders. The TWG Coordinator will:

- Provide technical assistance and management of related stakeholders to achieve the expected outputs under each Component Activity of MTRE3 project.
- Ensure that required expertise and technical assistance is made available for the TWG.
- Identify sources of information related to technical issues.
- Identification and synthesis of best practices and lessons learned to be directly fed back to the TWG.
- Coordinate the work of WG and expected outputs with other strategic partners to seek synergy.
- Conduct relevant studies to collate research-based information, evidences and facts to enrich the achievement of development results.
- Contribute to advocacy networks at national level, knowledge networks and communities of practice.
- Contribute to high-impact advocacy campaigns implemented with key partners.

To provide management service of Working Group and assistance to the National Project Manager (NPM). The TWG Coordinator will:

- Assist NPM on the implementation of the project to ensure that the works and achievements of Technical Working Group satisfies project's objective and targets and meets the requirement of the MEMR as Implementing Partner and UNDP and GEF;
- Provide technical inputs in the review of outputs by consultants hired under the component activities;
- Manage Technical Working Groups (TWGs), various teams and other ad-hoc task forces for the conduct of specific activities and tasks under the Project;
- Assist NPM in the preparation of annual work plan (AWP), annual project reports (APR), project implementation review (PIR), and others as required by the UNDP and Project Manager;
- Provide information on the TWG's work and achievements to the conduct of the project Mid-Term Review and Terminal Evaluation and follow up implementation of the Action Plan to address the Terminal Evaluation Recommendations;
- Act as PMU deputy in various meetings, workshops and other activities, as may be instructed by Project Manager;
- Take actions for resolving problems and barriers as they emerge in the process of implementation, especially in external communications, and in meeting various aspects of UNDP working processes and practice;
- Provide other support services as may be required by Project Manager.
- Conduct synthesis of the works, reports and other outputs of consultants and subcontractors involved in the Project;
- Assist PMU to organize and arrange international workshops, conferences and surveys.

Qualifications and Proven Experience:

- Having a Bachelor or Master degree in engineering, economics or other scientific/technical field with relevance to RE/EE;
- More than 5 years working experience in RE/EE issues;
- Proven technical knowledge on RE/EE planning/ monitoring/implementation.
- Having extensive experience in managing working groups, RE/EE program development and implementation;
- Demonstrated experience in capacity development initiatives in developing countries, notably at the systemic, institutional and technical levels;
- Understanding well the countries' economic, energy and environment situation;
- Strong existing relationships with RE/EE institutions at national and local level.
- Strong coordinative capability with national experts;
- Good spoken and written English;
- Familiarity with projects supported by UNDP/GEF would be an advantage.

Position: Provincial Liaison Officer

(4 national posts)

Level: SC-7 for 5 years duration

Scope of Work:

To provide stakeholder liaison expertise between national and provincial/district level governments and other relevant key stakeholders in Pilot Province. The Provincial Liaison Officer will:

- Provide liaison assistance and coordination management of stakeholders to the Project Management Unit (PMU) to achieve the expected outputs of MTRE3 project.
- Ensure good coordination with TWGs Coordinators in the implementation of planned activities at pilot provinces.
- Actively identify and collect best practices and lessons learned to be directly fed back to the stakeholders.
- Contribute to building networks at provincial level, knowledge networks and communities of practice.

- Assist Project Manager as representative of the PMU at pilot province. Actively participate in local various meetings, workshops and other activities, as may be instructed by Project Manager;
- Assist NPM on the implementation of the project at pilot provinces to ensure that the works satisfies project's objective and targets and meets the requirement of the MEMR as Implementing Partner and UNDP and GEF;
- Provide other support services as may be required by Project Manager;
- Provide regular reporting of progress and obstacles in project implementation to Project Manager;
- Assist PMU to organize and arrange workshops, conferences and surveys.

Qualifications and Proven Experience:

- Having a Bachelor or Master degree in social science, engineering, economics or other scientific/technical field with relevance to energy/climate change;
- More than 3 years working experience in energy/climate change issues;
- Demonstrated experience in network building, liaison functions;
- Acquire existing relationships with RE/EE institutions at national and local level.
- Good coordinative capability with local and national experts;
- Good spoken and written English;
- Familiarity with projects supported by UNDP/GEF would be an advantage.

Annex F: Co-financing Commitment Letters

1. Ministry of Energy and Mineral Resources

MINISTRY OF ENERGY AND MINERAL RESOURCES
DIRECTORATE GENERAL OF NEW RENEWABLE ENERGY AND ENERGY CONSERVATION
JALAN PEGANGSAAN TIMUR NO. 1A CIKINI JAKARTA 10320

Telepon : (021) Faksimile: (021) website : www.ebtke.esdm.go.id email: info@ebtke.esdm.go.id

Our ref : 663/05/DJE/2014 Jakarta, 14 October 2014
Subject : Confirmation of Co-financing and Support for the Market Transformation for Renewable Energy and Energy Efficiency (MTRE3) Project

To:
CEO of Global Environment Facility (GEF)
GEF Secretariat
1818 H Street, NW, Mail Stop P4 – 400
Washington, DC 20433 USA

Dear Dr.Naoko Ishii,

With this letter, we wish to confirm the formal intention and commitment of **Directorate General of New Renewable Energy and Energy Conservation, Ministry of Energy and Mineral Resources** of Indonesia to support the proposal to GEF for funding of the full-size project on Market Transformation for Renewable Energy and Energy Efficiency (MTRE3) as it pursues the objective of improving energy efficiency in commercial buildings and encouraging renewable energy investment (mini-hydro and biomass) in Indonesia.

In order to manifest our cooperation to further develop and subsequently implement the MTRE3 project, we will provide US$ 1,000,000 in-kind and US$ 7,000,000 in cash as our co-financing inputs and active participation through various activities and technical working groups meeting regarding MTRE3's project components which are very relevant to us.

We are looking forward to the commencement of this laudable project.

Yours sincerely,

Rida Mulyana
Director General of New Renewable Energy and Energy Conservation

Cc:
1. GEF Indonesia Focal Point
2. UNDP Indonesia Country Director

2. PT. Multi Fabrindo Gemilang

Jakarta, October 2nd 2014

Reference No : L-161/2014/MF-DIR
Subject : Confirmation of Co-financing and Support
for the Market Transformation for
Renewable Energy and Energy Efficiency (MTRE3) Project

To:
CEO of Global Environment Facility (GEF)
GEF Secretariat
1818 H Street, NW, Mail Stop P4 – 400
Washington, DC 20433 USA

Dear Dr.Naoko Ishii,

With this letter, we wish to confirm the formal intention and commitment of PT Multi Fabrindo Gemilang to support the Government of Indonesia's Project proposal to UNDP/GEF for funding of the full-size project on Market Transformation for Renewable Energy and Energy Efficiency (MTRE3) as it pursues the objective of improving energy efficiency in commercial buildings and encouraging renewable energy investment (mini-hydro and biomass) in Indonesia.

In order to manifest our cooperation to further develop and subsequently implement the MTRE3 project, we will invest US$ 2,000,000.00 in-cash as our co-financing inputs and active participation through various activities and technical working groups meeting regarding MTRE3's project components which are very relevant to us.

We are looking forward to the commencement and successful implementation of this laudable project.

Yours sincerely,

Ir. Boy Iswandi, MBA.
Director

cc.
1. Directorate General of Renewable Energy and Conservation Energy (EBTKE), MEMR
2. GEF Focal Point
3. UNDP Country Director

PT. MULTI FABRINDO GEMILANG Medco Building II, 3rd Fl, Jl. Ampera Raya No. 20, Cilandak - Jakarta 12560, Indonesia. Ph : 62.21.7800824, Fax: 62.21.7801560 E-mail:multifab@cbn.net.i
Member of MEDCO Group

3. PT. Pasadena Engineering Indonesia

PEI | PASADENA ENGINEERING INDONESIA

Jakarta, 09/30/2014

Reference No : PEI/RHW/14/09-275
Subject : Confirmation of Co-financing and Support for the Market Transformation for Renewable Energy and Energy Efficiency (MTRE3) Project

To:
CEO of Global Environment Facility (GEF)
GEF Secretariat
1818 H Street, NW, Mail Stop P4 – 400
Washington, DC 20433 USA

Dear Dr.Naoko Ishii,

With this letter, we wish to confirm the formal intention and commitment of PT. Pasadena Engineering Indonesia to support the Government of Indonesia's Project proposal to UNDP/GEF for funding of the full-size project on Market Transformation for Renewable Energy and Energy Efficiency (MTRE3) as it pursues the objective of improving energy efficiency in commercial buildings and encouraging renewable energy investment (biogas and biomass power plants) in Indonesia.

In order to manifest our cooperation to further develop and subsequently implement the MTRE3 project, and to extend our involvement in renewable energy projects in Indonesia, we intend to establish at least another 4 MW POME Biogas plants within 4 years time from 2015 to 2018 (a total investment of around USD 10,000,000) in addition to the other similar projects we have constructed previously.

We are looking forward to the commencement and successful implementation of this laudable project.

Yours sincerely,

R. Hikmawan Wargakusumah, MH.
President Director

cc.
1. Directorate General of Renewable Energy and Conservation Energy (EBTKE), MEMR
2. GEF Focal Point
3. UNDP Country Director

PT PASADENA ENGINEERING INDONESIA
Wisma Bumiputera 9th Floor, Jalan Jendral Sudirman Kav.75, Jakarta 12910 Ph.+62-21 570 1637 / 43 Facs.+62-21 526 0675 E-mail: pei@pasadenagroups.biz www.pasadena-engineering.com

4. PT. Daun Biru Engineering

PT Daun Biru Engineering
Maintenance Technology Specialist
Jl Raya Pekapuran No. 41, Sukamaju Baru – Tapos,
Kota Depok, Indonesia
Ph +62-21-87743652, +62-21-87743615
Fax +62-21-87743634
www.daunbiru.com

Jakarta, October 01 2014

Reference No : 063/DBE-GEF/X/2014
Subject : Confirmation of Co-financing and Support for the Market Transformation for Renewable Energy and Energy Efficiency (MTRE3) Project

To:
CEO of Global Environment Facility (GEF)
GEF Secretariat
1818 H Street, NW, Mail Stop P4 – 400
Washington, DC 20433 USA

Dear Dr.Naoko Ishii,

With this letter, we wish to confirm the formal intention and commitment of PT Daun Biru Engineering group to support the Government of Indonesia's Project proposal to UNDP/GEF for funding of the full-size project on Market Transformation for Renewable Energy and Energy Efficiency (MTRE3) as it pursues the objective of improving energy efficiency in commercial buildings and encouraging renewable energy investment (mini-hydro and biomass) in Indonesia.

In order to manifest our support to further develop and subsequently implement the MTRE3 project, we currently propose to develop some mini-hydro projects with total capacity 20 MW and this will be cost amount about US$ 40,000,000 (forty million US Dollars).

We are looking forward to the commencement and successful implementation of this laudable project.

Yours sincerely,

Rudy Andriyana, ST
Director

Page **1** of **2**

5. UNDP

United Nations Development Programme

Empowered lives.
Resilient nations.

20 June 2015

Dear Ms. Adriana,

Subject: **UNDP Indonesia Co-financing confirmation for the *Market Transformation through Design and Implementation of Appropriate Mitigation Actions in the Energy Sector* Project (PIMS # 4673)**

I am pleased to confirm the USD 100,000 (in-cash) as co-financing from UNDP Indonesia to support ***Market Transformation through Design and Implementation of Appropriate Mitigation Actions in the Energy Sector*** Project to meet its objective. The co-financing will support the implementation of the project activities and its management cost.

We thank you for your continuous cooperation.

Sincerely yours,

Budhi Sayoko
Assistant Country Director/
Head of Environment Unit

Ms. Adriana Dinu
Executive Coordinator
UNDP-Global Environment Facility
304 East, 45th Street, 9th Floor
New York, NY 10017, USA.

3. Jalan M. H. Thamrin (or : P.O. Box 2338), Jakarta 10250, Tel : (62-21) 3141308, Fax : (62-21) 3145251, E-mail : registry.id@undp.org

Annex G. Social and Environmental Screening Template

The completed template, which constitutes the Social and Environmental Screening Report, must be included as an annex to the Project Document. Please refer to the Social and Environmental Screening Procedure and Toolkit for guidance on how to answer the 6 questions.

Project Information

Project Information	
1. Project Title	**Market Transformation through Design and Implementation of Appropriate Mitigation Actions in the Energy Sector (MTRE3)**
2. Project Number	PIMS 4673
3. Location (Global/Region/Country)	INDONESIA

Part A. Integrating Overarching Principles to Strengthen Social and Environmental Sustainability

QUESTION 1: How Does the Project Integrate the Overarching Principles in order to Strengthen Social and Environmental Sustainability?
Briefly describe in the space below how the Project mainstreams the human-rights based approach
The project will also be designed and implemented with due consideration of human rights principles, through engagement of the concepts of equality in the dissemination of knowledge and sharing of benefits. MTRE3 will implement activities that will transform market in renewable energy-based power generation in four pilot provinces with low electrification ratio, thus the project will contribute in increasing access to clean electricity for people living in those areas. The project is also promoting energy efficiency activities in high-energy consuming sector, such as commercial buildings, thus more energy can be saved and distributed to more households/industries. By targeting the areas with low access to electricity, the project has been by design applying human-right approach. Furthermore, in the process of selection and implementation of renewable energy and energy efficiency projects that will be demonstrated in the pilot province, participatory approach and free, prior and informed consent (FPIC) process will be conducted by involving stakeholders that will be impacted by the projects. The project will also be designed and implemented with due consideration of human rights principles, through engagement of the concepts of equality in the dissemination of knowledge and sharing of benefits. The implementing partner as well as the project partners acknowledge human rights practices under international law and the application of human rights-related standards in the design and implementation of the project. The project is designed to enhance the availability, accessibility and quality of benefits and services for all relevant target groups including those that are potentially marginalized individuals and groups.
Briefly describe in the space below how the Project is likely to improve gender equality and women's empowerment
The project will demonstrate implementation of 15 MW of renewable energy-based power generation and energy efficiency interventions in commercial buildings. Gender problems related to energy project include limited participation of women in the planning and decision making process in the project activities and limited access of women in technical trainings provided by the project. Taking into account these gender problems, the MTRE3 project will involve women groups in targeted area during consultation process and every stage of demonstration activities implementation and ensuring at least 30% representation of women in technical trainings that will be conducted by the project. Further, the proposed project will ensure that women will gain opportunity to access RE/EE technical knowledge, operation/maintenance knowledge from capacity building activities that the project will conduct. Rural households will gain access to electricity from RE-based energy systems. Thereby, it is expected that the increase of access to modern energy will stimulate women groups to generate productive economy activities, which in long term will empower them economically. Women groups will be encouraged to build feasible home industries by utilizing the access to electricity. These benefits will reduce gender inequality and enhance women's empowerment.
Briefly describe in the space below how the Project mainstreams environmental sustainability

The climate change mitigation programs and administrative functions that will be supported under the MTRE3 Project are geared towards promoting and supporting renewable energy and energy efficiency implementation as key elements for the satisfactory achievement of the country's environmental and development agenda. These interventions will be designed in such a way that proper evaluation of the potential impacts to the natural environment done. While promoting RE-based energy generation and energy efficiency technology applications these energy conserving and energy cost saving actions/measures should not have negative impacts on the surrounding environment as well as to the people who are also present in such environment. In that regard, the MEMR and the provincial governments will be coordinating closely with the Department of Environment the siting, design, development and implementation of the RE and EE projects that will be carried out directly by the project, and coordinate also the replications that are expected to follow during the tail end of the MTRE3 project implementation and during the influence period. This may involve, for projects such as micro/mini-hydropower facilities, the conduct of environmental impact assessments. For EE projects, building construction codes/standards and applicable occupational safety requirements will be complied with. More detailed social and environmental impact assessment will be conducted for each RE-based power generation project that will be implemented and facilitated by the MTRE3 project following the Government of Indonesia's requirements for investment.
The magnitude of potential impacts to the natural environment shall be made one of the important criterion for the selection of the demo RE and EE projects in the pilot provinces, as well as the RE/EE projects that will be eligible for financial assistance from the SEF.

Part B. Identifying and Managing Social and Environmental Risks

QUESTION 2: What are the Potential Social and Environmental Risks? *Note: Describe briefly potential social and environmental risks identified in Attachment 1 – Risk Screening Checklist (based on any "Yes" responses). If no risks have been identified in Attachment 1 then note "No Risks Identified" and skip to Question 4 and Select "Low Risk". Questions 5 and 6 not required for Low Risk Projects.*	**QUESTION 3: What is the level of significance of the potential social and environmental risks?** *Note: Respond to Questions 4 and 5 below before proceeding to Question 6*			**QUESTION 6: What social and environmental assessment and management measures have been conducted and/or are required to address potential risks (for Risks with Moderate and High Significance)?**
Risk Description	***Impact and Probability (1-5)***	***Significance (Low, Moderate, High)***	***Comments***	***Description of assessment and management measures as reflected in the Project design. If ESIA or SESA is required note that the assessment should consider all potential impacts and risks.***
Likelihood that the proposed Project would have adverse impacts on gender equality and/or the situation of women and girls.	P= 1 I= 2	Low	The Project would bring about increased opportunities for women to gains access to technical knowledge of RE/EE and about operation/maintenance of RE systems. Reduced gender inequality and enhance women's empowerment in rural socio-economic development activities. All projects will be designed to ensure gender equity	The project M&E will include a number of human development indicators, and among them can be on gender equity, like the number of trained and employed women in new RE-based power generation facilities. By implementing professional job training in this area, for both men and women will improve and promote gender equality and improve the lot of qualified and capable Indonesian women in technical areas such as the energy generation sector.

RE-based power generation project construction, operation, or decommissioning pose potential safety risks to local communities.	P= 1 I= 2	Low	This could happen in RE-based generation project constructions (e.g., in remote areas) that do not comply with set regulations for occupational health and safety.	One of the main criterion for the selection of RE/EE projects that will be used for the demo activities, as well as those that will be eligible for SEF financing is the safety aspect (general and occupational) of the project. EIA obligations (where required by GOI) shall be emphasized in the selection of projects.
RE-based power generation project potentially result in the release of pollutants to the environment due to routine or non-routine circumstances with the potential for adverse local, regional, and/or trans-boundary impacts.	P= 1 I= 2	Low	This could happen in cases such as: (1) biomass-based power generation does not properly address waste management issues; (2) hydropower project does not take into account impacts on downstream uses of the water resource.	Most of the RE-based power generation projects are of standard design that all aspects from RE resource preparation, utilization, and the handling of resulting waste or effluents have their general design requirements and guides that have to be complied with. Unless there are specific characteristics of the RE resource, project site, etc., that will require explicit designs, the construction and operation of such facilities would not cause any release of pollutants (e.g., undigested waste matter, high BOD effluents, etc.). The EIA of such facilities (as per the GOI requirement) will also ensure that the design will not bring about release of materials that would detrimental to the natural environment accidentally or during disposal.
Social and climate-related risks impacts the sustainability of renewable energy projects implementation	P=3 I=4	*High*	• Climate change hinders full performance of RE technologies due to disturbance to supply of renewable energy resources and impacts of climate events like flood/drought/landslide. • Low level of social acceptance by local communities of renewable energy projects due to benefit-sharing issues.	• Climate factors and climate scenario will be taken into account in the feasibility studies that will be conducted in the potential RE/EE demonstration projects, as well as in the design and engineering of the selected RE/EE technology application demos. • The design of the demonstration projects will be as such that climate-related (direct and indirect) risks will be considered for proper mitigation, including getting insurance coverage, if necessary. • Free, Prior Informed Consent (FPIC) principle will be implemented for RE projects as part of social and environmental safeguard measure.

QUESTION 4: What is the overall Project risk categorization?

Select one (see SESP for guidance)		Comments
Low Risk	☐	
Moderate Risk	x	Strengthen communication with the relevant GOI agencies (MEMR, KLH), RE/EE technology hardware manufacturers and suppliers, community leaders, demo hosts (e.g., IPPs and commercial building owners). PMU and other pertinent entities, to ensure proper evaluation, engineering, construction and operation of demo RE/EE projects,
High Risk	☐	

	QUESTION 5: Based on the identified risks and risk categorization, what requirements of the SES are relevant?		
	Check all that apply		**Comments**
	Principle 1: Human Rights	☐	
	Principle 2: Gender Equality and Women's Empowerment	☒	Gender equality and women's active and gainful participation in rural development has been advocated by Indonesia. The project has been designed and will be implemented with the idea of raising the level of gender equality and enhancing women's participation in the design and implementation of CCM actions in Indonesia's energy and energy end use sectors.
	1. Biodiversity Conservation and Natural Resource Management	☐	
	2. Climate Change Mitigation and Adaptation	☐	
	3. Community Health, Safety and Working Conditions	X	Indonesia has existing policies and regulations on occupational health and safety, and these shall be fully used in the construction and operation of RE/EE technology installations that the project will support.
	4. Cultural Heritage	☐	
	5. Displacement and Resettlement	☐	
	6. Indigenous Peoples	☐	
	7. Pollution Prevention and Resource Efficiency	☒	The EIA requirements (where required), and in particular proper engineering design principles and codes/standards shall be emphasized in the design and operation of the RE/EE technology installations that will be supported by the project.

Final Sign Off

Signature	***Date***	***Description***
QA Assessor Verania Andria	01 August 2015	UNDP Programme Manager. Final signature confirms they have "checked" to ensure that the SESP is adequately conducted.
QA Approver Stephen Rodrigues	10 August 2015	UNDP Deputy Country Director (DCD).
PAC Chair		UNDP chair of the PAC. In some cases, PAC Chair may also be the QA Approver. Final signature confirms that the SESP was considered as part of the project appraisal and considered in recommendations of the PAC.

SESP Attachment G.1. Social and Environmental Risk Screening Checklist

Checklist Potential Social and Environmental Risks		Answer (Yes/No)
Principles 1: Human Rights		
1.	Could the Project lead to adverse impacts on enjoyment of the human rights (civil, political, economic, social or cultural) of the affected population and particularly of marginalized groups?	No
2.	Is there a likelihood that the Project would have inequitable or discriminatory adverse impacts on affected populations, particularly people living in poverty or marginalized or excluded individuals or groups? [25]	No
3.	Could the Project potentially restrict availability, quality of and access to resources or basic services, in particular to marginalized individuals or groups?	No
4.	Is there a likelihood that the Project would exclude any potentially affected stakeholders, in particular marginalized groups, from fully participating in decisions that may affect them?	No
5.	Is there a risk that duty-bearers do not have the capacity to meet their obligations in the Project?	No
6.	Is there a risk that rights-holders do not have the capacity to claim their rights?	No
7.	Have local communities or individuals, given the opportunity, raised human rights concerns regarding the Project during the stakeholder engagement process?	No
8.	Is there a risk that the Project would exacerbate conflicts among and/or the risk of violence to project-affected communities and individuals?	No
Principle 2: Gender Equality and Women's Empowerment		
1.	Is there a likelihood that the proposed Project would have adverse impacts on gender equality and/or the situation of women and girls?	Yes
2.	Would the Project potentially reproduce discriminations against women based on gender, especially regarding participation in design and implementation or access to opportunities and benefits?	No
3.	Have women's groups/leaders raised gender equality concerns regarding the Project during the stakeholder engagement process and has this been included in the overall Project proposal and in the risk assessment?	No
4.	Would the Project potentially limit women's ability to use, develop and protect natural resources, taking into account different roles and positions of women and men in accessing environmental goods and services? *For example, activities that could lead to natural resources degradation or depletion in communities who depend on these resources for their livelihoods and well being*	No
Principle 3: Environmental Sustainability: Screening questions regarding environmental risks are encompassed by the specific Standard-related questions below		
Standard 1: Biodiversity Conservation and Sustainable Natural Resource Management		
1.1	Would the Project potentially cause adverse impacts to habitats (e.g. modified, natural, and critical habitats) and/or ecosystems and ecosystem services? *For example, through habitat loss, conversion or degradation, fragmentation, hydrological changes*	No
1.2	Are any Project activities proposed within or adjacent to critical habitats and/or environmentally sensitive areas, including legally protected areas (e.g. nature reserve, national park), areas proposed for protection, or recognized as such by authoritative sources and/or indigenous peoples or local communities?	No
1.3	Does the Project involve changes to the use of lands and resources that may have adverse impacts on habitats, ecosystems, and/or livelihoods? (Note: if restrictions and/or limitations of access to lands would apply, refer to Standard 5)	No
1.4	Would Project activities pose risks to endangered species?	No
1.5	Would the Project pose a risk of introducing invasive alien species?	No
1.6	Does the Project involve harvesting of natural forests, plantation development, or reforestation?	No
1.7	Does the Project involve the production and/or harvesting of fish populations or other aquatic species?	No
1.8	Does the Project involve significant extraction, diversion or containment of surface or ground water? *For example, construction of dams, reservoirs, river basin developments, groundwater extraction*	No
1.9	Does the Project involve utilization of genetic resources? (e.g. collection and/or harvesting, commercial development)	No
1.10	Would the Project generate potential adverse transboundary or global environmental concerns?	No
1.11	Would the Project result in secondary or consequential development activities which could lead to adverse social and environmental effects, or would it generate cumulative impacts with other known existing or planned activities in the area? *For example, a new road through forested lands will generate direct environmental and social impacts (e.g. felling of trees, earthworks, potential relocation of inhabitants). The new road may also facilitate encroachment on lands by illegal settlers or generate unplanned commercial development along the route,*	No

[25] Prohibited grounds of discrimination include race, ethnicity, gender, age, language, disability, sexual orientation, religion, political or other opinion, national or social or geographical origin, property, birth or other status including as an indigenous person or as a member of a minority. References to "women and men" or similar is understood to include women and men, boys and girls, and other groups discriminated against based on their gender identities, such as transgender people and transsexuals.

	potentially in sensitive areas. These are indirect, secondary, or induced impacts that need to be considered. Also, if similar developments in the same forested area are planned, then cumulative impacts of multiple activities (even if not part of the same Project) need to be considered.	
Standard 2: Climate Change Mitigation and Adaptation		
2.1	Will the proposed Project result in significant[26] greenhouse gas emissions or may exacerbate climate change?	No
2.2	Would the potential outcomes of the Project be sensitive or vulnerable to potential impacts of climate change?	No
2.3	Is the proposed Project likely to directly or indirectly increase social and environmental vulnerability to climate change now or in the future (also known as maladaptive practices)? *For example, changes to land use planning may encourage further development of floodplains, potentially increasing the population's vulnerability to climate change, specifically flooding*	No
Standard 3: Community Health, Safety and Working Conditions		
3.1	Would elements of Project construction, operation, or decommissioning pose potential safety risks to local communities?	Yes
3.2	Would the Project pose potential risks to community health and safety due to the transport, storage, and use and/or disposal of hazardous or dangerous materials (e.g. explosives, fuel and other chemicals during construction and operation)?	No
3.3	Does the Project involve large-scale infrastructure development (e.g. dams, roads, buildings)?	No
3.4	Would failure of structural elements of the Project pose risks to communities? (e.g. collapse of buildings or infrastructure)	No
3.5	Would the proposed Project be susceptible to or lead to increased vulnerability to earthquakes, subsidence, landslides, and erosion, flooding or extreme climatic conditions?	No
3.6	Would the Project result in potential increased health risks (e.g. from water-borne or other vector-borne diseases or communicable infections such as HIV/AIDS)?	No
3.7	Does the Project pose potential risks and vulnerabilities related to occupational health and safety due to physical, chemical, biological, and radiological hazards during Project construction, operation, or decommissioning?	No
3.8	Does the Project involve support for employment or livelihoods that may fail to comply with national and international labor standards (i.e. principles and standards of ILO fundamental conventions)?	No
3.9	Does the Project engage security personnel that may pose a potential risk to health and safety of communities and/or individuals (e.g. due to a lack of adequate training or accountability)?	No
Standard 4: Cultural Heritage		
4.1	Will the proposed Project result in interventions that would potentially adversely impact sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture (e.g. knowledge, innovations, practices)? (Note: Projects intended to protect and conserve Cultural Heritage may also have inadvertent adverse impacts)	No
4.2	Does the Project propose utilizing tangible and/or intangible forms of cultural heritage for commercial or other purposes?	No
Standard 5: Displacement and Resettlement		
5.1	Would the Project potentially involve temporary or permanent and full or partial physical displacement?	No
5.2	Would the Project possibly result in economic displacement (e.g. loss of assets or access to resources due to land acquisition or access restrictions – even in the absence of physical relocation)?	No
5.3	Is there a risk that the Project would lead to forced evictions?[27]	No
5.4	Would the proposed Project possibly affect land tenure arrangements and/or community based property rights/customary rights to land, territories and/or resources?	No
Standard 6: Indigenous Peoples		
6.1	Are indigenous peoples present in the Project area (including Project area of influence)?	No
6.2	Is it likely that the Project or portions of the Project will be located on lands and territories claimed by indigenous peoples?	No
6.3	Would the proposed Project potentially affect the human rights, lands, natural resources, territories, and traditional livelihoods of indigenous peoples (regardless of whether indigenous peoples possess the legal titles to such areas, whether the Project is located within or outside of the lands and territories inhabited by the affected peoples, or whether the indigenous peoples are recognized as indigenous peoples by the country in question)? *If the answer to the screening question 6.3 is "yes" the potential risk impacts are considered potentially severe and/or critical and the Project would be categorized as either Moderate or High Risk.*	No

[26] In regards to CO_2, 'significant emissions' corresponds generally to more than 25,000 tons per year (from both direct and indirect sources). [The Guidance Note on Climate Change Mitigation and Adaptation provides additional information on GHG emissions.]

[27] Forced evictions include acts and/or omissions involving the coerced or involuntary displacement of individuals, groups, or communities from homes and/or lands and common property resources that were occupied or depended upon, thus eliminating the ability of an individual, group, or community to reside or work in a particular dwelling, residence, or location without the provision of, and access to, appropriate forms of legal or other protections.

6.4	Has there been an absence of culturally appropriate consultations carried out with the objective of achieving FPIC on matters that may affect the rights and interests, lands, resources, territories and traditional livelihoods of the indigenous peoples concerned?	No
6.5	Does the proposed Project involve the utilization and/or commercial development of natural resources on lands and territories claimed by indigenous peoples?	No
6.6	Is there a potential for forced eviction or the whole or partial physical or economic displacement of indigenous peoples, including through access restrictions to lands, territories, and resources?	No
6.7	Would the Project adversely affect the development priorities of indigenous peoples as defined by them?	No
6.8	Would the Project potentially affect the physical and cultural survival of indigenous peoples?	No
6.9	Would the Project potentially affect the Cultural Heritage of indigenous peoples, including through the commercialization or use of their traditional knowledge and practices?	No
Standard 7: Pollution Prevention and Resource Efficiency		
7.1	Would the Project potentially result in the release of pollutants to the environment due to routine or non-routine circumstances with the potential for adverse local, regional, and/or transboundary impacts?	Yes
7.2	Would the proposed Project potentially result in the generation of waste (both hazardous and non-hazardous)?	No
7.3	Will the proposed Project potentially involve the manufacture, trade, release, and/or use of hazardous chemicals and/or materials? Does the Project propose use of chemicals or materials subject to international bans or phase-outs? *For example, DDT, PCBs and other chemicals listed in international conventions such as the Stockholm Conventions on Persistent Organic Pollutants or the Montreal Protocol*	No
7.4	Will the proposed Project involve the application of pesticides that may have a negative effect on the environment or human health?	No
7.5	Does the Project include activities that require significant consumption of raw materials, energy, and/or water?	No